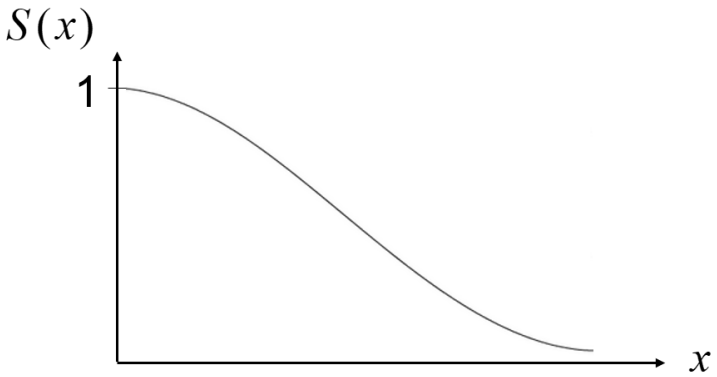
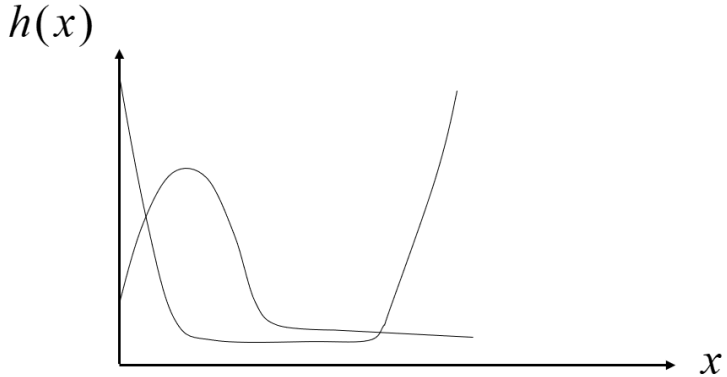
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Analysis of Survival Data

WELCOME!





Lecture 1: Introduction

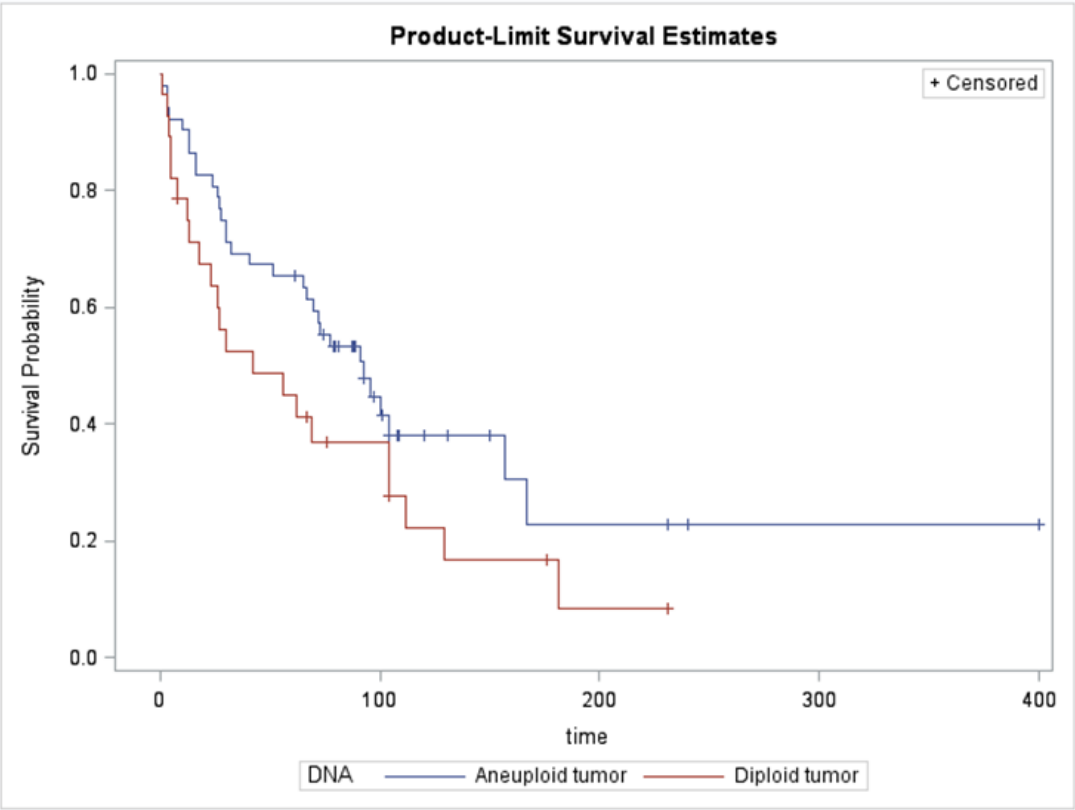
Inger Persson

Program L1

- **Survival analysis - introduction**
  - What is this course about?
  - The “flipped classroom”
- **Basic quantities and models**
  - Survival data
  - Basic concepts in Survival analysis
  - Common parametric models for survival data
  - Regression models for survival data

What is this course about?

- **Estimating survival probabilities**



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What is this course about?

- **Proportional hazards regression**

$$h(t \mid \mathbf{Z}) = h_0(t)c(\boldsymbol{\beta}^t \mathbf{Z})$$

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Learning outcomes

After this course you should:

- have basic knowledge of the most common methods for the analysis of survival data
- be able to apply these methods on data from the medical field and other subject areas
- be able to assimilate the content of scientific articles based on survival data
- have the ability to both orally and in written form account for results of analyses based on methods for survival data

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Why SAS?

We'll be using the SAS software, and there are two main resons for doing so:

- Future employers in biostatistics use SAS (as it is the only "validated" statistical software and as such fulfils requirements of e.g. medical agencies)
- Some of the functions we'll be using are not (yet) implemented in R

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Examination

- **Home assignments (using SAS)**

  1) Estimation of basic quantities and hypothesis testing
  2) Cox proportional hazards regression
  3) Article review
  4) Oral presentation

- **Final exam (pen and paper)**

  See Ladok for date and registration

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Grades

You will get credits for the home assignment part and the in-class examination separately, but the course will be marked as complete only if you pass all parts.

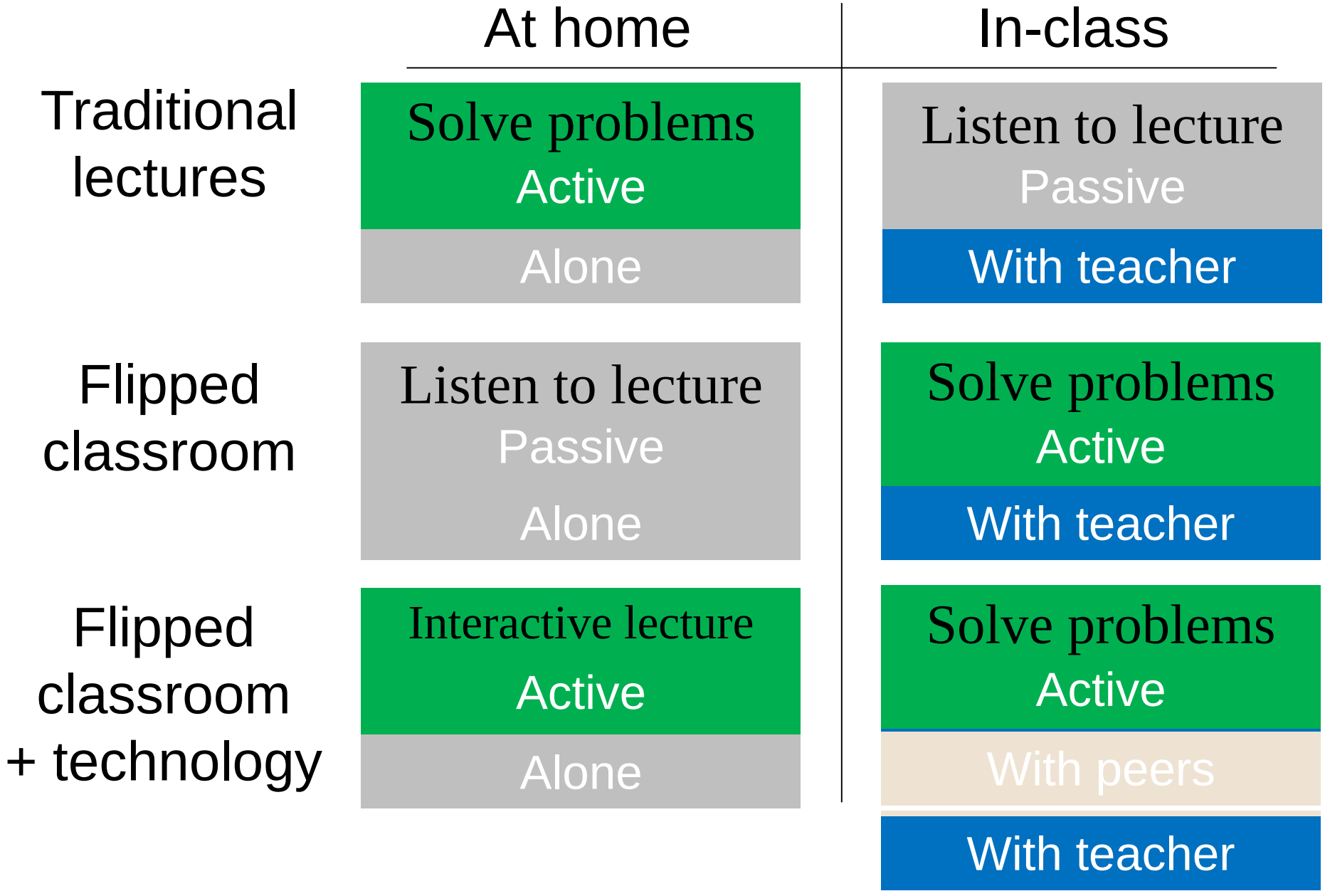
To **pass** with distinction (VG), you need to score 90% on the exam (60% for G), including any bonus points from home assignments and attendance.

Bonus points from home assignments:
If you pass on the first try you get 2 bonus points (a total of 4/6* bonus points if you pass all assignments on the first try)

Any bonus points will be added to your exam score.

* Math students: 2 home assignments (4 possible bonus points). One less task on the exam.
* Stat students: 3 home assignments (6 possible bonus points).

Flipping the classroom



Presentation borrowed from David Black-Schaffer, IT dept. UU

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”Interactive lecture”

Short lecture segments + self-assessment quizzes

Max 5-10 minutes video, then a quiz

Online lectures posted as Assignments at Studium (most web browsers except Safari should work)

The ”flipped classroom”

This only works if you prepare, and show up in class!

Bonus points for attendance:

- Watch all online lectures on time: 2 bonus points
- Attend 6/7 of the in-class meetings: 2 bonus points

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Why?

How much do we remember?



Lectures

Remarkable lectures

Video lectures (stop, rewind, repeat)

Video lectures with built-in quizzes

Discuss the content with someone else

Explain the content to someone else

In-class sessions

- **Online lecture follow-up**
  Any parts from the online lectures that you had trouble understanding
- **Review questions**
  Based on the content from online lecture and book
  IPA (individually, peers, all)
- **Exercises**
  Pen and paper or SAS

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Basic rules

- **Be on time!**

  I will start on time, and I expect you to be here when we start.

  Our time is limited and valuable, it is very disturbing to me and to your fellow class mates if you drop in late.

- **Keep quiet (when I talk)**

  When you have questions, raise them out loud.

- **Interrupt at any time**

  Questions are always welcome!

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Survival data

Data involving time to the occurrence of a certain event.

Examples of events:

- Death (which accounts for the name)
- The appearance of a tumor
- The development of some disease
- Conception (pregnancy)
- Cessation of smoking (quitting)

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Survival data

Applications of statistical methods for survival data analysis have been extended beyond biomedical research to other fields, e.g.:

- Lifetime of electronic devices (reliability engineering)
- Felon's time to parole (criminology)
- Duration of first marriage (sociology)
- Length of app subscription (marketing)
- Worker's compensation claims (insurance)
- etc…

Survival analysis

Survival analysis encompasses a wide variety of methods for analyzing the timing of events.

Terminology varies:

- **Survival analysis** in biostatistics, which has the richest tradition in this area
- **Failure-time analysis** (or reliability analysis) in engineering
- **Event-history analysis** in sociology
- **Time-to-event analysis** more generally

Survival analysis

X = time to some specified event

$X \geq 0$

Different ways of describing X:

1) Survival function $S(x)$
2) Probability density function $f(x)$
3) Hazard function (rate) $h(x)$
4) Cumulative Hazard function $H(x)$
5) Mean Residual Life and Median lifetime

1) Survival function

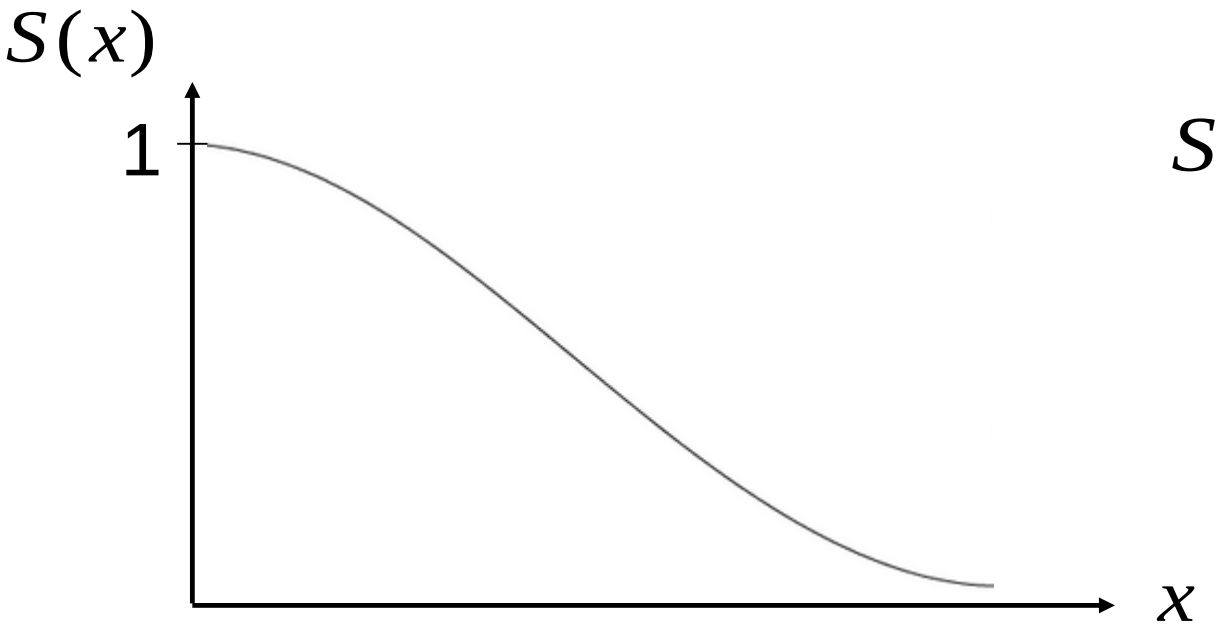
$$S(x) = \Pr(X > x) = 1 - F(x)$$

$$F(x) = \Pr(X \leq x)$$

The probability to experience the event beyond a certain time, x.

E.g. the probability to survive beyond time x.

1) Survival function



$$S(x) = \int_x^\infty f(t)dt$$

Probability density function

A non-increasing function, equal to 1 at time 0 and equal to 0 as time approaches infinity.

2) Probability density function

$$f(x) = -\frac{dS(x)}{d(x)}$$

A nonnegative function, with the area under *f(x)* equal to 1.

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$S(x) = \int_{x}^{\infty} f(t)dt$$

Example: All cause mortality in the US

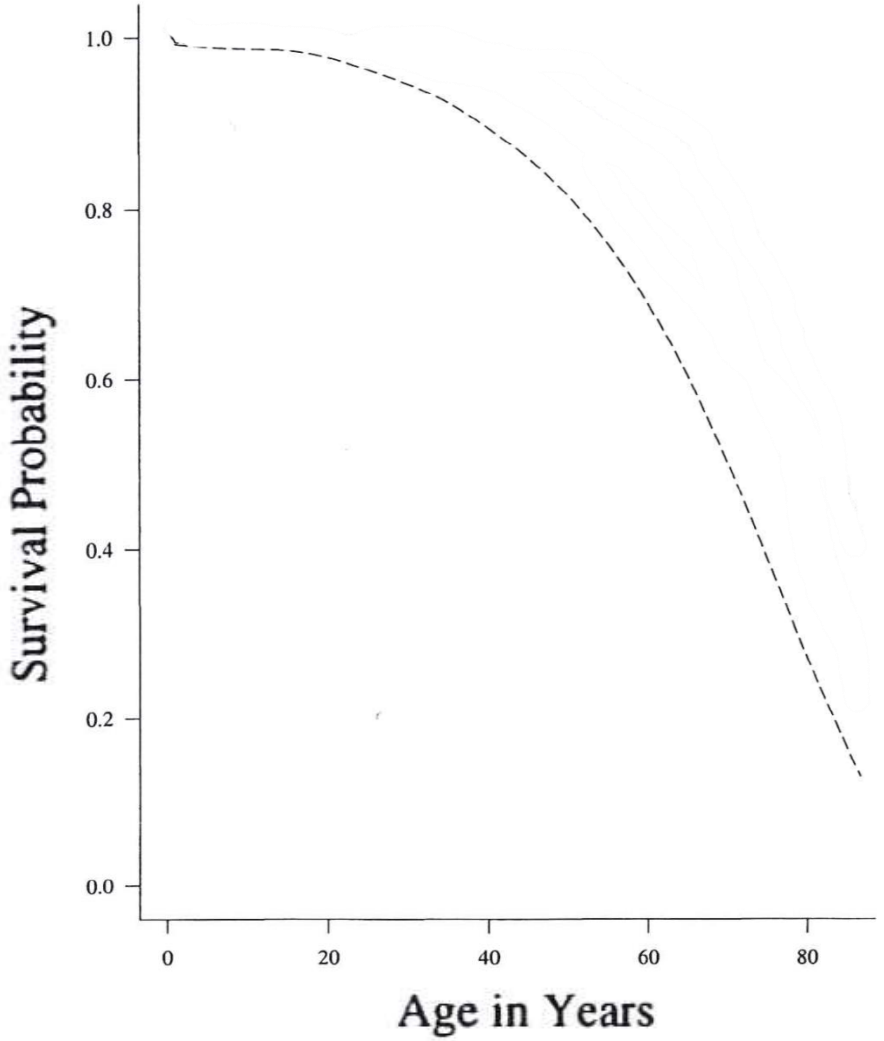


Figure 2.2 *Survival Functions for all cause mortality for the US population in 1989. White males (———); white females (·····); black males (------); black females (— — —).*

Survival function for discrete X

The previous definitions hold when X is a continuous random variable.

When X is a discrete random variable:

$$S(x) = \Pr(X > x) = \sum_{x_j > x} p(x_j)$$

Probability mass function

$$p(x_j) = \Pr(X = x_j)$$

3) Hazard function (intensity function)

$$h(x) = \lim_{\Delta x \to 0} \frac{\Pr(x \leq X < x + \Delta x \mid X \geq x)}{\Delta x}$$

Conditional on the fact that you still haven't experienced the event at time x.

$h(x)\,\Delta x$ = "approximate" probability of an individual to experience the event in the next instant $(x + \Delta x)$.

Describes how the probability of experiencing the event changes over time.

3) Hazard function (intensity function)

$$h(x) = \lim_{\Delta x \to 0} \frac{\Pr(x \le X < x + \Delta x \mid X \ge x)}{\Delta x}$$

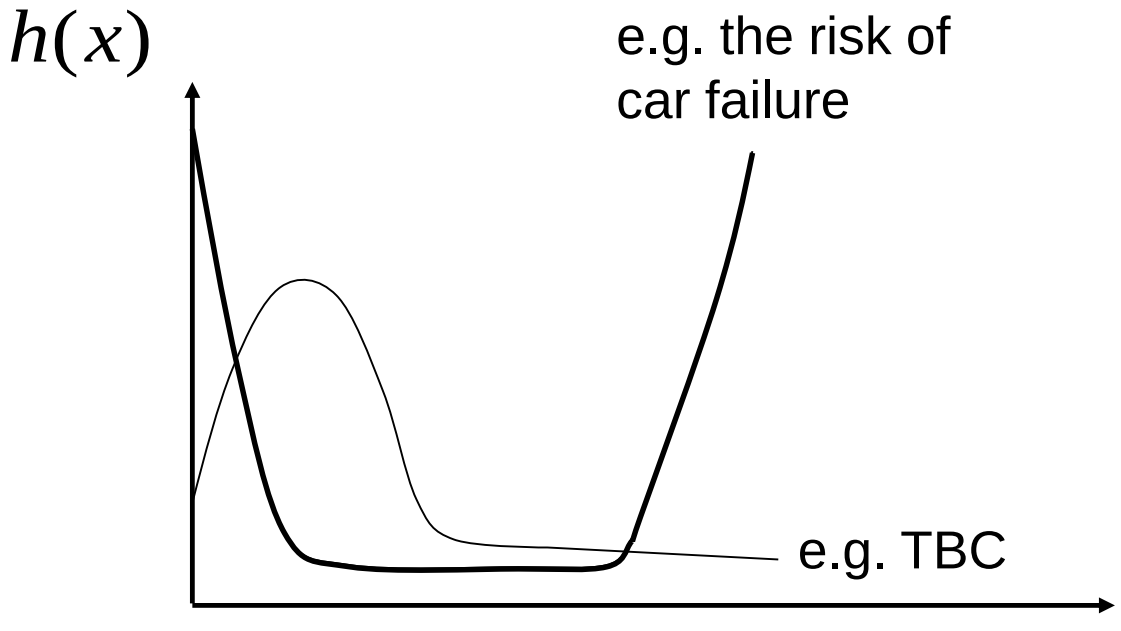
Conditional on the fact that you still haven't experienced the event at time x.

Compare:

$$f(x) = \lim_{\Delta x \to 0} \frac{\Pr(x \le X < x + \Delta x)}{\Delta x}$$

$$h(x) = \frac{f(x)}{S(x)} = -\frac{d \ln(S(x))}{dx}$$

3) Hazard function (intensity function)



$\int_a^b h(x)dx$ can be > 1 (!)

Only one restriction: $h(x) \geq 0$

4) Cumulative Hazard function

$$H(x) = \int_0^x h(t)dt \; = -\ln\big(S(x)\big)$$

$$S(x) = e^{-H(x)} \; = e^{-\int_0^x h(t)dt}$$ (for continuous lifetimes)

Denotes the total amount of risk accumulated up to a certain time point, i.e. the number of times we would expect to observe events over a given time period, if events were repeatable.

Example: Survival vs. Hazard function

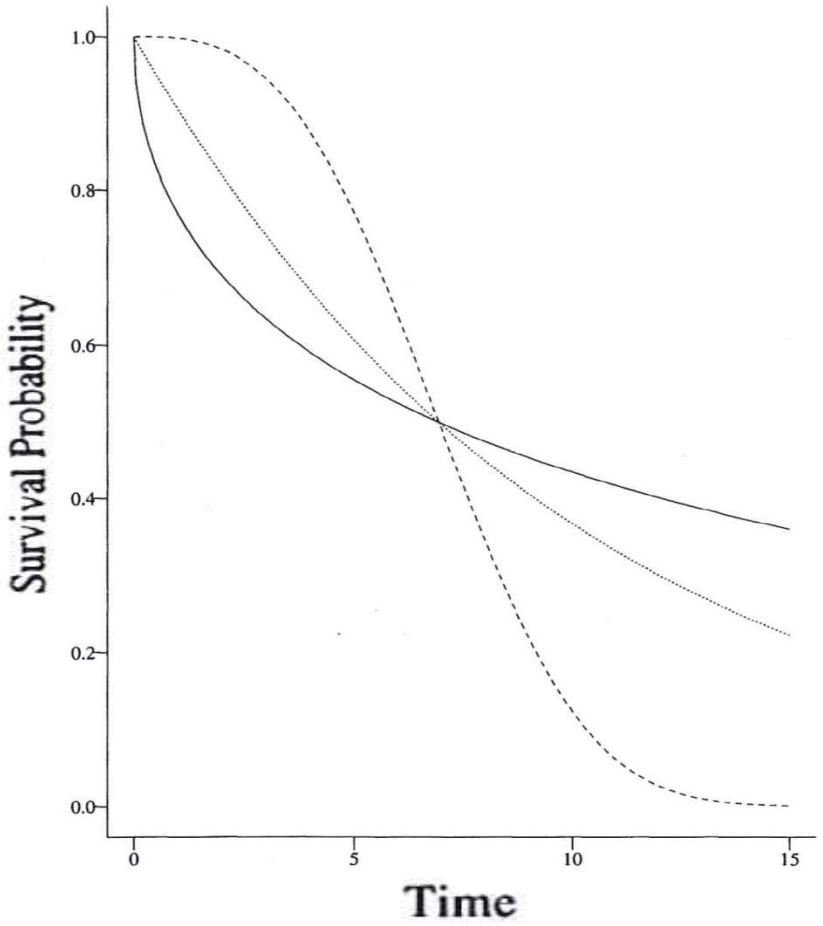


Figure 2.1 *Weibull Survival functions for* $\alpha = 0.5$, $\lambda = 0.26328$ (————); $\alpha = 1.0$, $\lambda = 0.1$ (·····); $\alpha = 3.0$, $\lambda = 0.00208$ (------).

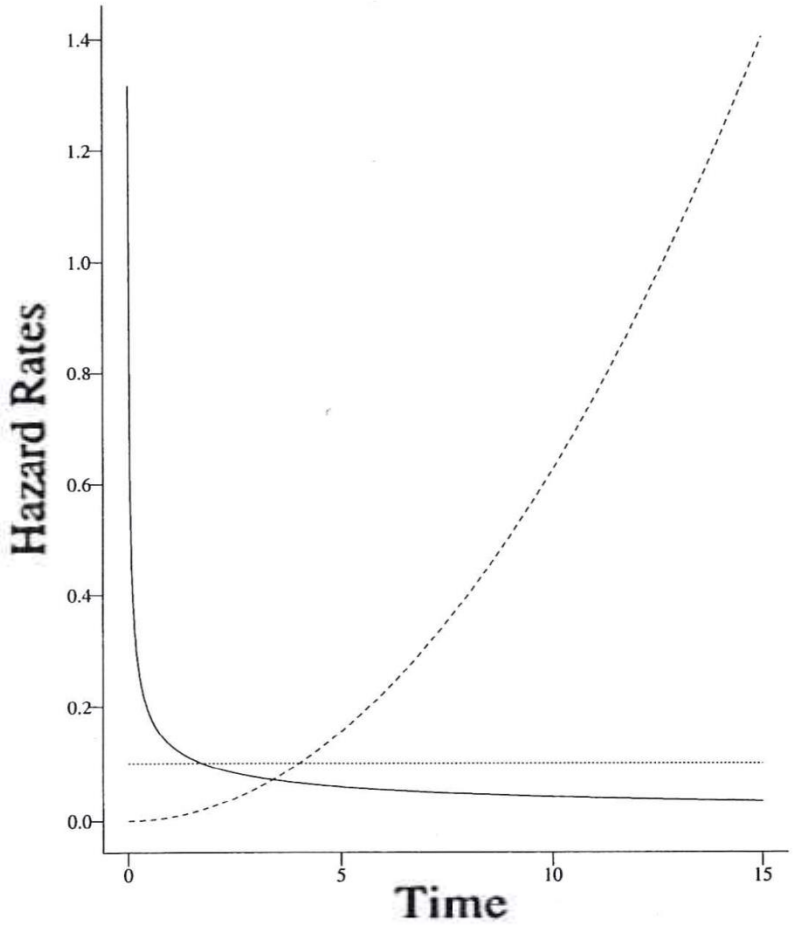


Figure 2.5 *Weibull hazard functions for* $\alpha = 0.5$, $\lambda = 0.26328$ (————); $\alpha = 1.0$, $\lambda = 0.1$ (------); $\alpha = 3.0$, $\lambda = 0.00208$ (— — —).

5) Mean Residual Life

$$\text{mrl}(x) = E(X - x \mid X > x)$$

The expected remaining lifetime at time x.

$\text{mrl}(0) = \mu$ = average lifetime (from birth)

Quantiles of the survival data distribution

The pth quantile (or the 100pth percentile) of the distribution of X is the smallest x_p that fulfills

$$S(x_p) \leq 1 - p$$

The time where 100p% have experienced the event.

Example: All cause mortality in the US

$$S(x_{0.2}) \leq 0.8$$

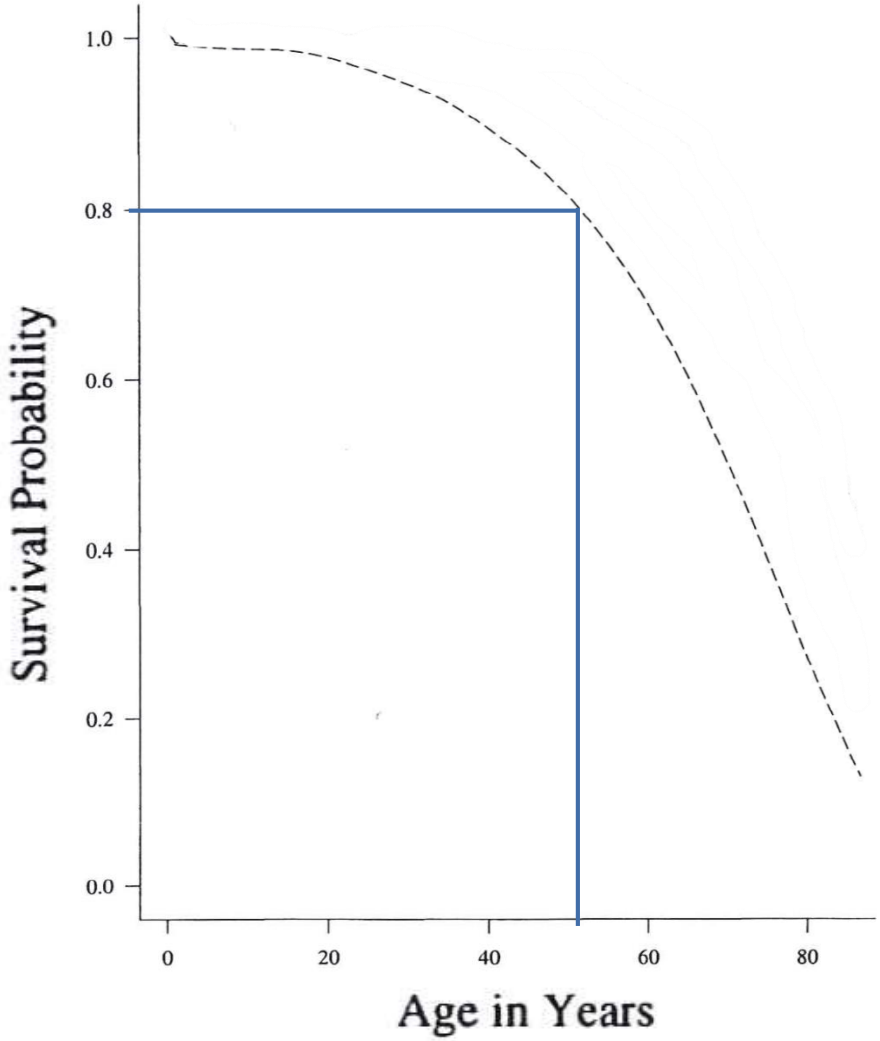


Figure 2.2 *Survival Functions for all cause mortality for the US population in 1989. White males (———); white females (·····) black males (------) females (— — —).*

The 20th percentile of the survival time distribution is approximately 50 years

Median Lifetime

The **median lifetime** is the time at which half of the individuals have experienced the event.

The median lifetime is the 50th percentile of the distribution of X.

In other words, the median lifetime is the value $x_{0.5}$ that fulfills

$$S(x_{0.5}) \leq 0.5$$

Example: All cause mortality in the US

$$S(x_{0.5}) \leq 0.5$$

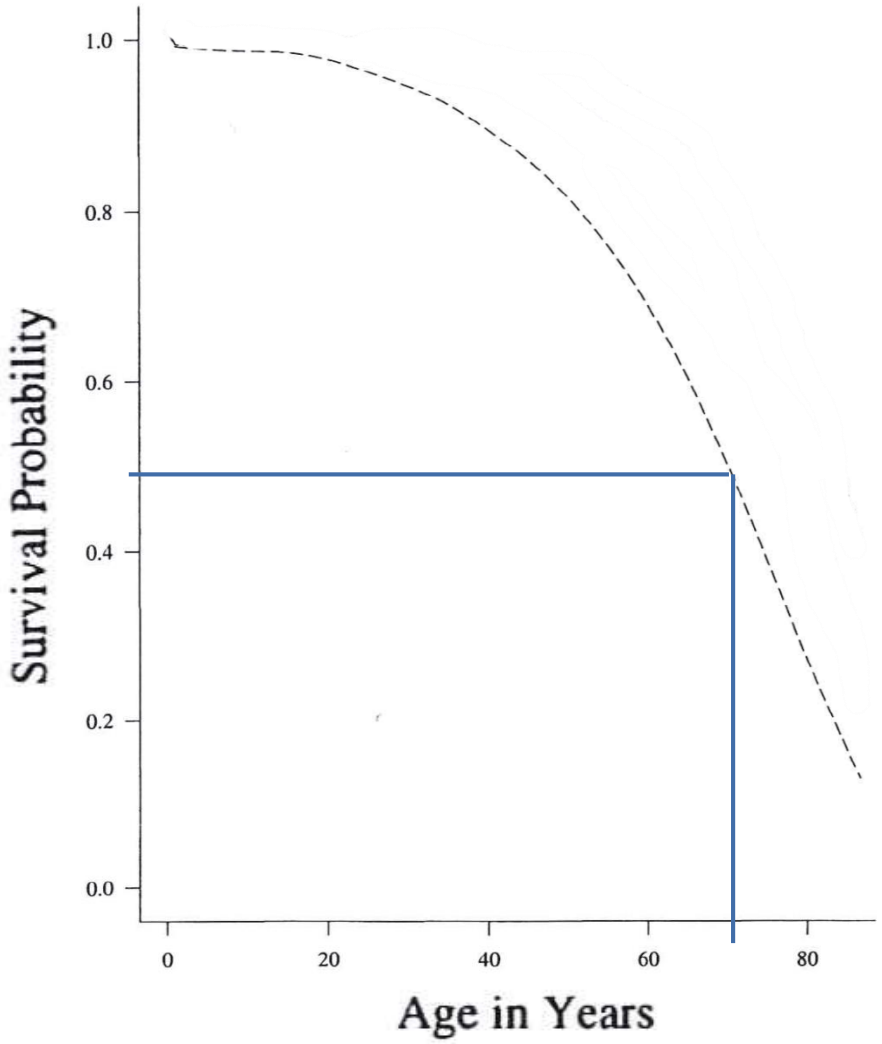


Figure 2.2 *Survival Functions for all cause mortality for the US population in 1989. White males (———); white females (·····); black males (------); black females (— — —).*

The median lifetime is approximately 70 years

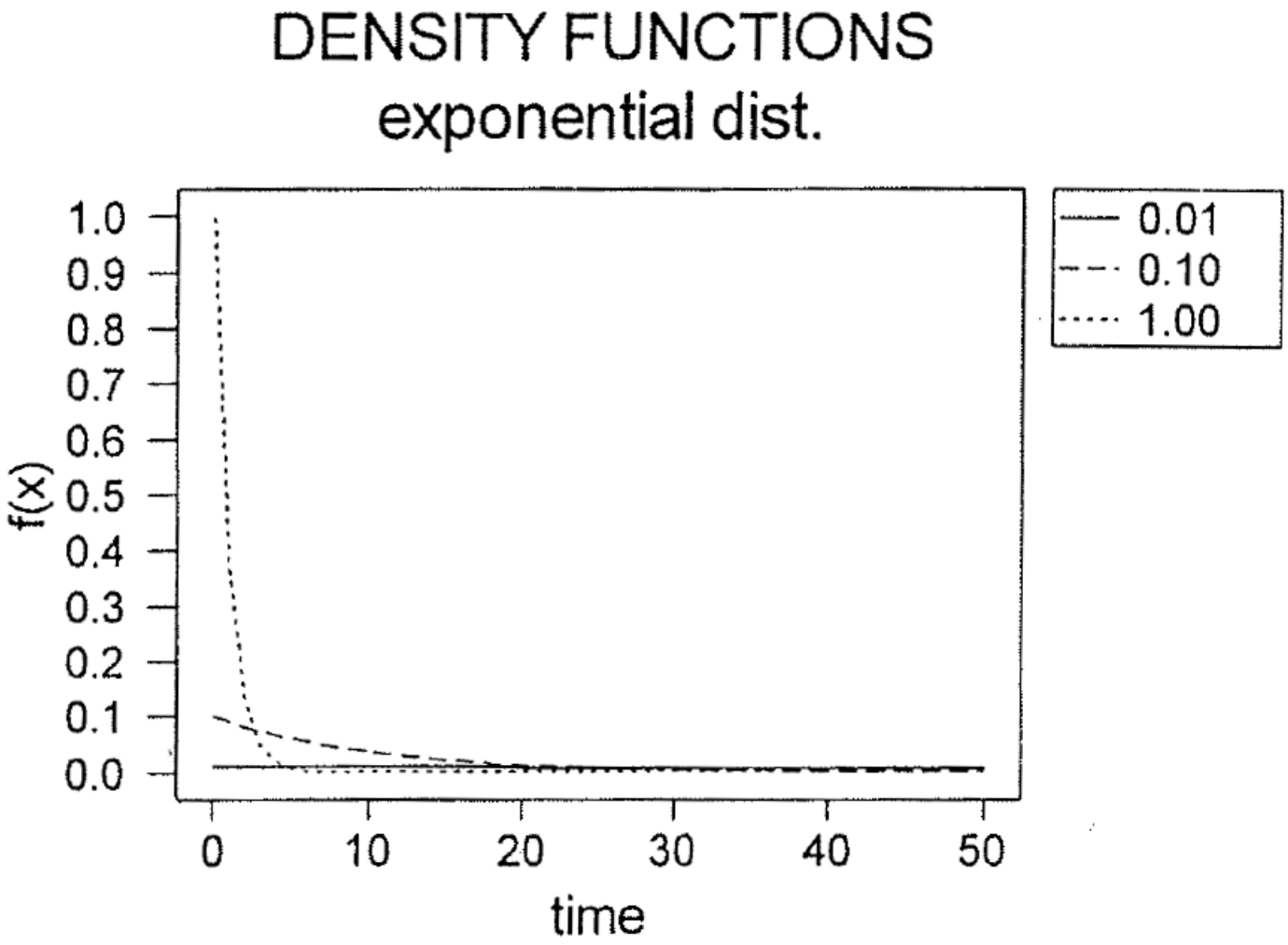
Common parametric models

Survival data can follow a number of parametric distributions, e.g.

- Exponential
- Weibull
- Gamma
- Log normal
- Log logistic
- etc..

Exponential distribution

$$f(x) = \lambda e^{(-\lambda x)}$$

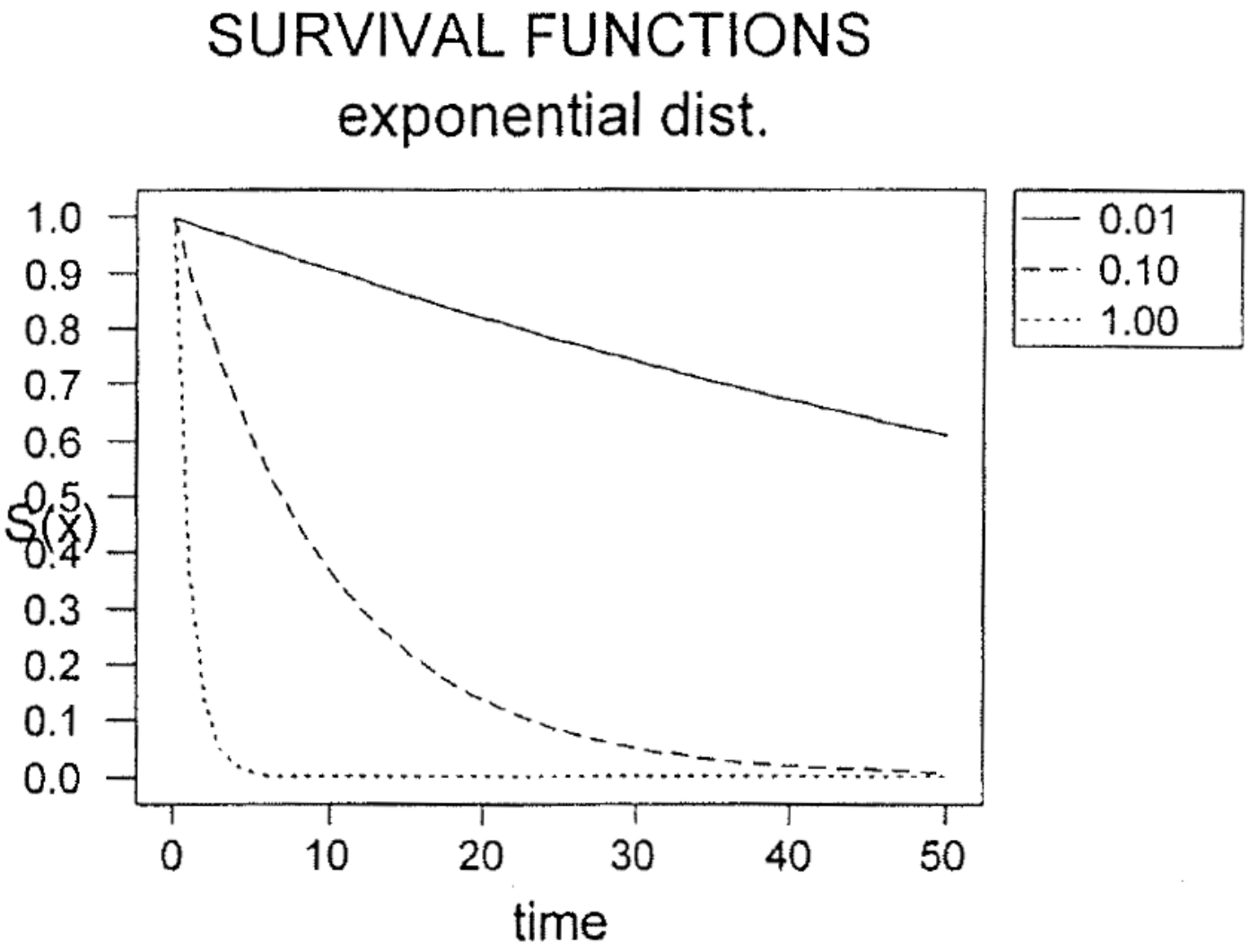


Exponential distribution

$$S(x) = \int_x^\infty f(t)dt$$

$$= \int_x^\infty \lambda e^{(-\lambda t)} dt$$

$$\boxed{S(x) = e^{(-\lambda x)}}$$



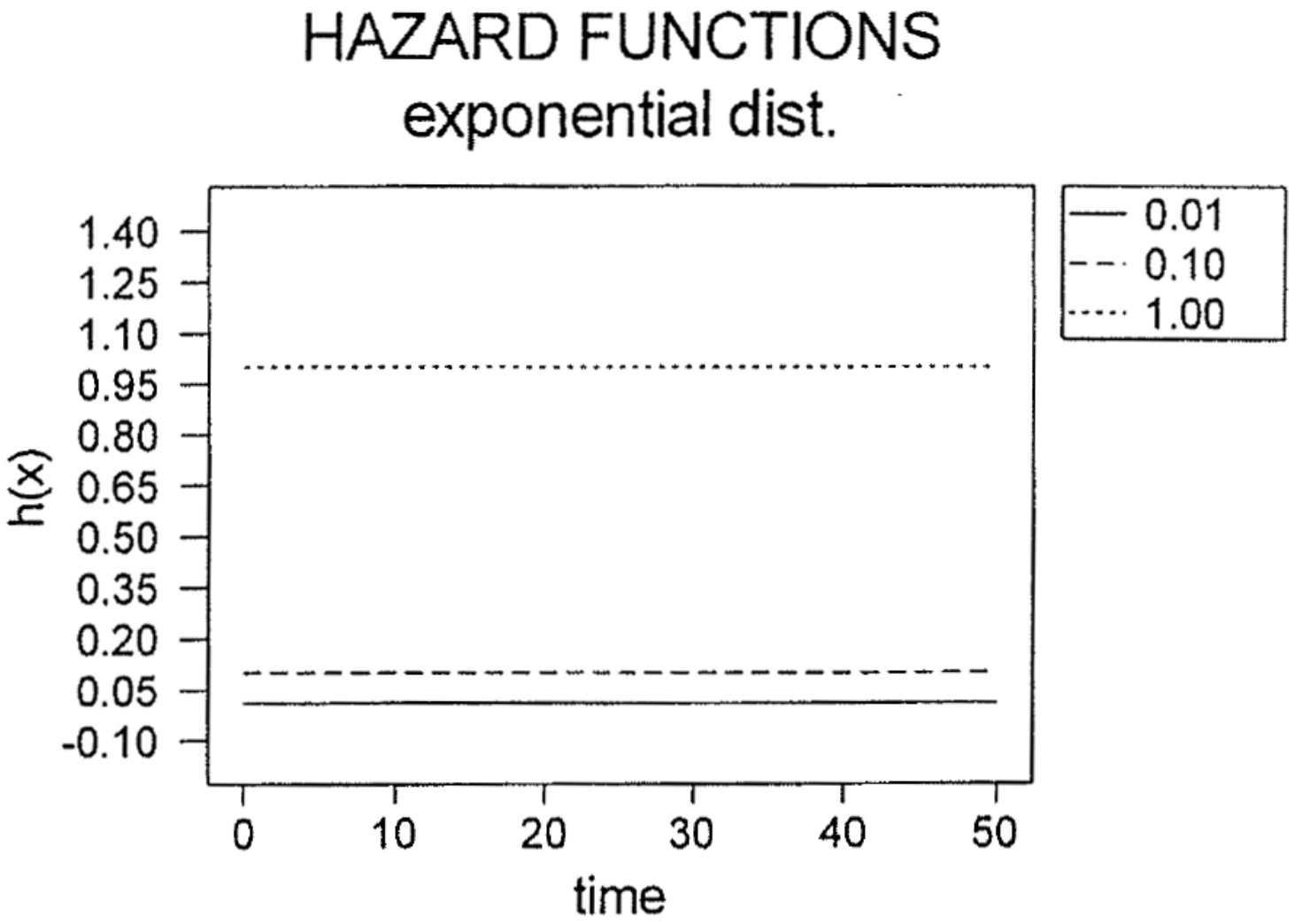
Exponential distribution

$$h(x) = \frac{f(x)}{S(x)}$$

$$= \frac{\lambda e^{(-\lambda x)}}{e^{(-\lambda x)}}$$

$$h(x) = \lambda$$

Constant hazard function



Exponential distribution

A constant hazard rate means that the probability to experience the event at any time t, does not depend on t.

The constant hazard rate also means that the mean residual life (the expected remaining lifetime) is constant.

Weibull distribution

$$f(x) = \alpha\lambda x^{\alpha-1} e^{(-\lambda x^{\alpha})}$$

$$S(x) = \int_x^{\infty} f(t)dt = \int_x^{\infty} \alpha\lambda t^{\alpha-1} e^{(-\lambda t^{\alpha})} dt$$

$$S(x) = e^{(-\lambda x^{\alpha})}$$

$$h(x) = \frac{f(x)}{S(x)} = \frac{\alpha\lambda x^{\alpha-1} \cancel{e^{(-\lambda x^{\alpha})}}}{\cancel{e^{(-\lambda x^{\alpha})}}}$$

$$h(x) = \alpha\lambda x^{\alpha-1}$$

Weibull distribution

Weibull hazard rates can be increasing ($\alpha > 1$), decreasing ($\alpha < 1$), or constant ($\alpha = 1$),

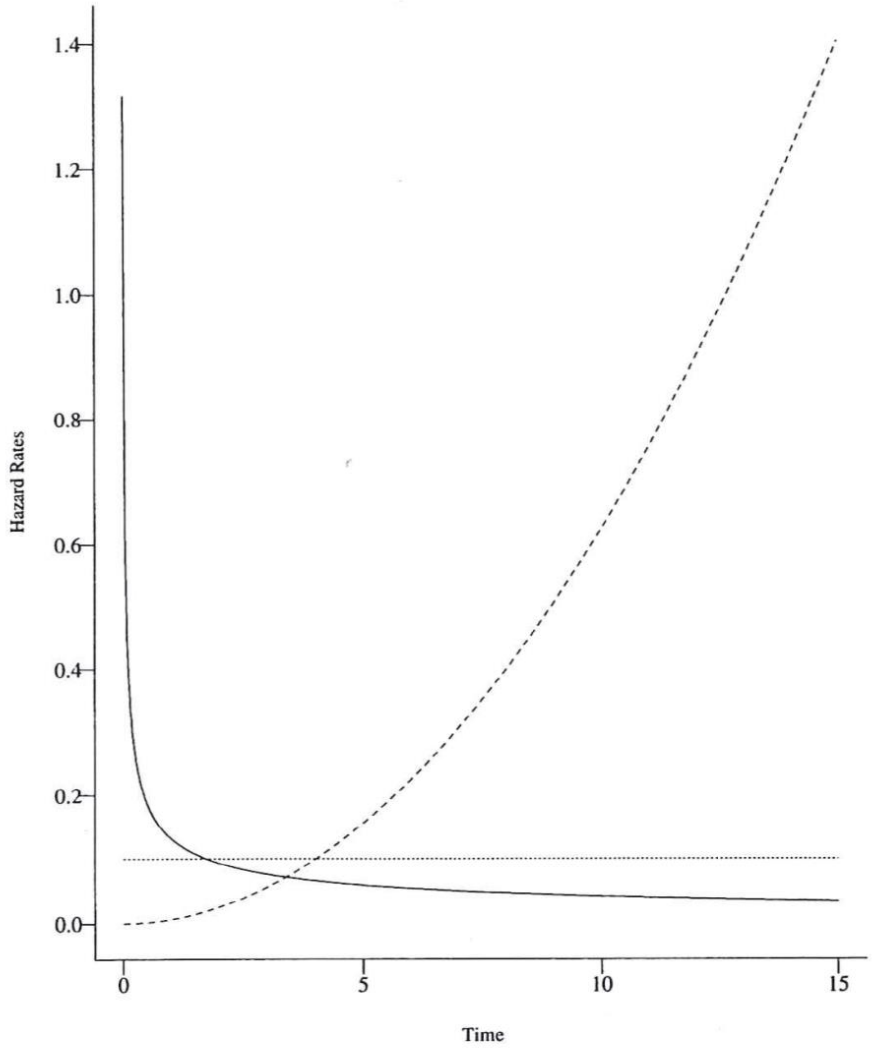


Figure 2.5 *Weibull hazard functions for* $\alpha = 0.5$, $\lambda = 0.26328$ (————); $\alpha = 1.0$, $\lambda = 0.1$ (------); $\alpha = 3.0$, $\lambda = 0.00208$ (— — —).

Log normal distribution

$$f(x) = \frac{e^{\left(-\frac{1}{2}\left(\frac{\ln x - \mu}{\sigma}\right)^2\right)}}{x(2\pi)^{1/2}\sigma} = \phi\left(\frac{\ln x - \mu}{\sigma}\right) / x$$

Density function of a standard Normal variable.

$$S(x) = 1 - \Phi\left(\frac{\ln x - \mu}{\sigma}\right)$$

Cumulative distribution function of a standard Normal variable.

$$h(x) = \frac{f(x)}{S(x)}$$

Log normal distribution

Log normal hazard rates are hump-shaped

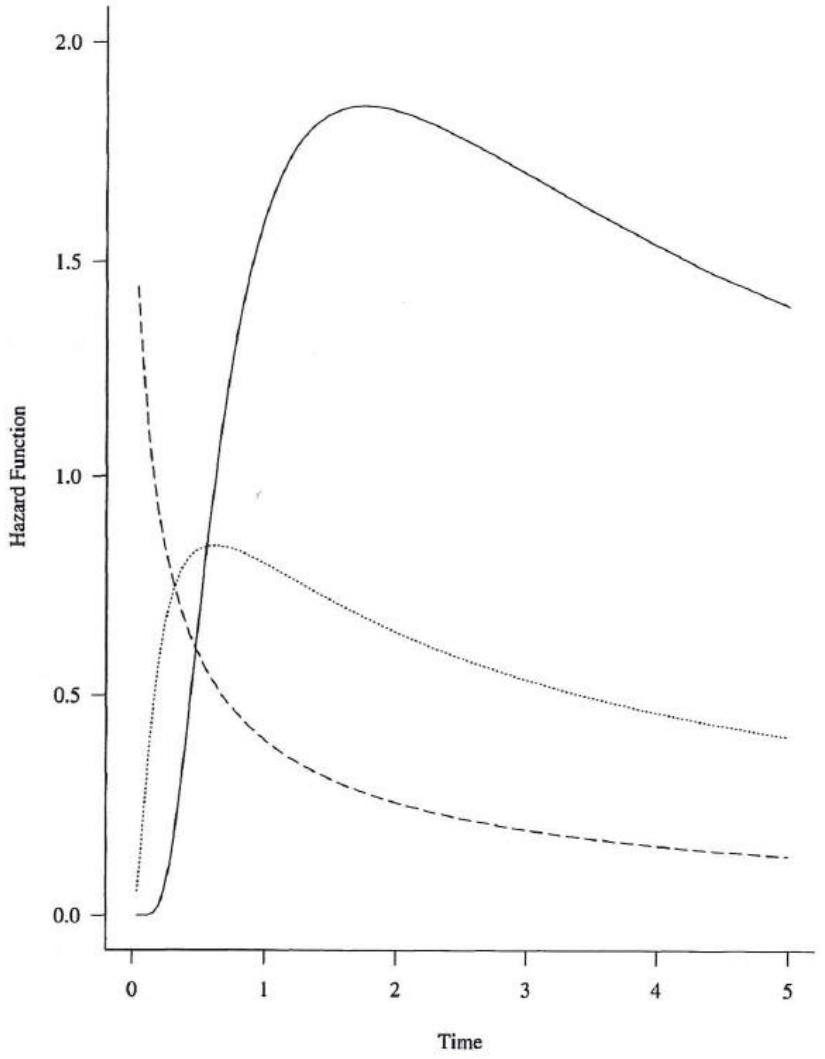


Figure 2.8 *Log normal hazard rates.* $\mu = 0, \sigma = 0.5$ (———); $\mu = 0, \sigma = 0.1$ (------); $\mu = 0, \sigma = 2.0$ (— — —)

Log logistic distribution

Log logistic hazard rates can be monotone decreasing ($1/\sigma \leq 1$), or hump-shaped ($1/\sigma > 1$)

Hazard rates similar to the log normal, but with simpler hazard and survival functions.

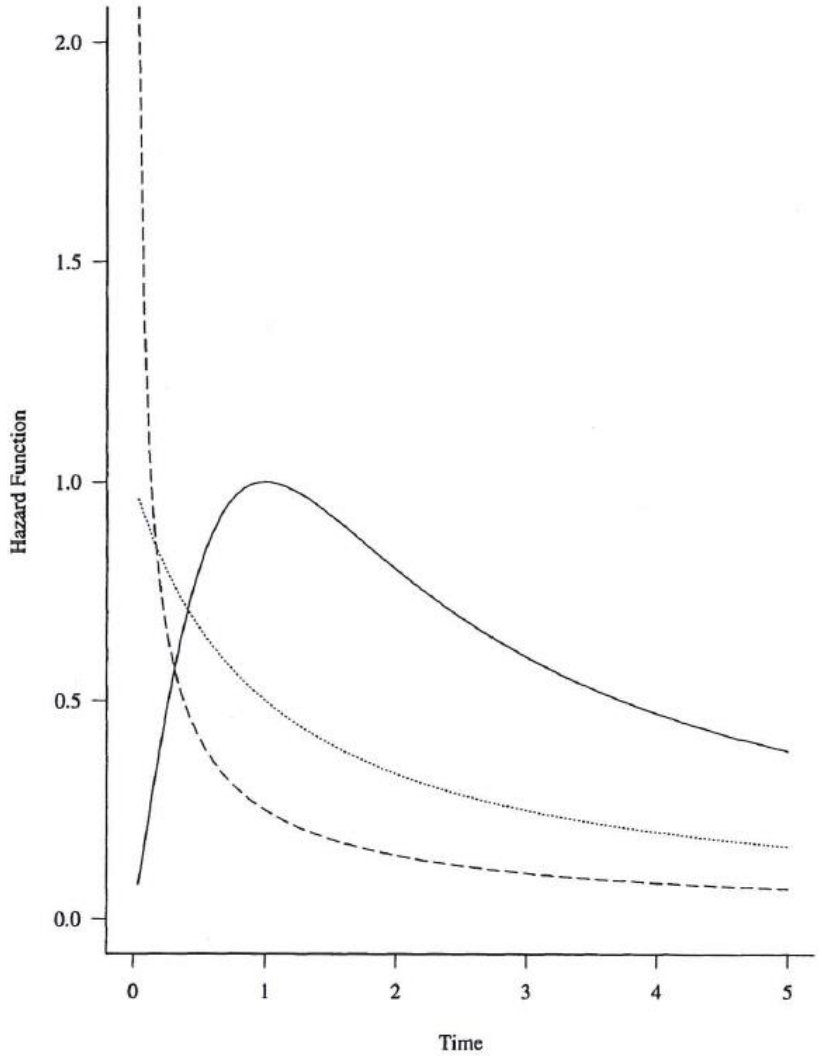


Figure 2.9 *Log logistic hazard rates.* $\lambda = 1, \sigma = 0.5$ (———); $\lambda = 1, \sigma = 1.0$ (·····); $\lambda = 1, \sigma = 2.0$ (------)

Gamma distribution

Gamma hazard rates can be monotone increasing ($\beta > 1$), monotone decreasing ($\beta < 1$), or constant ($\beta = 1$)

Properties similar to the Weibull distribution but mathematically more complicated.

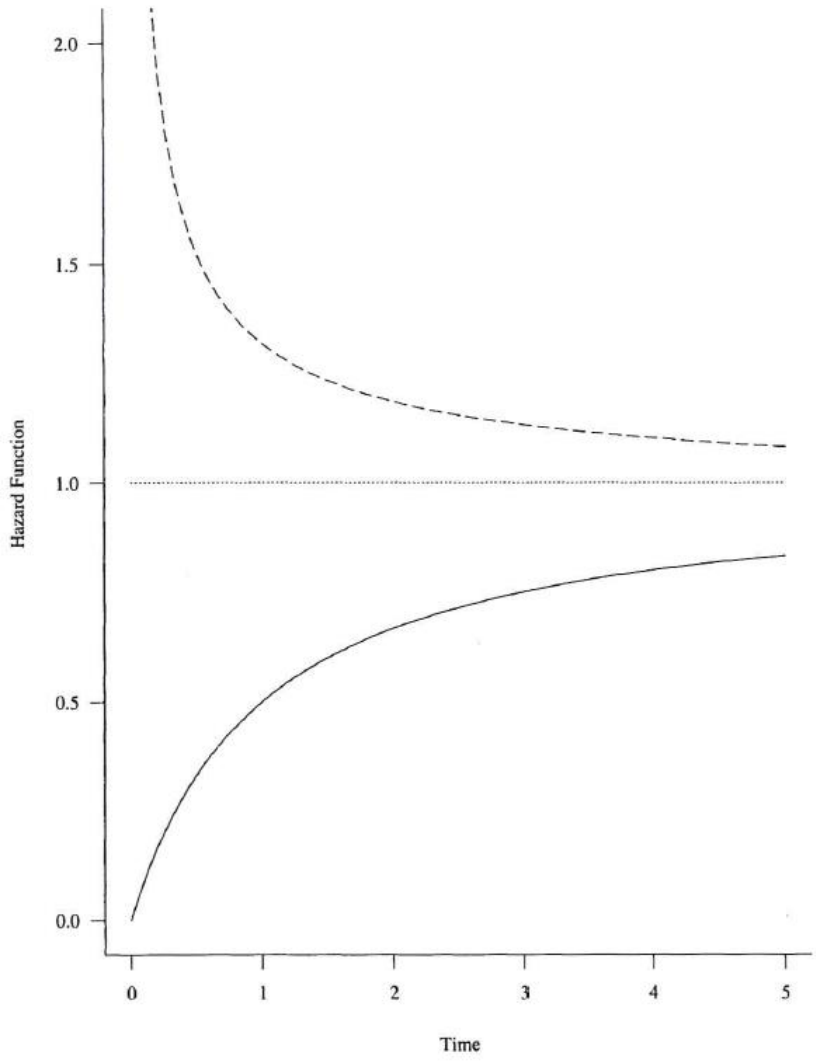


Figure 2.10 *Gamma hazard rates.* $\lambda = 1, \beta = 2.0$ (————); $\lambda = 1, \beta = 1.0$ (------); $\lambda = 1, \beta = 0.5$ (— — —)

Estimating parametric model parameters

Once a parametric distribution is known the model parameters (λ, α, σ, etc) can be estimated by maximum likelihood methods.

There are also methods of estimating the fit of a certain distribution (e.g. Kolmogorov-Smirnov test).

Regression models for survival data

As for other types of data, you might wish to investigate the relationship between the survival time X and one or more explanatory variables.

X = survival time (time to event)

$\mathbf{Z}^t = (Z_1, \ldots, Z_p)$ = vector of explanatory variables associated with X.

Z_i can be quantitative, qualitative, and/or time-dependent.

Time-dependent explanatory variables

Variables that depend on the survival time are said to be **time-dependent (**the effect is not constant over time).

A time-dependent variable can e.g. be:

- a variable that denotes if some intermediate event has occurred by time x (e.g. development or recovery of some other disease, cessation of smoking, a transplant, change in blood pressure, etc.)
- The amount of time passed since some intermediate event

Time-dependent explanatory variables

If $\mathbf{Z}^t$ includes time-dependent variables:

$$\mathbf{Z}^t(x) = [Z_1(x), \ \ldots \ , Z_p(x)].$$

Modelling of covariate effects on survival

Two popular approaches:

1) Accelerated failure-time model, analogous to the classical linear regression approach.
2) Modelling the conditional hazard rate as a function of the covariates (which is the major approach).

Accelerated failure-time model

$Y = \ln(X)$ is modelled.

A linear model is assumed for Y:

$$Y = \mu + \boldsymbol{\gamma}^{t}\mathbf{Z} + \sigma W$$

$\boldsymbol{\gamma}^{t} = (\gamma_1, \ldots, \gamma_p)$ = vector of regression coefficients

W = error distribution.

The regression coefficients are estimated using maximum likelihood methods.

Accelerated failure-time model

$X = e^{Y}$

The effect of the explanatory variables in the original time scale is accelerated (or degraded) by a constant factor, thereof the name of the model (further explanation in the book).

The use of the accelerated failure-time model is restricted by the error distributions one can assume.

Modelling the conditional hazard rate

Two classes of models:

- Multiplicative hazard rate models
- Additive hazard rate models

Multiplicative hazard rate models

$$h(x|\mathbf{z}) = h_0(x)c(\boldsymbol{\beta}^t\mathbf{z})$$

Baseline hazard rate

Function of covariates

$h_0(x)$ is the underlying hazard function, describing how the risk of event per time unit changes over time at baseline levels of covariates.

$h_0(x)$ can have a specified parametric form, or be an arbitrary nonnegative function.

Example: Exponential distribution

$$h(x|\mathbf{z}) = h_0(x)c(\boldsymbol{\beta}^t\mathbf{z})$$

$h_0(x) = \lambda$

$c(\boldsymbol{\beta}^t\mathbf{z}) = \exp(\boldsymbol{\beta}^t\mathbf{z})$

Commonly used function

Risk of experiencing the event at time x for covariate values $\mathbf{z}$ is then

$$h(x \mid \mathbf{z}) = \lambda \exp(\boldsymbol{\beta}^t\mathbf{z})$$

Estimates of $\boldsymbol{\beta}$ can be obtained by maximum likelihood estimation.

Arbitrary baseline hazard function

If the interest lies in comparing hazard rates (or survival rates) between different groups, e.g. individuals with different covariate values, the baseline hazard is of less importance and doesn't have to be specified.

Multiplicative hazard rate models
Proportional hazard rates

Consider two individuals with covariate values $\mathbf{z}_1$ and $\mathbf{z}_2$.

The relationship between the hazard rates of these individuals:

$$\frac{h(x \mid \mathbf{z}_1)}{h(x \mid \mathbf{z}_2)} = \frac{\cancel{h_0(x)}c(\boldsymbol{\beta}^t \mathbf{z}_1)}{\cancel{h_0(x)}c(\boldsymbol{\beta}^t \mathbf{z}_2)} = \frac{c(\boldsymbol{\beta}^t \mathbf{z}_1)}{c(\boldsymbol{\beta}^t \mathbf{z}_2)}$$

The ratio is constant, independent of time.

This means that the hazard rates are proportional.

Additive hazard rate models

$$h(x|\mathbf{z}) = h_0(x) + \sum_{j=1}^{p} z_j(x)\beta_j(x)$$

Baseline hazard rate

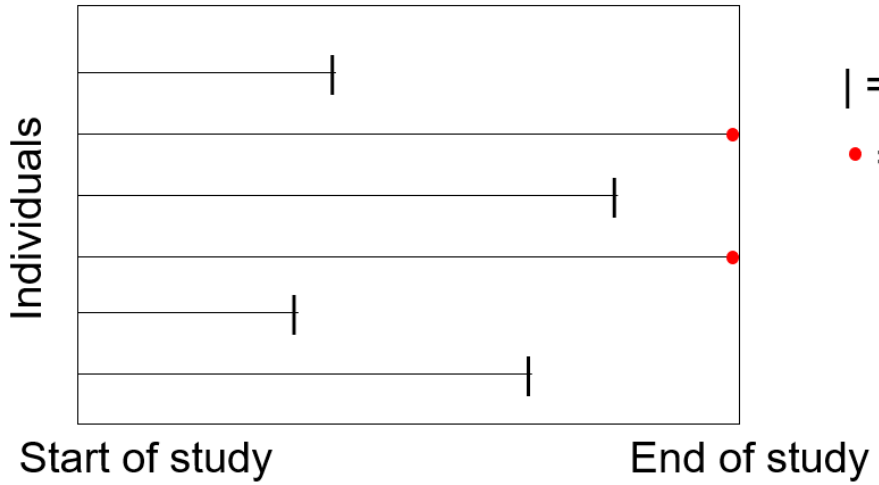
Additive models are usually estimated by nonparametric (weighted) least-squares methods.

We'll focus on the multiplicative models in this course.

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Analysis of Survival Data

Lecture 2: Censoring and truncation



| = event experienced

• = event not (yet) experienced
censored observation

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Program L2

- **Censoring and truncation**
  - Right, left and interval censoring
  - Right and left truncation
  - Likelihood for censored and truncated data

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Censored observations

Typical for survival data:
not all individuals experience the event of interest.

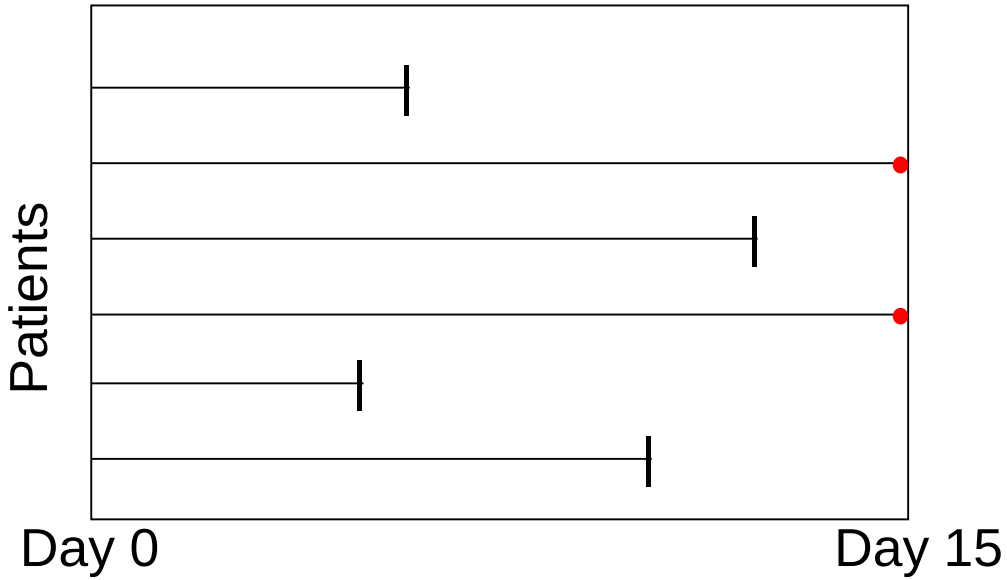
When the study is closed, some patients have experienced the event and others have not.

Or, some patients may have experienced the event but the exact time point is unknown.

The observation for an individual who has not experienced the event, or for whom the event time point is unknown, is said to be a **censored observation.**

Example: Time to treatment response

Patients with a certain disease are being followed for 15 days after starting a new treatment to see how long it takes before they respond to the treatment.



| = response

• = no response (yet)
censored observation

Survival analysis

Survival analysis takes censoring into account, and calculates likelihoods based on the combination of the probability to experience the event at a certain time with the probability to experience the event at a certain time *or later*.

Censoring categories

Different categories of censoring:

- Right censoring
- Left censoring
- Interval censoring

Each type of censoring will lead to a different likelihood function.

Right censoring

Right censoring occurs when the event is observed only if it occurs before a certain time, e.g. the predetermined end of a study.

Starting times and censoring times may be fixed or vary from individual to individual.

- Type I censoring
- Progressive type I censoring
- Generalized type I censoring
- Type II censoring
- Random censoring

Type I censoring (right censoring)

The starting point is the same for all individuals.

Censoring times may vary from individual to individual.

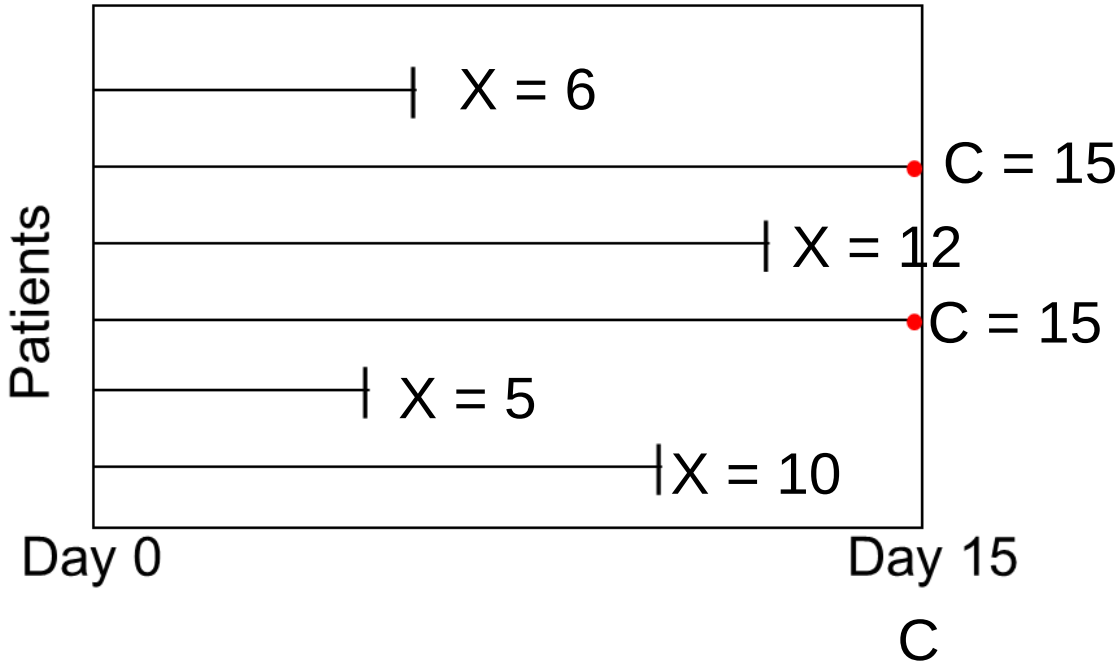
Example: Time to treatment response

X = time to response

C = censoring time

T = min (X, C)

$$\delta = \begin{cases} 1 & \text{If event occurred} \\ 0 & \text{If censored} \end{cases}$$



T	δ
6	1
15	0
12	1
15	0
5	1
10	1

Progressive Type I censoring

Progressive type I censoring is when some individuals are censored at a certain time point, and others are further observed until a second time point.

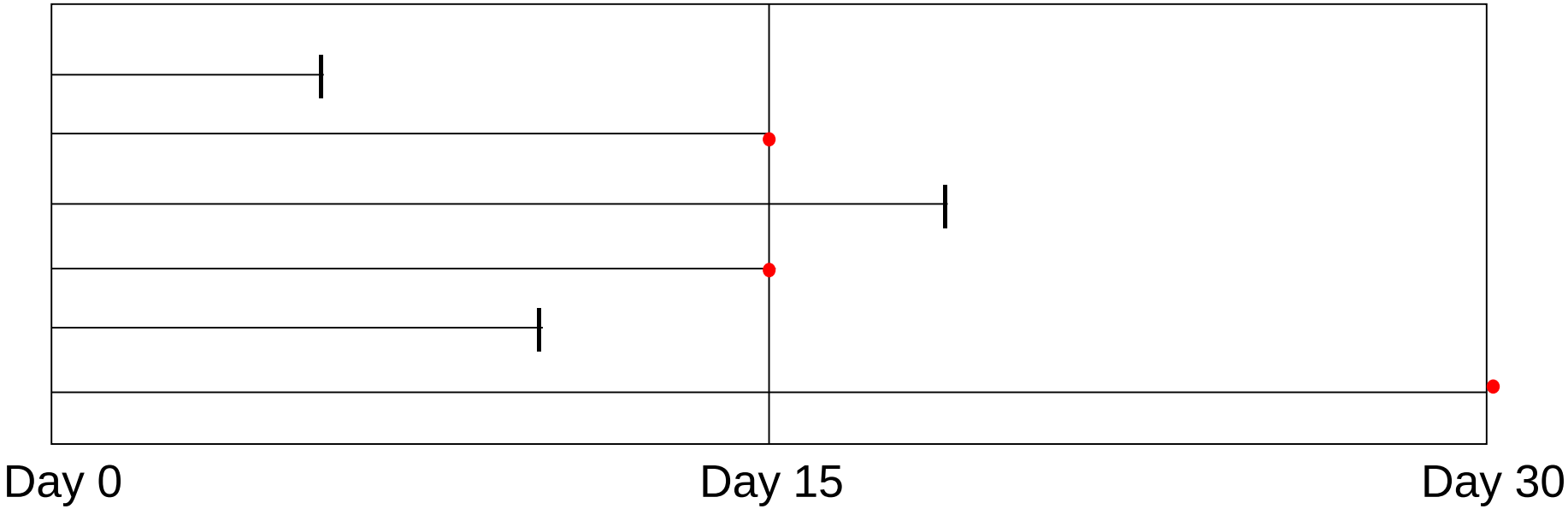
A type of study usually performed by practical and economical reasons.

Example: Time to treatment response

X= time to response of the new treatment.
Patients are followed for 15 days after treatment start.

In addition, lagged response effects are of interest.

Some patients are followed for another 15 days, after stopping the treatment.



Generalized Type I censoring

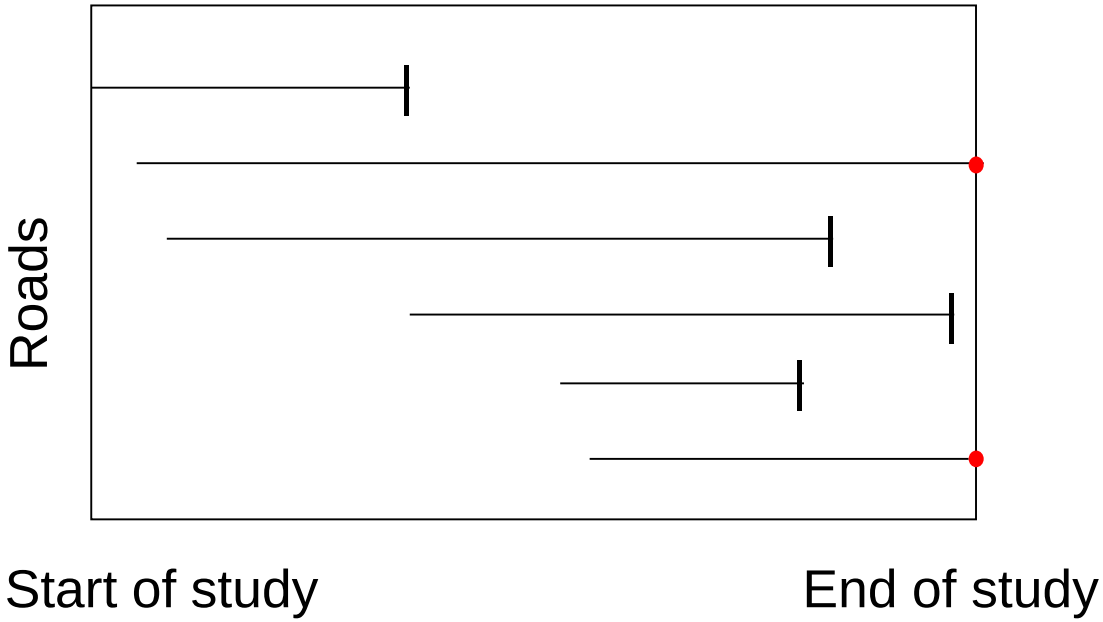
Generalized type I censoring is when individuals enter the study at different times, and the end of the study is predetermined.

Censoring times are known when individuals enter the study.

A very common type of censoring in patient studies, where patients are included e.g. at the time of diagnosis.

Example: Road maintenance

X= time to when a road in the Swedish road network needs to be maintained (e.g. with new pavement)

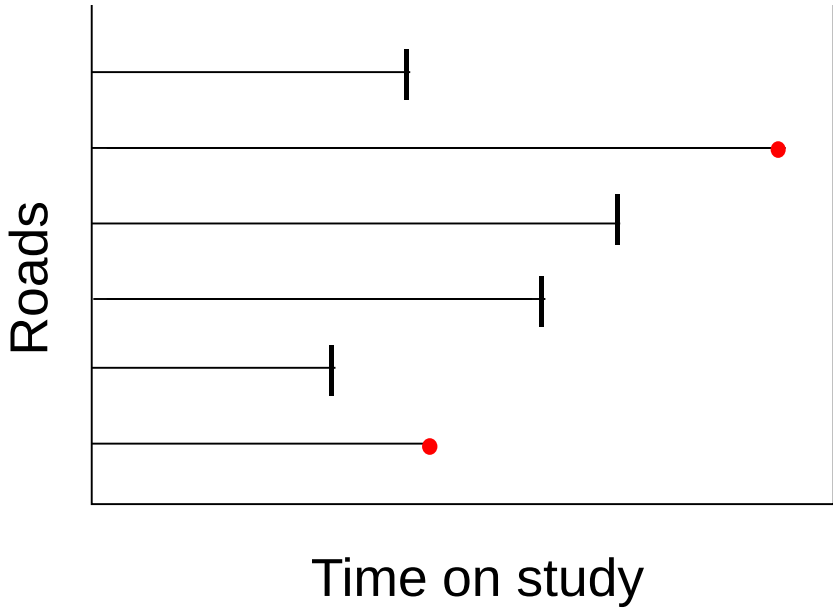


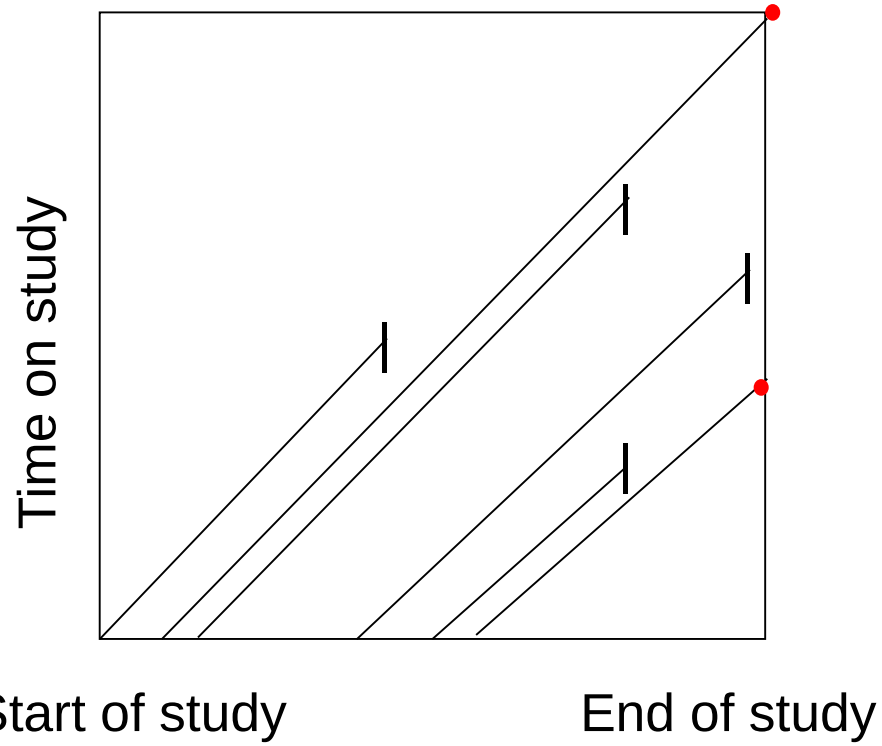
Generalized Type I censoring

Two common ways of representing data with generalized type 1 censoring:

1) Shifting each individual's starting time to 0
2) Lexis diagram

Example: Road maintenance





Each starting time shifted to 0

Lexis diagram

Type II censoring (right censoring)

Type II censoring is also a type of right censoring.

The study is terminated when r individuals have experienced the event ($r < n$, predetermined numbers).

Advantages:

- You can be sure to include a certain number of events in your data (e.g. sample size calculations are based on the number of events).
- Data consist of the r shortest survival times, thus the theory of order statistics can be used (simpler).

Competing risks

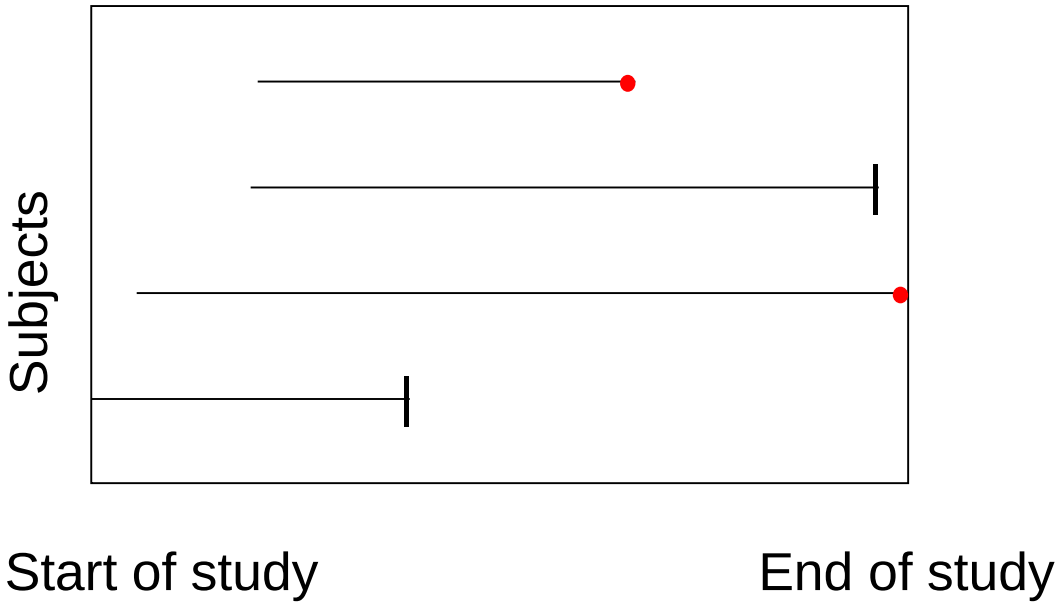
Sometimes the individuals in the study may fail due to **competing risks**.

A competing risk is something that if it occurs, it prevents the event of interest from being observed.

E.g. death in the study of time to response of a medical treatment, or death from heart-failure when the lifetime of cancer patients is being investigated, etc.

Special cases of competing risks can be handled by random censoring.

Random censoring (special case of competing risks censoring)



E.g. a study of how long marriages last. If one of the spouses dies, that couple's observation is censored.

Another example is when people in the study move abroad, or no longer want to participate.

Left censoring

Left censoring occurs when the event has happened prior to the study start.

The event is known to have happened but the exact time is unknown.

Example: Children learning to read

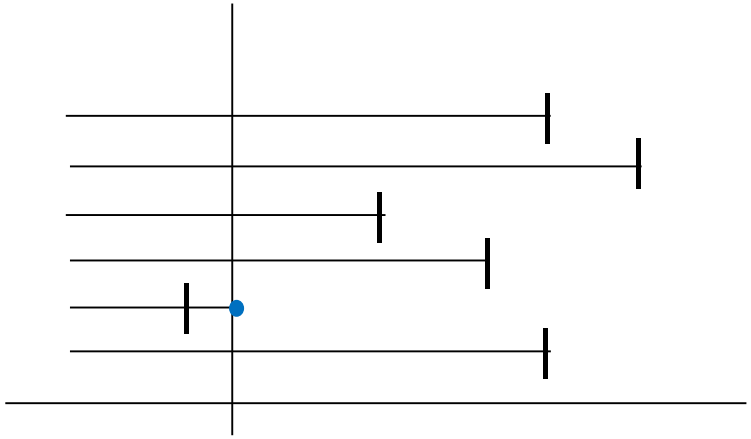
X = The time it takes for children in school to learn to read

C = School start

T = max (X, C)

$$\varepsilon = \begin{cases} 1 & \text{If event is observed} \\ 0 & \text{If left censored} \end{cases}$$

• = left censored observation



School start
= study start
(C=0)

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Double censoring

Double censoring occurs when both left and right censoring are present in the study.

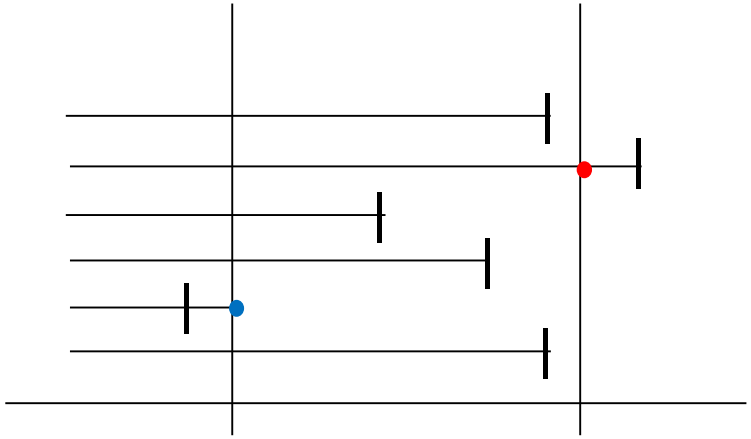
Example: Children learning to read

X = The time it takes for children in school to learn to read

C = School start

T = max (X, C)

$$\delta = \begin{cases} 1 & \text{If event is observed} \\ 0 & \text{If right censored} \\ -1 & \text{If left censored} \end{cases}$$



School start = study start (C=0)

End of study

• = left censored observation

• = right censored observation

Interval censoring

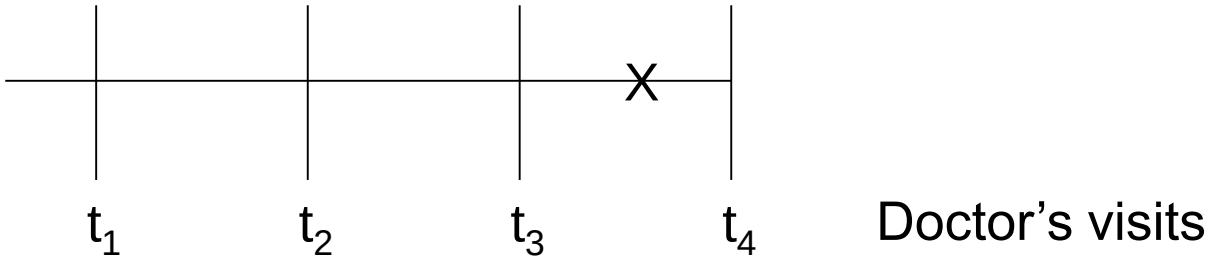
Interval censoring occurs when the event is known to happen only within an interval of time.

Happens in e.g. clinical trials or longitudinal studies when patients have periodic follow-up and the exact event date is not recorded between follow-up visits.

You know that the event occurred sometime between time points A and B.

Example: Doctor's visits

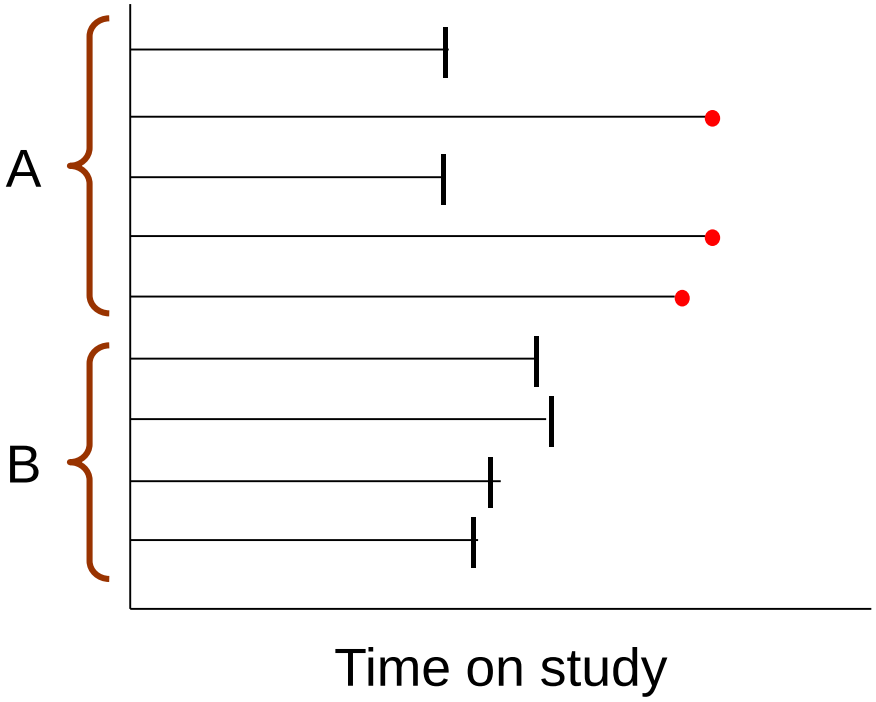
X = Age at first symptom of hearing loss



At a certain doctor's visit it is discovered that the patient has some hearing loss, but the exact time when it started is unknown.

How to deal with censored observations?

Example: a study of two different treatments, A and B.



B patients have the longest survival times, if you ignore the censored observations.

If you include the censored observations the A patients have the longest survival times.

How to deal with the censored times?

Program L2

- **Censoring and truncation**
  - Right, left and interval censoring
  - Right and left truncation
  - Likelihood for censored and truncated data

Truncation

Sometimes only individuals with a certain characteristic are included in the study.

There is a risk of missing individuals whom you have an interest of including in the study.

Right truncation

Right truncation is when only subjects who experience the event within a certain observational window (e.g. the study period) are included in the study.

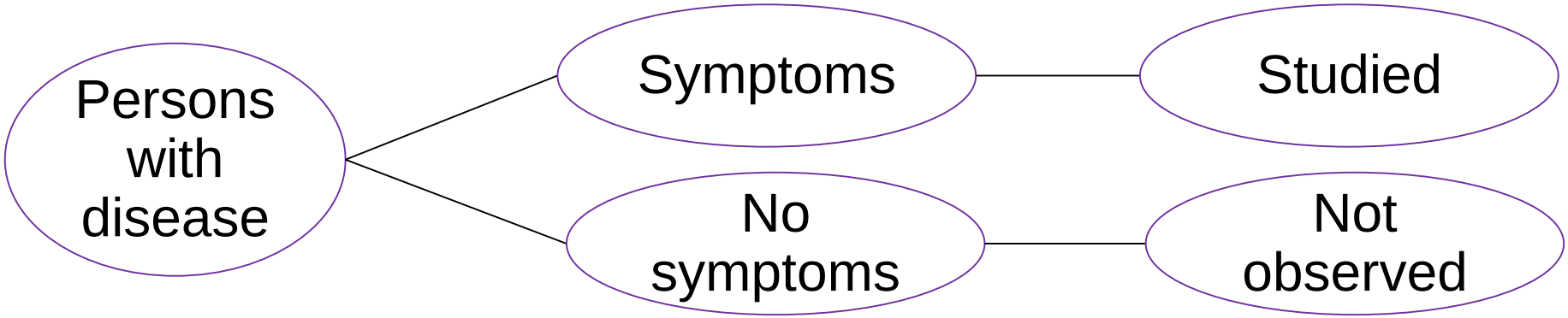
E.g. failure time of electronic devices. If they don’t break down within the limited guarantee time they won’t be handed in for service and will thus not be observed.

Or, any study based on death records.

Left truncation

Left truncation is when there is a selection before the event, you must have experienced something earlier.

Example: survival for individuals with a certain disease:



Individuals who die before the truncation time (e.g. study start) are not observed, and the survival is overestimated.

Program L2

- **Censoring and truncation**
  - Right, left and interval censoring
  - Right and left truncation
  - Likelihood for censored and truncated data

Likelihood construction

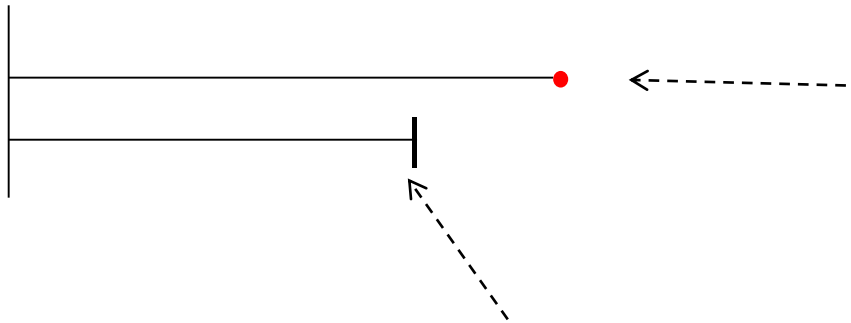
No censored observations:

$$L = \prod_{i=1}^{n} f(x_i) \qquad \text{(Sample } (x_1, \ldots x_n))$$

Probability to get the observed data.

Likelihood construction for censored data

For right censored data:



Information provided by the survival function evaluated at the time on study.

Provides information on the probability of the event occurring at this time

The different categories of information are to be combined.

Likelihood construction: right censored data

$$L = \prod_{i \in E} f(x_i) \prod_{i \in R} S(C_i)$$

E = event R = right censored

$$L = \prod_{i=1}^{n} f(t_i)^{\delta} S(t_i)^{1-\delta} \qquad \delta = \begin{cases} 1 & \text{If event occurred} \\ 0 & \text{If censored} \end{cases}$$

Critical assumption:

Lifetimes and censoring times are independent.

Example: Exponential distribution

Four observed survival times:

2, 3, 4+, 4+ (+ denotes right censoring)

Assume that the survival times follow an exponential distribution with hazard rate λ.

Find the maximum likelihood estimator of λ.

Example: Exponential distribution

$$f(x) = \lambda e^{-\lambda x}$$

$$S(x) = e^{-\lambda x}$$

$$L = \lambda e^{-\lambda 2} \lambda e^{-\lambda 3} e^{-\lambda 4} e^{-\lambda 4}$$

$$= \lambda^2 e^{-\lambda(2+3+4\cdot 2)} = \lambda^2 e^{-13\lambda}$$

To be maximized
How do we do that?

The derivative of the likelihood (or the logarithm of the likelihood) gives the maximum

Example: Exponential distribution

$$\ln L = \ln \lambda^2 e^{-13\lambda} = 2\ln\lambda - 13\lambda$$

$$\frac{\partial \ln L}{\delta \lambda} = \frac{2}{\lambda} - 13$$

$$\frac{2}{\hat{\lambda}} - 13 = 0$$

$$\frac{2}{\hat{\lambda}} = 13 \qquad \hat{\lambda} = \frac{2}{13} \approx 0.15$$

Likelihoods for different types of censoring

$$L = \prod_{i \in E} f(x_i) \prod_{i \in R} S(C_i) \prod_{i \in L} [1 - S(C_i)] \prod_{i \in I} [S(L_i) - S(R_i)]$$

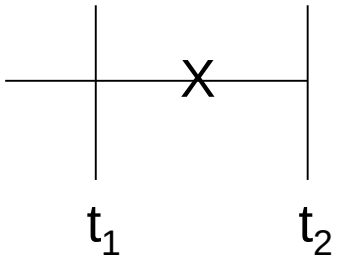
E = event

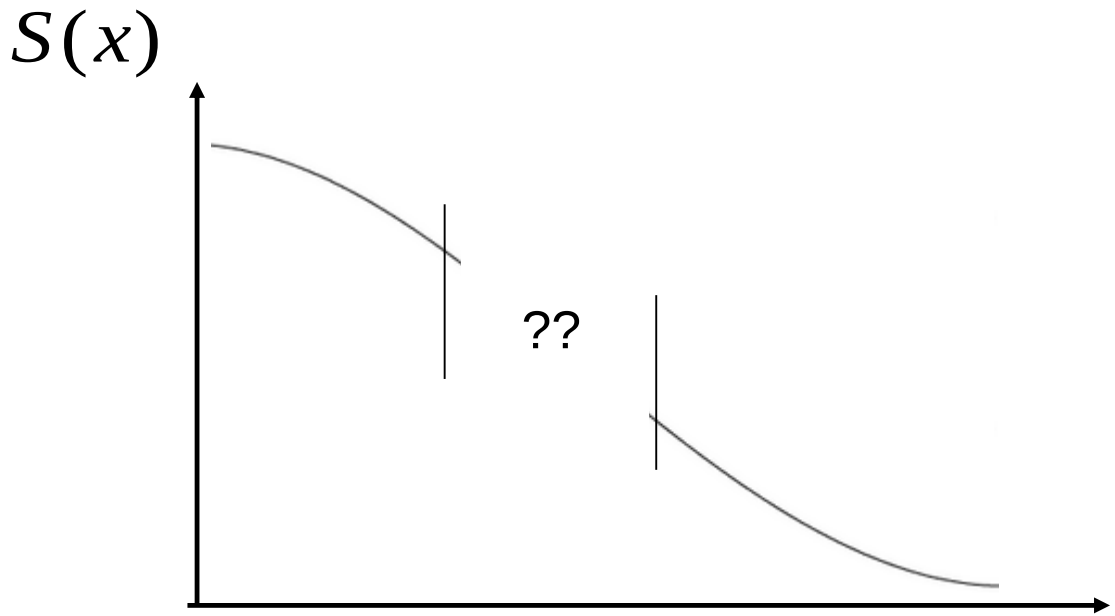
R = right censored

L = left censored

I = interval censored
L=Left interval time
R=Right interval time

Example: Interval censoring vs survival





$$S(t_1) > S(t_2)$$

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Likelihood construction: right truncated data

Only events are observed (all times, T_i, are event times, X_i).

The probability to observe an event at T_i given that $T_i \leq Y_i$:

$$f(T_i|T_i \leq Y_i) = \frac{f(T_i)}{P(T_i \leq Y_i)} = \frac{f(T_i)}{1 - S(Y_i)} = \frac{f(X_i)}{1 - S(Y_i)}$$

$$L_i = \prod_i \frac{f(x_i)}{1 - S(Y_i)}$$

Likelihood construction: left truncated data

The observed time T_i is left truncated at Y_i.

Then we have to consider the conditional distribution of T_i given that $T_i \geq Y_i$:

$$f(T_i | T_i \geq Y_i) = \frac{f(t_i)}{P(T_i \geq Y_i)} = \frac{f(t_i)}{S(Y_i)}$$

Likelihood construction: left truncated data

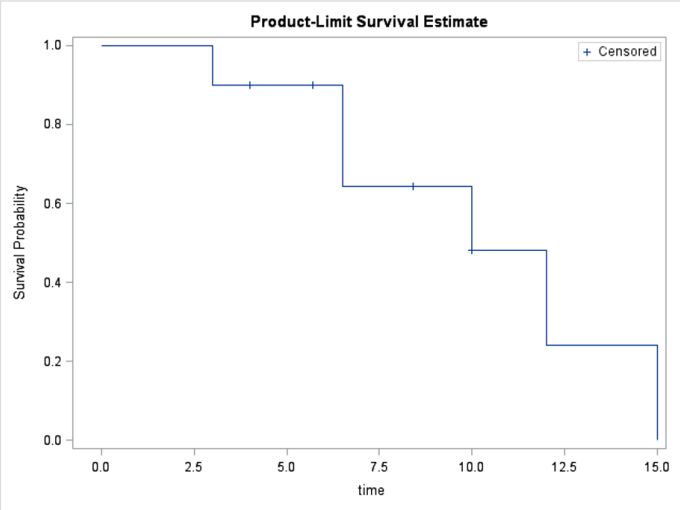
Replace $f(x_i)$ by $\dfrac{f(x_i)}{S(Y_i)}$ ← Y = truncation time

and replace $S(C_i)$ by $\dfrac{S(C_i)}{S(Y_i)}$

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Analysis of Survival Data

Lecture 3:
Nonparametric estimation of basic quantities



$$\hat{S}(t) = \begin{cases} 1 & \text{if } t < t_1 \\ \prod_{t_i \le t} \left(1 - \frac{d_i}{Y_i}\right) & \text{if } t_1 \le t \end{cases}$$

Inger Persson

Program L3

- **Nonparametric estimation of basic quantities**
  - Survival estimates
    - Kaplan-Meier estimator
    - Nelson-Aalen estimator
    - Assumptions
    - Left-truncated data
  - Pointwise confidence intervals for $S(t)$
  - Confidence bands
  - Mean and median survival time, with confidence intervals

Estimation of basic quantities

Events occur at D distinct times, $t_1 < t_2 < \cdots < t_D$

At time t_i there are d_i events.

Y_i = number of individuals at risk at t_i (number of individuals who have not yet experienced the event or experience the event at t_i).

$Y_i = n$ - number of individuals who have experienced the event or been censored up to t_{i-1}.

Example: Time to treatment response (h)

t_i	d_i	Y_i
3	1	11
4+	0	10
5.7+	0	9
6.5	2	8
8.4+	0	6
10	1	5
10+	0	
12	2	3
15	1	1

n = 11 (6 events)

Nonparametric estimation of $S(t)$ for right-censored data

Common approaches to estimate $S(t)$:

1) Kaplan-Meier estimator (= Product-Limit estimator)
2) Nelson-Aalen estimator

The history of the Kaplan-Meier (Product-Limit) estimator

In a paper published in the *Journal of the American Statistical Association* in June 1958, Edward Kaplan and Paul Meier put forth a new, efficient method for estimating patient survival rates, taking into account the fact that some patients may have died during a research trial while others will survive beyond the end of the trial.

The history of the Kaplan-Meier (Product-Limit) estimator

The method, called the Kaplan-Meier estimator (also known as the Product-Limit estimator), is based on information from those who have died (experienced the event) and those who have survived (censored observations) to estimate the proportion of patients alive at any point during the study.

The resulting curve, plotted over time, is called the Kaplan-Meier curve - a series of horizontal steps of declining magnitude that, when a large enough sample is taken, approaches the true survival function for that population.

Example: Time to treatment response (h)

t_i	d_i	Y_i	$\hat{S}(t)$
3	1	11	
4+	0	10	
5.7+	0	9	
6.5	2	8	
8.4+	0	6	
10	1	5	
10+	0		
12	2	3	
15	1	1	

$$\hat{S}(t) = \begin{cases} 1 & \text{if } t < t_1 \\ \prod_{t_i \le t} \left(1 - \frac{d_i}{Y_i}\right) & \text{if } t_1 \le t \end{cases}$$

Example: Time to treatmen

t_i	d_i	Y_i
3	1	11
4+	0	10
5.7+	0	9
6.5	2	8
8.4+	0	6
10	1	5
10+	0	
12	2	3
15	1	1

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{Y_i}\right)$$

$$\hat{S}(3) = \left(1 - \frac{d_1}{Y_1}\right) = \left(1 - \frac{1}{11}\right) = 0.91$$

$$\hat{S}(4) = \prod_{t_i \le 4} \left(1 - \frac{d_i}{Y_i}\right) = 0.91\left(1 - \frac{0}{10}\right) = 0.91$$

$$\hat{S}(6.5) = \prod_{t_i \le 6.5} \left(1 - \frac{d_i}{Y_i}\right) = 0.91\left(1 - \frac{2}{8}\right) = 0.68$$

Example: Time to treatment response (h)

t_i	d_i	Y_i	$\hat{S}(t)$
3	1	11	0.91
4+	0	10	(0.91)
5.7+	0	9	
6.5	2	8	0.68
8.4+	0	6	
10	1	5	0.54
10+	0		
12	2	3	0.18
15	1	1	0

$\hat{S}(t)$ usually not presented for censored times

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Example: Time to treatment response (h)

The LIFETEST Procedure

Product-Limit Survival Estimates

resptime		Survival	Failure	Survival Standard Error	Number Failed	Number Left
0.0000		1.0000	0	0	0	11
3.0000		0.9091	0.0909	0.0867	1	10
4.0000	*	.	.	.	1	9
5.7000	*	.	.	.	1	8
6.5000		.	.	.	2	7
6.5000		0.6818	0.3182	0.1536	3	6
8.4000	*	.	.	.	3	5
10.0000		0.5455	0.4545	0.1731	4	4
10.0000	*	.	.	.	4	3
12.0000		.	.	.	5	2
12.0000		0.1818	0.8182	0.1593	6	1
15.0000		0	1.0000	.	7	0

Note: The marked survival times are censored observations.

In Sas: proc lifetest

Example: Time to treatment response (h)

Summary Statistics for Time Variable resptime

Quartile Estimates				
Percent	Point Estimate	95% Confidence Interval		
		Transform	[Lower	Upper)
75	12.0000	LOGLOG	10.0000	.
50	12.0000	LOGLOG	6.5000	.
25	6.5000	LOGLOG	3.0000	12.0000

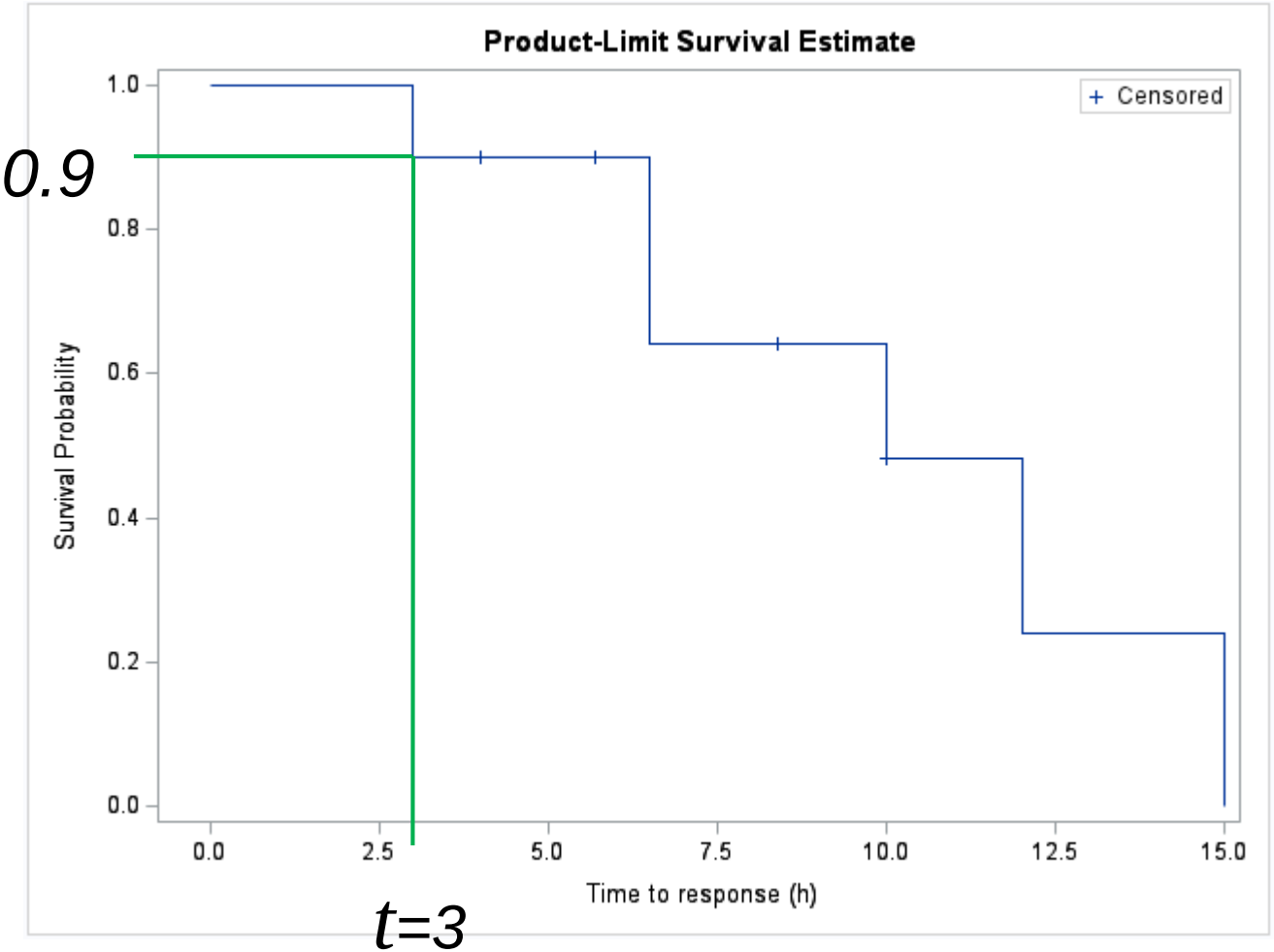
Median survival --->

Mean	Standard Error
10.2045	1.3133

Mean survival --->

Example: Time to treatment response (h)

$S(3) \approx 0.9$



t=3

In Sas: proc lifetest

Kaplan-Meier (Product-Limit) estimator

The standard estimator of $S(t)$ is the **Kaplan-Meier estimator**.

$$\hat{S}(t) = \begin{cases} 1 & \text{if } t < t_1 \\ \prod_{t_i \le t} \left(1 - \frac{d_i}{Y_i}\right) & \text{if } t_1 \le t \end{cases}$$

A step function, jumps at the observed event times.

$$\hat{V}\left[\hat{S}(t)\right] = \hat{S}(t)^2 \sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$$

Greenwod's estimator

Example: Time to treatmen

t_i	d_i	Y_i
3	1	11
4+	0	10
5.7+	0	9
6.5	2	8
8.4+	0	6
10	1	5
10+	0	
12	2	3
15	1	1

$$\sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$$

$$t_1: \quad \frac{1}{11(11-1)} = \frac{1}{90} = 0.009$$

$$t_4: \quad 0.009 + \frac{2}{8(8-2)} = 0.009 + \frac{2}{48} = 0.051$$

$$t_6: \quad 0.051 + \frac{1}{5(5-1)} = 0.051 + \frac{1}{20} = 0.101$$

Example: Time to treatment response (h)

t_i	d_i	Y_i	$\hat{S}(t)$	$\sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$
3	1	11	0.91	0.009
4+	0	10	(0.91)	
5.7+	0	9		
6.5	2	8	0.68	0.051
8.4+	0	6		
10	1	5	0.54	0.101
10+	0			
12	2	3	0.18	0.768
15	1	1	0	

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Example: Time to

$$\hat{V}\left[\hat{S}(t)\right] = \hat{S}(t)^2 \sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$$

t_i	d_i	Y_i	$\hat{S}(t)$	$\sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$
3	1	11	0.91	0.009
4+	0	10	(0.91)	
5.7+	0	9		
6.5	2	8	0.68	0.051
8.4+	0	6		
10	1	5	0.54	0.101
10+	0			
12	2	3	0.18	0.768
15	1	1	0	

t_1: $0.91^2\ 0.009 = 0.007$

t_4: $0.68^2\ 0.051 = 0.024$

t_6: $0.54^2\ 0.101 = 0.029$

Example: Time to treatment response (h)

t_i	d_i	Y_i	$\hat{S}(t)$	$\sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$	$\hat{V}\left[\hat{S}(t)\right]$
3	1	11	0.91	0.009	0.007
4+	0	10	(0.91)		
5.7+	0	9			
6.5	2	8	0.68	0.051	0.024
8.4+	0	6			
10	1	5	0.54	0.101	0.029
10+	0				
12	2	3	0.18	0.768	0.025
15	1	1	0		

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The curve that changed the world

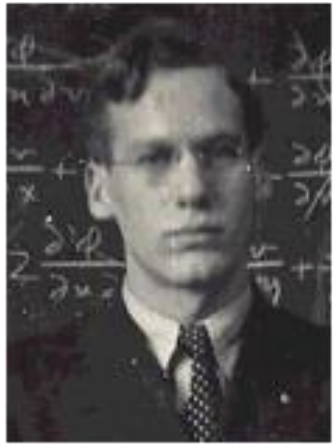
By John Easton, Science Life 2011

"Let's start with a statistic: almost 2,000 citations a year."

The 1958 paper "has been cited more often, by a wide margin, than any other paper in the field of statistics. At last count it was the fifth most cited research paper of all time, in any field." It has been "cited by another scientific publication about once, on average, for every day of Meier's long life time."

Paul Meier
1924-2011





Edward L Kaplan
1920-2006

Nelson-Aalen estimator

An alternate estimator of the cumulative hazard rate $H(t)$, is the **Nelson-Aalen estimator**.

$$\tilde{H}(t) = \begin{cases} 0 & \text{if } t < t_1 \\ \sum_{t_i \le t} \left(\frac{d_i}{Y_i} \right) & \text{if } t_1 \le t \end{cases}$$

Has better small-sample properties than $\hat{H}(t) = -\ln\left[\hat{S}_{KM}(t)\right]$

$$\hat{V}\left[\tilde{H}(t)\right] = \sum_{t_i \le t} \frac{d_i}{Y_i^2}$$

Example: Time to treatment response (h)

t_i	d_i	Y_i	$\tilde{H}(t)$
3	1	11	
4+	0	10	
5.7+	0	9	
6.5	2	8	
8.4+	0	6	
10	1	5	
10+	0		
12	2	3	
15	1	1	

$$\tilde{H}(t) = \begin{cases} 0 & \text{if } t < t_1 \\ \sum_{t_i \le t} \left(\frac{d_i}{Y_i} \right) & \text{if } t_1 \le t \end{cases}$$

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Example: Time to treatmen

t_i	d_i	Y_i
3	1	11
4+	0	10
5.7+	0	9
6.5	2	8
8.4+	0	6
10	1	5
10+	0	
12	2	3
15	1	1

$$\hat{H}(t) = \sum_{t_i \le t} \left(\frac{d_i}{Y_i} \right)$$

$$\hat{H}(t_1) = \left(\frac{d_1}{Y_1} \right) = \frac{1}{11} = 0.09$$

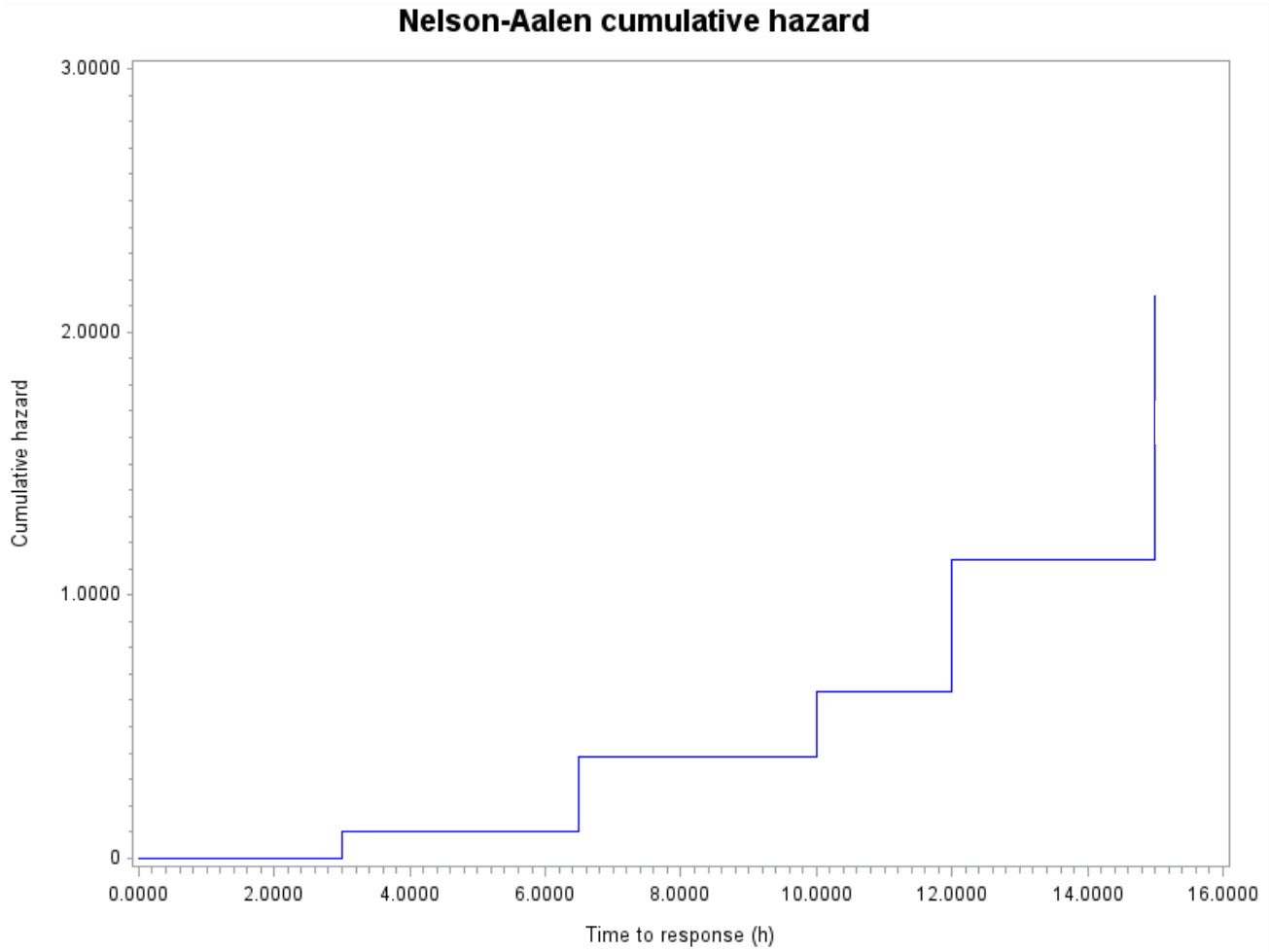
$$\hat{H}(t_4) = 0.09 + \left(\frac{2}{8} \right) = 0.34$$

Example: Time to treatment response (h)

t_i	d_i	Y_i	$\tilde{H}(t)$	$\hat{S}_{NA}(t) = e^{-\tilde{H}(t)}$	$\hat{S}_{KM}(t)$
3	1	11	0.09	0.91	0.91
4+	0	10			
5.7+	0	9			
6.5	2	8	0.34	0.71	0.68
8.4+	0	6			
10	1	5	0.54	0.58	0.54
10+	0				
12	2	3	1.21	0.30	0.18
15	1	1	2.21	0.11	0

The estimate for the last survival time is not always accurate

Example: Time to treatment response (h)



The slope of the Nelson-Aalen estimator gives a crude estimate of the hazard rate $h(t)$.

In Sas: proc lifetest

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Which estimator to choose?

The Kaplan Meier and the Nelson Aalen estimators can both be used to estimate survival and/or cumulative hazard.

The Kaplan-Meier estimator is biased at high degrees of censoring[1] (tends to overestimate survival probabilities).

The Nelson-Aalen estimator generally yields higher survival probabilities than the Kaplan-Meier[2], except at small sample sizes where it has better properties than K-M.

[1] Ramadurai & Ponnuraja (2011), Non-parametric estimation of the survival probability of children affected by TB meningitis, International Refereed Research Journal, 2 (2), 216-228.
[2] Bohoris (1994), Comparison of the Cumulative-Hazard and Kaplan-Meier Estimators of the Survivor Function, IEEE Transactions on Reliability, 43 (2), 230 - 232.

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Which estimator to choose?

There is also a weighted version[1] of the Kaplan-Meier estimator with good properties at high degrees of censoring in some situations[2]. This estimator is however not (yet) implemented in SAS (or R).

[1] *Jan et al. (2005), Weighted kaplan-meier estimation of survival function in heavy censoring. Pakistan Journal of Statistics, 21(1), 55–63.*

[2] *Malmquist (2025), Determination of the censoring rate that introduces problematic bias in the Kaplan-Meier estimator for small samples. Master thesis, Dept. of Statistics, Uppsala university.*

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Assumptions

Assumptions of the Kaplan Meier survival estimator*:

- random sample(s)
- independent samples (if >1 group)
- noninformative censoring
- right censored data

**Kaplan and Meier, Nonparametric Estimation from Incomplete Observations, JASA, 1958;*

Assumption of noninformative censoring

Both the Kaplan-Meier (Product-Limit) and the Nelson-Aalen estimators are based on an assumption of **noninformative censoring**, meaning that knowledge of a person's censoring time gives no information about this person's future event time (had the person continued to be studied).

When this assumption is violated, both estimators provide wrong estimates and can be misleading.

Informative censoring

Examples of **informative** censoring:

- Patients where the physician believes that a progression of the disease is upcoming, and withdraws the patient (=censoring) for initiation of another treatment.
- In HIV, patients lost to follow-up are usually the most severe cases. They have many side effects and don't come voluntarily to study visits.
- With asthma, it can be the other way around. Patients lost to follow-up are usually patients who’s asthma is well under control and they don't have the need to visit the physician.

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Other important considerations

If there are **implicit factors** unaccounted for in the analysis that affect survival and/or censoring times, then the Kaplan-Meier calculations may not give useful estimates for survival.

Small samples make it more difficult to detect possible dependencies between censoring and survival, or the presence of implicit factors.

Heavy censoring may affect the reliability of the Kaplan-Meier estimates.

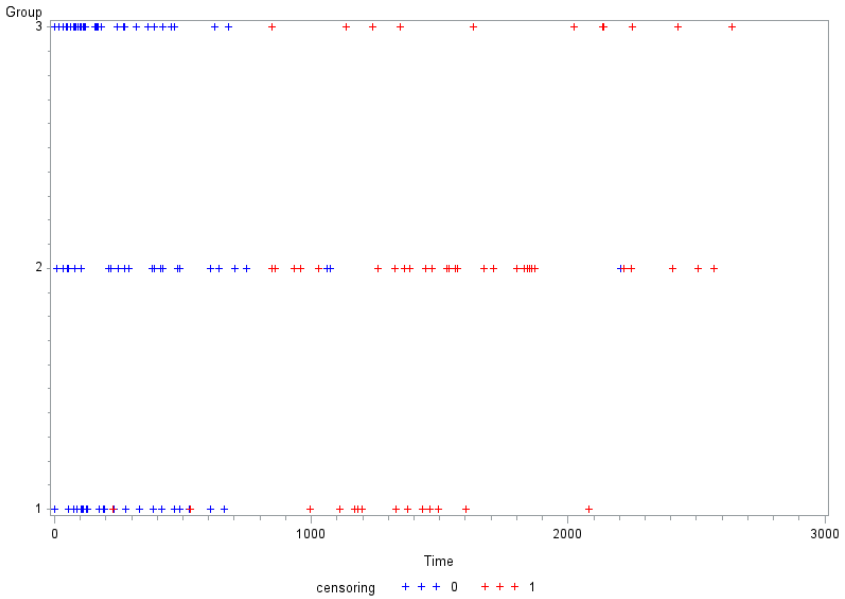
How investigate the assumption of noninformative censoring?

It is very difficult to spot any relationships between censoring and survival as a statistician, we need to discuss this with the researcher who knows the data.

We can investigate the pattern of censoring, as one way to support the reasoning of noninformative censoring. The censoring pattern alone isn't enough as a "test" of informative censoring, but it's what we can contribute with to the discussion.

Plot censoring values vs time

If the assumptions for the censoring and survival distributions are correct, then a plot of either the censored or the noncensored values (or both together) against time should show no particular patterns, and the patterns should be similar across the various groups.



Check that there are no obvious differences in the censoring pattern in different groups. But remember that this alone is no proof of noninformative censoring!

Program L3

- **Nonparametric estimation of basic quantities**
  - Survival estimates
    - Kaplan-Meier estimator
    - Nelson-Aalen estimator
    - Assumptions
    - Left-truncated data
  - Pointwise confidence intervals for $S(t)$
  - Confidence bands
  - Mean and median survival time, with confidence intervals

Left truncated data

The survival function $S(t)$ can be modified to handle left-truncated data (together with right censoring).

Redefine Y_i = number of individuals *who entered the study* prior to t_i .

The Kaplan-Meier estimator $\hat{S}(t)$ now describes $P(X>t \mid$ *entering the study*$)$,

i.e. $P(X>t \mid X>L)$

L = time of study entry

Example: Survival of dogs

X = lifetime for dogs (time to death)

L = age at registration (maximum 4 months)

Say that the minimum age of registration is 1 month.

Impossible to estimate the survival probability < 1 month.

Only possible to estimate the conditional probability
$P(X>t \mid X>1)$

Example: Diagnosed patients

X = time to treatment response

L = time of diagnosis

Estimates must condition on being diagnosed.

Common to define X = time from diagnosis, then the conditionality is natural.

Program L3

- **Nonparametric estimation of basic quantities**
  - Survival estimates
    - Kaplan-Meier estimator
    - Nelson-Aalen estimator
    - Assumptions
    - Left-truncated data
  - Pointwise confidence intervals for $S(t)$
  - Confidence bands
  - Mean and median survival time, with confidence intervals

Confidence intervals for $S_{KM}(t)$

Different approaches to estimate confidence intervals for the survival function $S(t)$:

1) Linear confidence interval (most commonly used)
2) Log-transformed confidence interval
3) Arcsine-Square root transformed confidence interval

The transformed confidence intervals provide better estimates, especially for small samples.

Linear confidence interval for $S(t)$

100×(1 - α)% linear confidence interval for $S(t)$ at time t:

$$\hat{S}(t) \pm Z_{1-\alpha/2}\sqrt{\hat{V}\left(\hat{S}(t)\right)}$$

where $$\hat{V}\left(\hat{S}(t)\right) = \hat{S}(t)^2 \sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$$

This interval covers, with confidence level (1-α), the population value of the survival function at a predetermined time t.

Example: Time to treatment response (h) (recap.)

t_i	d_i	Y_i	$\hat{S}(t)$	$\sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$	$\hat{V}\left[\hat{S}(t)\right]$
3	1	11	0.91	0.009	0.007
4+	0	10	(0.91)		
5.7+	0	9			
6.5	2	8	0.68	0.051	0.024
8.4+	0	6			
10	1	5	0.54	0.101	0.029
10+	0				
12	2	3	0.18	0.768	0.025
15	1	1	0		

Example: Time to treatment response (h)

95% linear confidence interval for $S(t)$ at $t = 3$:

$$\hat{S}(t) \pm Z_{1-\alpha/2}\sqrt{\hat{V}\left(\hat{S}(t)\right)}$$

$$0.91 \pm 1.96 \times \sqrt{0.007}$$

$$0.91 \pm 0.16 \quad \text{or} \quad [0.74,\ 1.0]$$

$S(t) \leq 1$

The interval 0.74 to 1.0 covers, with 95% confidence, the population value of the survival function at time $t = 3$ h.

Example: Time to treatment response (h)

95% linear confidence interval for $S(t)$ at $t = 6.5$:

$$0.68 \pm 1.96 \times \sqrt{0.024}$$

$$0.68 \pm 0.30 \quad \text{or} \quad [0.38,\ 0.98]$$

The interval 38% to 98% covers, with 95% confidence, the probability of not yet reaching a treatment response in the population at 6.5 hours from treatment start.

Example: Time to treatment response (h)

95% linear confidence intervals

Obs	resptime	_CENSOR_	SURVIVAL	SDF_LCL	SDF_UCL
1	0.0	.	1.00000	1.00000	1.00000
2	3.0	0	0.90909	0.73920	1.00000
3	4.0	1	0.90909	.	.
4	5.7	1	0.90909	.	.
5	6.5	0	0.68182	0.38075	0.98289
6	8.4	1	0.68182	.	.
7	10.0	0	0.54545	0.20611	0.88480
8	10.0	1	0.54545	.	.
9	12.0	0	0.18182	0.00000	0.49400
10	15.0	0	0.00000	.	.

In Sas: proc lifetest

Log-transformed confidence interval for $S_{KM}(t)$

Based on first finding a confidence interval for the log of the cumulative hazard function.

Sometimes called a log-log transformed interval, since

$$H(x) = -\ln\big(S(x)\big)$$

The default transformation in Sas.

Log-transformed confidence interval for $S_{KM}(t)$

100×(1 - α)% log-transformed confidence interval for $S(t)$ at time t:

$$\left[\hat{S}(t)^{1/\theta},\quad \hat{S}(t)^{\theta}\right]$$

where $$\theta = \exp\left\{\frac{Z_{1-\alpha/2}\sqrt{\sum_{t_i \le t}\frac{d_i}{Y_i(Y_i-d_i)}}}{\ln\left(\hat{S}(t)\right)}\right\}$$

This interval is not symmetric around $\hat{S}(t)$.

Example: Time to treatment response (h)

95% log-transformed confidence intervals (SAS default)

Obs	resptime	_CENSOR_	SURVIVAL	SDF_LCL	SDF_UCL
1	0.0	.	1.00000	1.00000	1.00000
2	3.0	0	0.90909	0.50808	0.98667
3	4.0	1	0.90909	.	.
4	5.7	1	0.90909	.	.
5	6.5	0	0.68182	0.29727	0.88611
6	8.4	1	0.68182	.	.
7	10.0	0	0.54545	0.18420	0.80479
8	10.0	1	0.54545	.	.
9	12.0	0	0.18182	0.00940	0.53652
10	15.0	0	0.00000	.	.

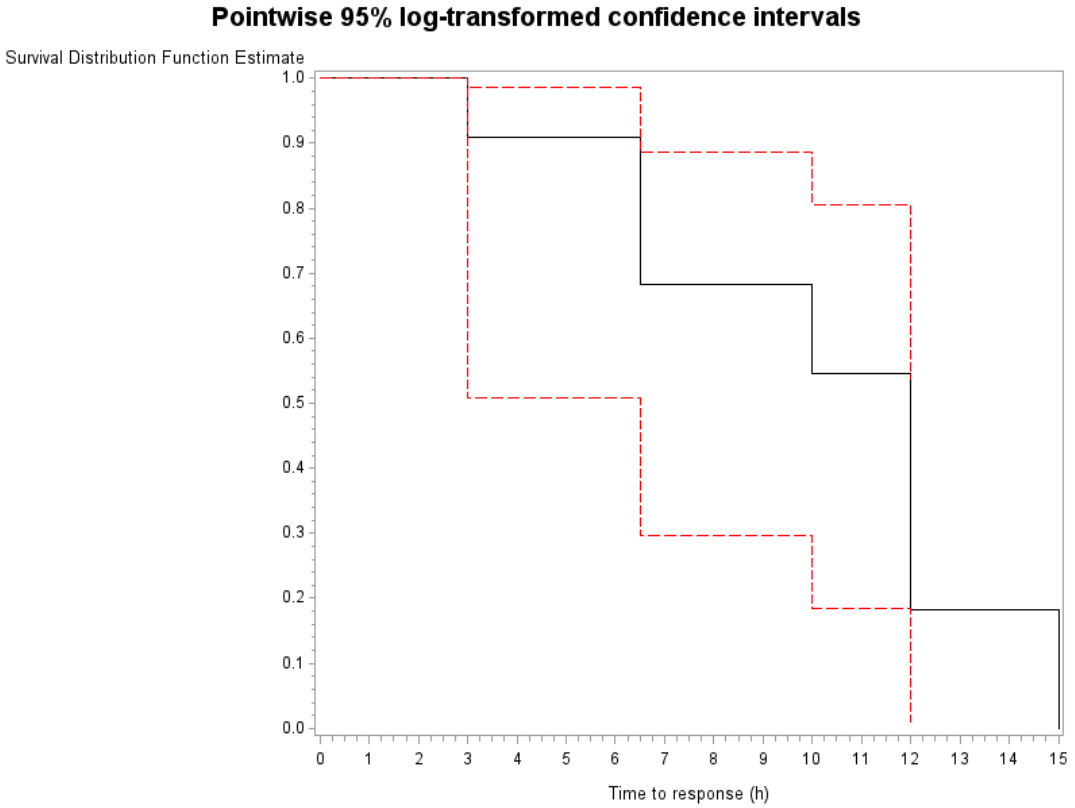
<---

Compare to the linear interval [0.74, 1].

Huge difference!

In Sas: proc lifetest

Example: Time to treatment response (h)



Note! These CI:s are for **single time points only**, NOT to be interpreted as having 95% confidence that the entire survival function lies within these intervals.

Arcsine-square root transformed confidence interval for $S_{KM}(t)$

Based on $g(x) = \arcsin\sqrt{x}$

which is variance stabilizing for the situation with no censoring.

Arcsine-square root transformed confidence interval for $S_{KM}(t)$

100×(1 - α)% arcsine-square root transformed confidence interval for $S(t)$ at time t:

$$\left[\sin^2 \left\{ \max \left[0, \quad \arcsin\left(\hat{S}(t)^{1/2}\right) - 0.5 Z_{1-\alpha/2} \sigma_s(t) \left(\frac{\hat{S}(t)}{1-\hat{S}(t)} \right)^{1/2} \right] \right\}, \right.$$

$$\left. \sin^2 \left\{ \min \left[\frac{\pi}{2}, \quad \arcsin\left(\hat{S}(t)^{1/2}\right) + 0.5 Z_{1-\alpha/2} \sigma_s(t) \left(\frac{\hat{S}(t)}{1-\hat{S}(t)} \right)^{1/2} \right] \right\} \right]$$

Example: Time to treatment response (h)

95% arcsine-square root transformed confidence intervals

Obs	resptime	_CENSOR_	SURVIVAL	SDF_LCL	SDF_UCL
1	0.0	.	1.00000	1.00000	1.00000
2	3.0	0	0.90909	0.67954	0.99988
3	4.0	1	0.90909	.	.
4	5.7	1	0.90909	.	.
5	6.5	0	0.68182	0.36460	0.92568
6	8.4	1	0.68182	.	.
7	10.0	0	0.54545	0.22162	0.84898
8	10.0	1	0.54545	.	.
9	12.0	0	0.18182	0.00128	0.55967
10	15.0	0	0.00000	.	.

Compare to the log-transformed [0.508, 0.987]

In Sas: proc lifetest

Linear vs. transformed confidence intervals for $S_{KM}(t)$

The log-transformed and arcsine-square root transformed confidence intervals perform better than the linear interval. Both transformed intervals give a correct coverage probability even for relatively small samples and as much as 50% censoring.

For very large samples, the linear and transformed confidence intervals are equivalent.

For very small samples, the log-transformed intervals have a coverage probability slightly larger than $(1 - \alpha)$, and the arcsine-square root transformed intervals have a coverage probability slightly smaller than $(1 - \alpha)$. *(Wrong in the book!)*

Assumptions for confidence intervals for $S_{KM}(t)$

Assumption (in addition to the assumptions for the estimator of *S(t)*):

- large sample(s) (intervals based on asymptotic normality)

Confidence bands for $S_{KM}(t)$

Confidence bands are confidence intervals that apply to the whole survival function.

Two approaches:

1) Equal probability bands (EP bands), proportional to the pointwise confidence intervals
2) Hall - Wellner confidence bands, not proportional to the pointwise confidence intervals.

Equal probability bands for $S_{KM}(t)$

1) Pick $t_L \geq t_1$ (lower limit)

$t_U \leq t_n$ (upper limit)

2) Calculate $a_L = \dfrac{n\sigma_s^2(t_L)}{1+n\sigma_s^2(t_L)}$ and $a_U = \dfrac{n\sigma_s^2(t_U)}{1+n\sigma_s^2(t_U)}$

where $\sigma_s^2(t) = \sum_{t_i \leq t} \dfrac{d_i}{Y_i(Y_i - d_i)}$

$0 < a_L < a_U < 1$

Equal probability bands for $S_{KM}(t)$

3) Find a confidence coefficient $c_\alpha(a_L, a_U)$ from Table C3.

4) Decide form for the confidence band; linear, log-transformed, or arcsine-square root transformed band.

Linear equal probability bands for $S_{KM}(t)$

100(1-α)% linear equal probability bands for $S(t)$ over the range $[t_L, t_U]$:

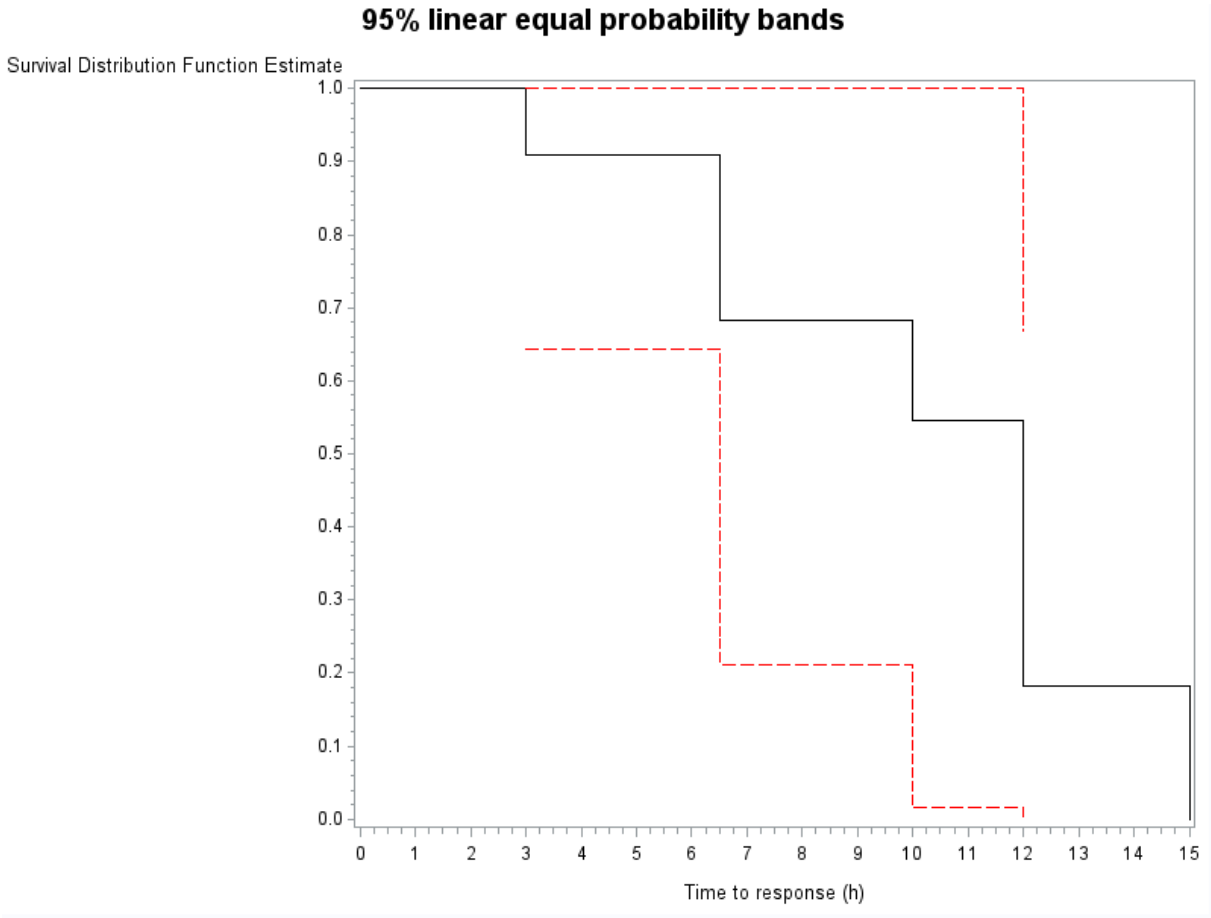
$$\hat{S}(t) \pm c_\alpha(a_L, a_U)\sqrt{\hat{V}\left(\hat{S}(t)\right)}$$

where $$\hat{V}\left(\hat{S}(t)\right) = \hat{S}(t)^2 \sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$$

cover, with confidence level (1-α), the survival function within the interval $[t_L, t_U]$.

The coverage probability is however considerably smaller than the target level for small sample sizes (< 200).

Example: Time to treatment response (h)



These bands cover, with a lot less than 95% confidence, the survival function from 3 to 12 hours.

Log-transformed equal probability bands for $S_{KM}(t)$

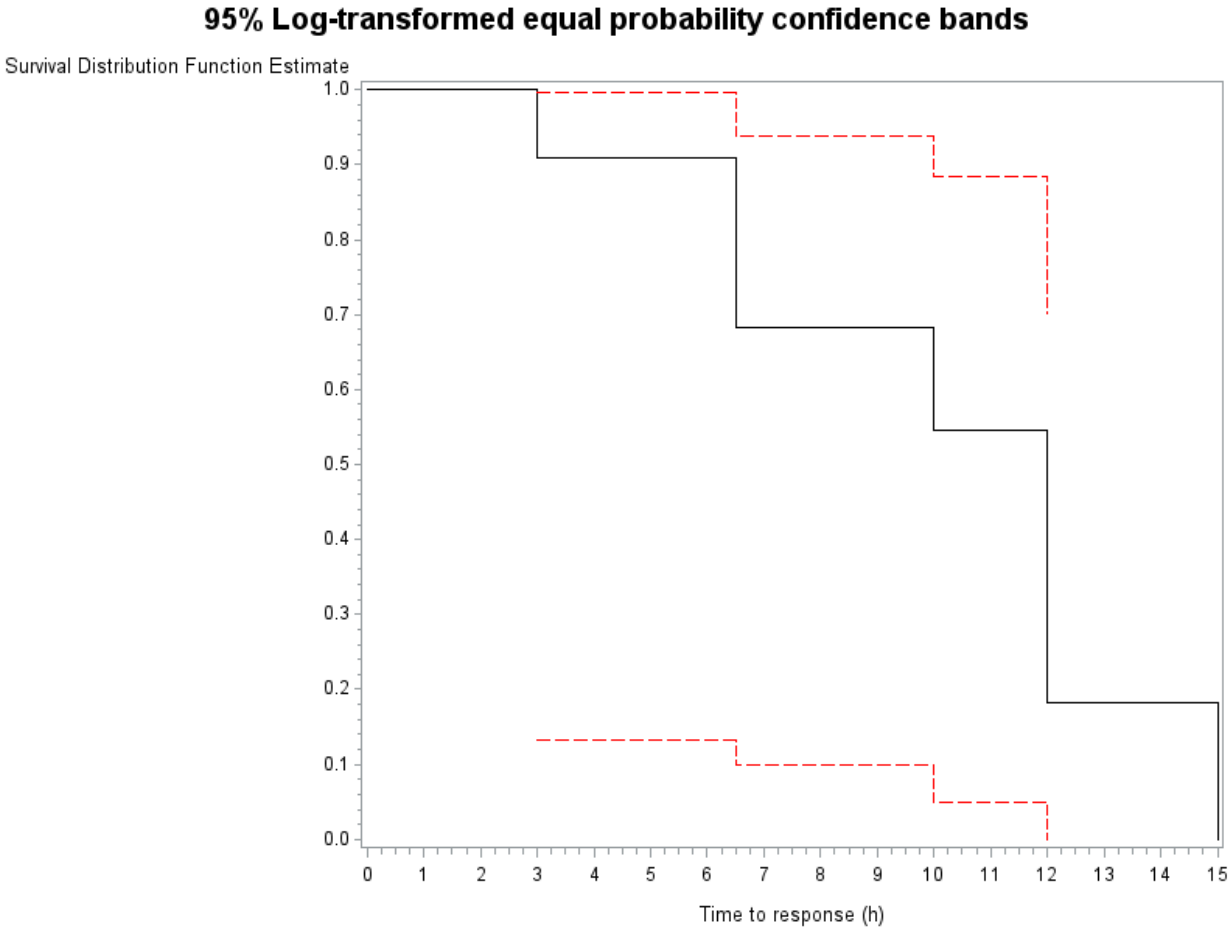
100(1-α)% log-transformed equal probability bands for $S(t)$ over the range $[t_L, t_U]$:

$$\left[\hat{S}(t)^{1/\theta}, \quad \hat{S}(t)^{\theta}\right]$$

where $$\theta = \exp\left\{\frac{c_\alpha(a_L, a_U)\sqrt{\sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}}}{\ln\left(\hat{S}(t)\right)}\right\}$$

cover, with approximate confidence level (1-α), the survival function within the interval $[t_L, t_U]$.

Example: Time to treatment response (h)



These bands cover, with approximately 95% confidence, the survival function within the interval [*3*, *12*] h.

Arcsine-square root transformed equal probability bands for $S_{KM}(t)$

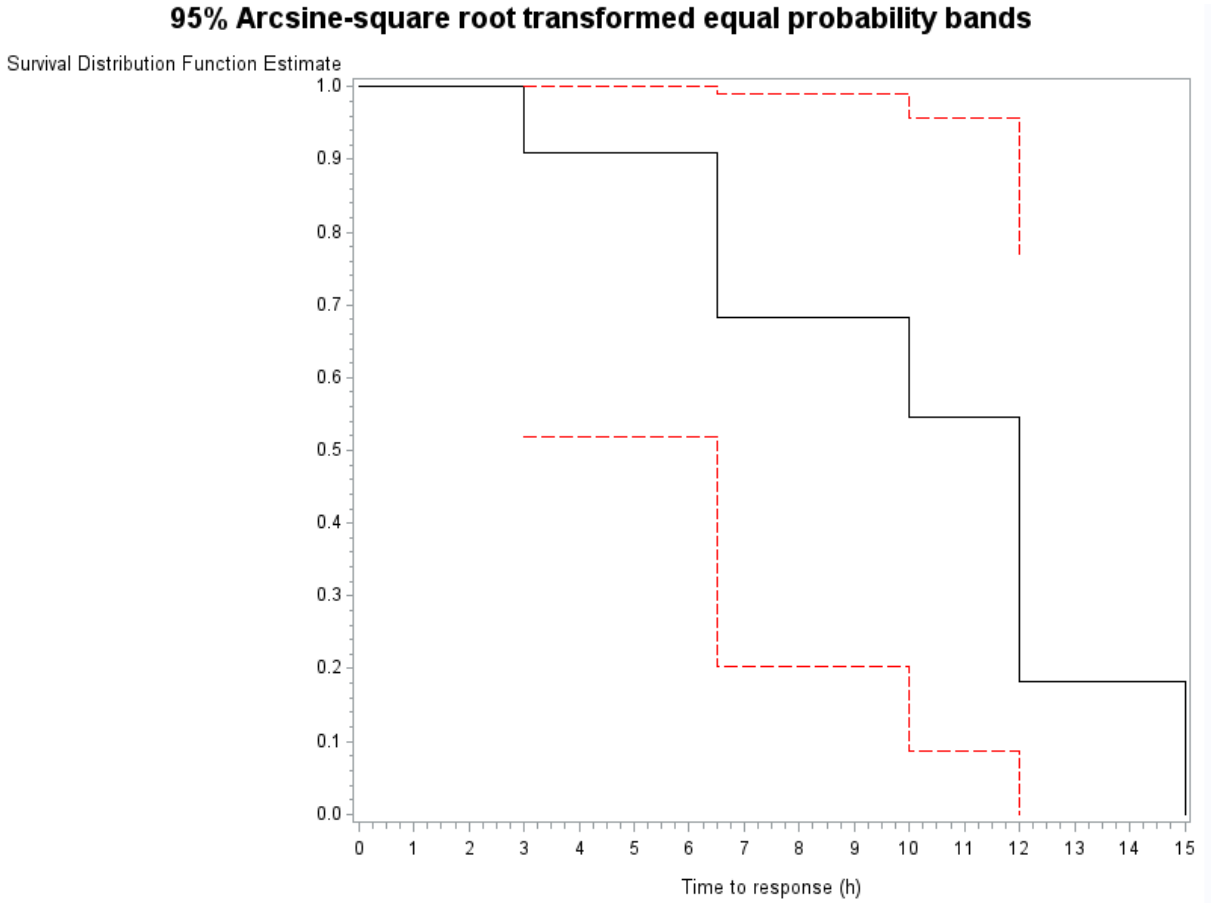
100(1-α)% arcsine-square root transformed equal probability bands for $S(t)$ over the range $[t_L, t_U]$:

$$\left[\sin^2 \left\{ \max \left[0, \quad \arcsin\left(\hat{S}(t)^{1/2}\right) - 0.5 c_\alpha(a_L, a_U)\sigma_s(t)\left(\frac{\hat{S}(t)}{1-\hat{S}(t)}\right)^{1/2} \right] \right\}, \right.$$

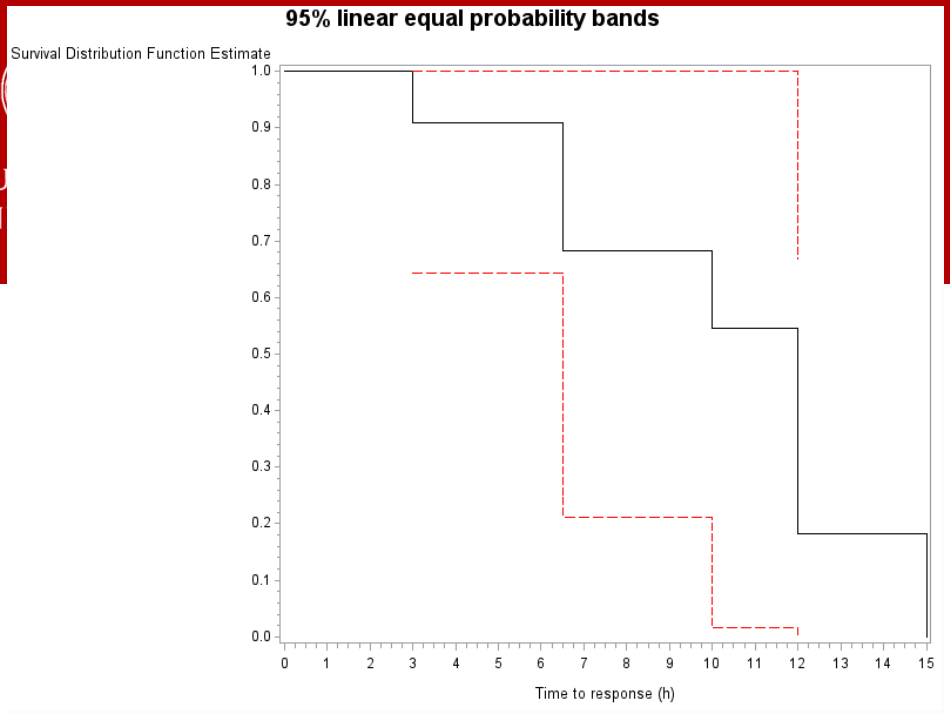
$$\left. \sin^2 \left\{ \min \left[\frac{\pi}{2}, \quad \arcsin\left(\hat{S}(t)^{1/2}\right) + 0.5 c_\alpha(a_L, a_U)\sigma_s(t)\left(\frac{\hat{S}(t)}{1-\hat{S}(t)}\right)^{1/2} \right] \right\} \right]$$

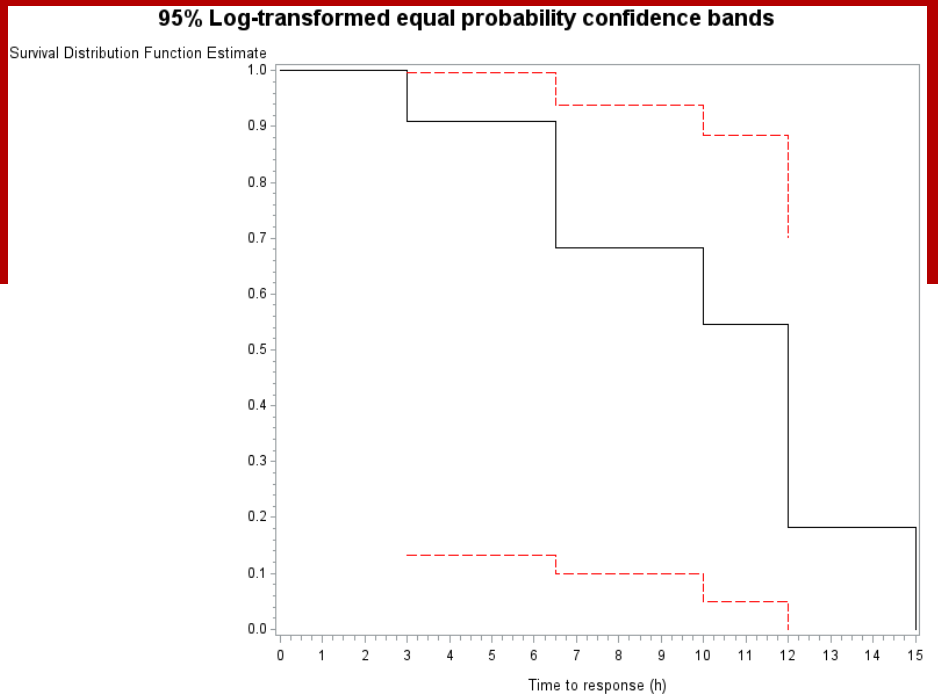
cover, with approximate confidence level (1-α), the survival function within the interval $[t_L, t_U]$.

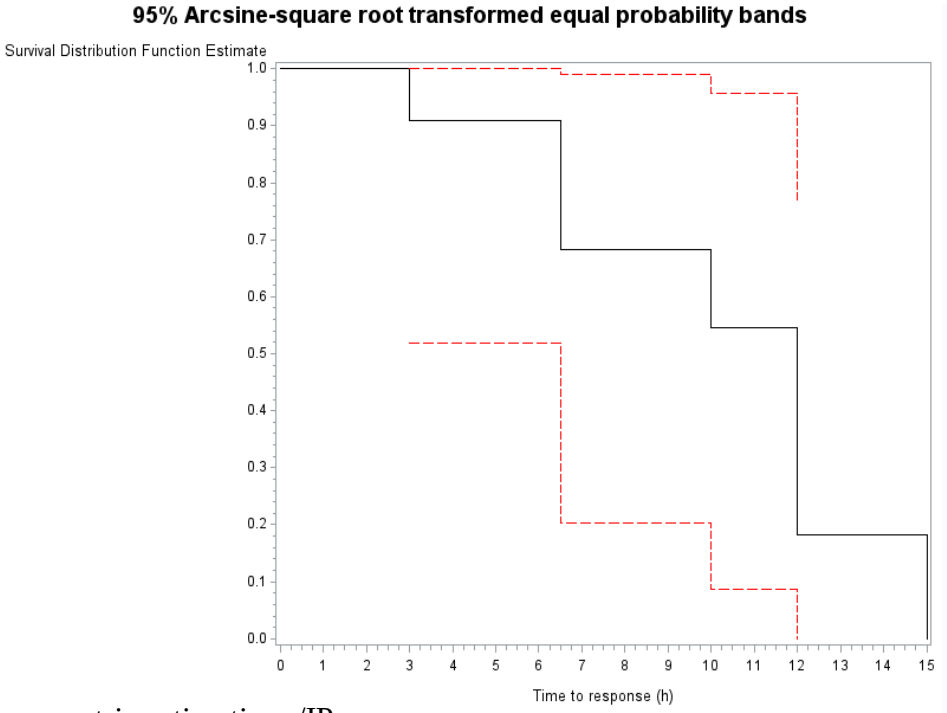
Example: Time to treatment response (h)



These bands cover, with approximately 95% confidence, the survival function within the interval [*3*, *12*] h.


95% linear equal probability bands
Survival Distribution Function Estimate
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Time to response (h)


95% Log-transformed equal probability confidence bands
Survival Distribution Function Estimate
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Time to response (h)


95% Arcsine-square root transformed equal probability bands
Survival Distribution Function Estimate
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Time to response (h)

Linear vs. transformed confidence bands for $S_{KM}(t)$

The log-transformed and arcsine-square root transformed confidence bands both perform better than the linear bands (reasonable results for >20 events).

The arcsine-square root transformed confidence bands perform a bit better than the log-transformed bands, thus the arcsine-square root transformed bands are recommended.

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Hall-Wellner confidence bands

The Hall-Wellner bands, not proportional to the pointwise confidence intervals, are wider than the equal probability bands for small t and narrower for large t.

The properties of the Hall-Wellner bands are comparable to the transformed equal probabilities bands.

Details of Hall-Wellner confidence bands left for self-studies.

Program L3

- **Nonparametric estimation of basic quantities**
  - Survival estimates
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  - Pointwise confidence intervals for $S(t)$
  - Confidence bands
  - Mean and median survival time, with confidence intervals

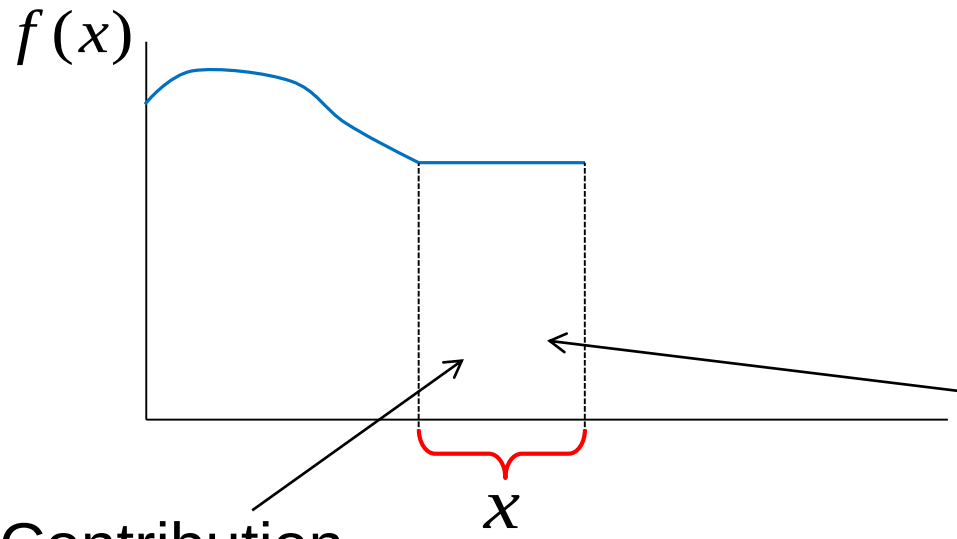
Mean survival time

$$\mu = \frac{\sum_{i=1}^{n} x_i}{n}$$

Arithmetic means cannot be used when some observations are censored.

Mean survival time

$$\mu_x = \int_{-\infty}^{\infty} x\ f(x)dx$$ (usual way of calculating the mean)



Contribution to the mean:

$x\ f(x)$

$$S(x) = \int_{x}^{\infty} f(t)dt$$

Since $x\ f(x)$ is already a part of $S(x)$, the contribution to the mean in terms of survival is $S(x)$

Mean survival time

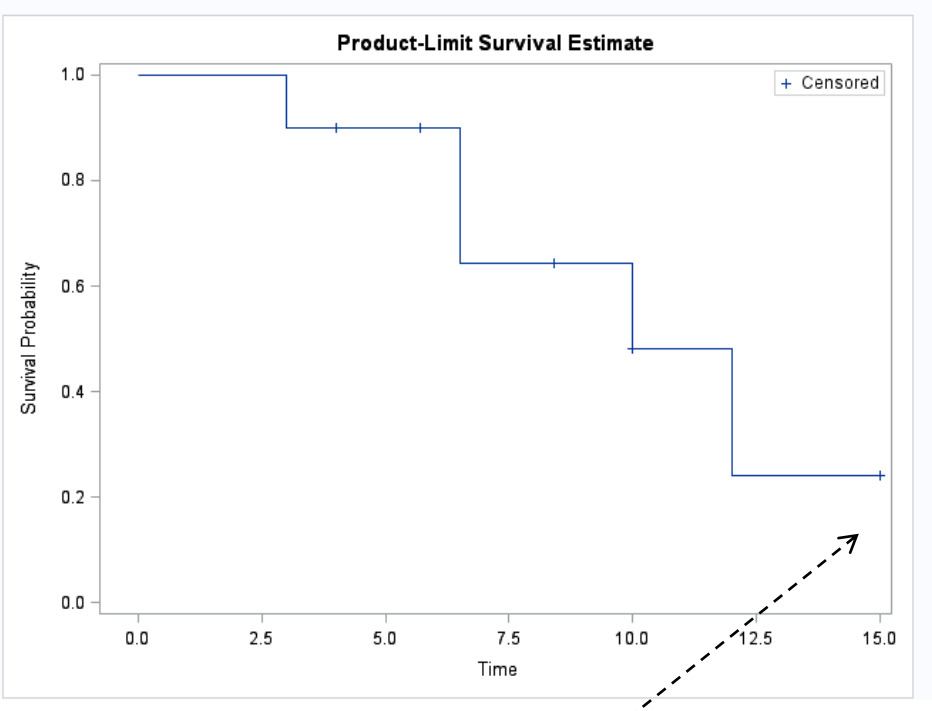
$$\mu = \int_0^\infty S(t)dt$$

can be used when the parametric survival function is known.

Nonparametric estimation of Mean survival time

A nonparametric estimator is obtained by using the Kaplan-Meier (Product-Limit) estimator $\hat{S}(t)$

$$\hat{\mu} = \int_0^{\infty} \hat{S}(t)dt$$



But if the last observation is censored, how do you calculate the area under the graph?

Nonparametric estimation of Mean survival time

Different ways of estimating the mean survival time when the last observation is censored:

1) Efron's tail correction to the Kaplan-Meier (Product-Limit) estimator, where the largest observed time is changed to an event if it is a censored observation.

2) Estimate the mean restricted to some preassigned interval $[0, \tau]$, where τ is chosen to be the "longest possible time to which anyone could survive".

Nonparametric estimation of Mean survival time

$$\hat{\mu}_\tau = \int_0^\tau \hat{S}(t)dt$$

where τ is either the longest observed time (event or censored), or preassigned by the investigator.

$$\hat{V}(\hat{\mu}_\tau) = \sum_{i=1}^{D} \left(\int_{t_i}^{\tau} \hat{S}(t)dt \right)^2 \frac{d_i}{Y_i(Y_i - d_i)}$$

Example: Time to treatment response (h)

If the last observation is an event:

Mean	Standard Error
10.2045	1.3133

If the last observation is censored:

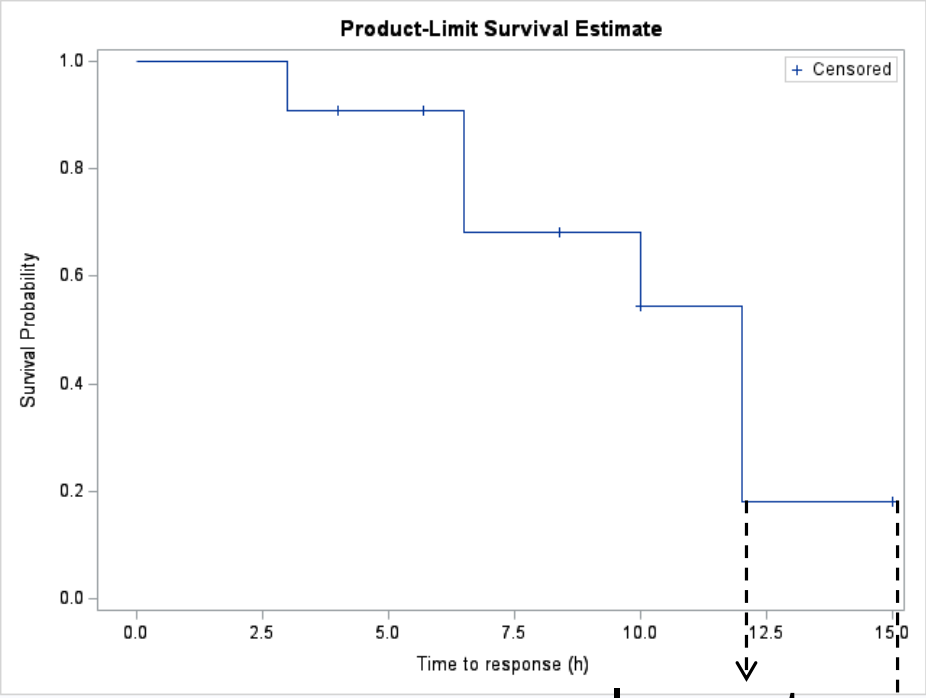
Mean	Standard Error
9.6591	1.0769

Note: The mean survival time and its standard error were underestimated because the largest observation was censored and the estimation was restricted to the largest event time.

In Sas: proc lifetest

Example: Time to treatment response (h)

Sas (proc lifetest) uses the last event time as default.



τ = largest event time

τ = largest observed time can be chosen (Efron's tail correction)

Example: Time to treatment response (h)

Efron's tail correction:

Mean	Standard Error
10.2045	1.3319

Note: The mean survival time and its standard error may have been underestimated because the estimation was restricted to the largest observation, which was censored.

(Better to use the median)

In Sas: proc lifetest

Confidence interval for the Mean survival time

100(1 - α)% confidence interval for the mean:

$$\hat{\mu}_\tau \pm Z_{1-\alpha/2}\sqrt{\hat{V}(\hat{\mu}_\tau)}$$ (variance shown previously)

Assumption:
(in addition to assumptions for S(t) estimator)

- large sample(s) (interval based on asymptotic normality)

Example: Time to treatment response (h)

95% confidence interval for the mean survival time:

$$\hat{\mu}_\tau \pm Z_{1-\alpha/2}\sqrt{\hat{V}(\hat{\mu}_\tau)}$$

Mean	Standard Error
10.2045	1.3133

$$10.2045 \pm 1.96 \cdot 1.3133$$

$$10.2045 \pm 2.5741$$

$$[7.63,\ 12.78]\ \text{h}$$

The interval 7.63 to 12.78 h covers, with 95% confidence, the mean survival time in the population.

Recap: Median Lifetime

The **median lifetime** is the time at which half of the individuals survive.

The median lifetime is the 50th percentile of the distribution of X.

In other words, the median lifetime is the value $x_{0.5}$ that fulfills

$$S(x_{0.5}) \leq 0.5$$

Nonparametric estimation of Median survival time

To estimate $x_{0.5}$, find the smallest time for which the Kaplan-Meier (Product-Limit) or Nelson-Aalen estimator is less than or equal to 0.5

$\hat{x}_{0.5}$ = median lifetime

Confidence interval, no censoring

Normally a confidence interval for the mean (or any other parameter) is created as follows:

$$\bar{x} \pm Z_{1-\alpha/2} SE$$

$$\bar{x} - Z_{1-\alpha/2} SE \leq x \leq \bar{x} + Z_{1-\alpha/2} SE$$

$$-Z_{1-\alpha/2} \leq \frac{x - \bar{x}}{SE} \leq Z_{1-\alpha/2}$$

Confidence interval, censoring present

When there is censoring, a confidence interval for the median is created as follows (linear approach):

$$-Z_{1-\alpha/2} \leq \frac{\hat{S}(t)-0.5}{\sqrt{\hat{V}\left(\hat{S}(t)\right)}} \leq Z_{1-\alpha/2}$$

The confidence interval is the set of all time points which satisfy this condition.

You simply have to try reasonable values!

Example: Time to treatment response (h) (recap.)

t_i	d_i	Y_i	$\hat{S}(t)$	$\sum_{t_i \le t} \frac{d_i}{Y_i(Y_i - d_i)}$	$\hat{V}\left[\hat{S}(t)\right]$
3	1	11	0.91	0.009	0.007
4+	0	10	(0.91)		
5.7+	0	9			
6.5	2	8	0.68	0.051	0.024
8.4+	0	6			
10	1	5	0.54	0.101	0.029
10+	0				
12	2	3	0.18	0.768	0.025
15	1	1	0		

$$S(\hat{x}_{0.5}) \le 0.5$$

Example: Time to treatment response (h)

Median time to response:

$$\hat{x}_{0.5} = 12 \text{ h}$$

95% confidence interval for median time to response:

$$-1.96 \leq \frac{\hat{S}(t) - 0.5}{\sqrt{\hat{V}\left(\hat{S}(t)\right)}} \leq 1.96$$

Try different times to find the interval.

Example: Time to treatment response (h)

$$\hat{S}(3) = 0.91 \qquad \hat{V}(\hat{S}(3)) = 0.007$$

$$\frac{\hat{S}(t) - 0.5}{\sqrt{\hat{V}(\hat{S}(t))}} = \frac{0.91 - 0.5}{\sqrt{0.007}} = 4.9$$

Not included in the interval.

Example: Time to treatment response (h)

$$\hat{S}(6.5) = 0.68 \qquad \hat{V}(\hat{S}(6.5)) = 0.024$$

$$\frac{\hat{S}(t) - 0.5}{\sqrt{\hat{V}(\hat{S}(t))}} = \frac{0.68 - 0.5}{\sqrt{0.024}} = 1.16$$

This means that t=6.5 is included in the interval
(6.5 is thus the lower limit of a 95% confidence interval).

Example: Time to treatment response (h)

Quartile Estimates				
Percent	Point Estimate	95% Confidence Interval		
		Transform	[Lower	Upper)
75	12.0000	LINEAR	10.0000	.
50	12.0000	LINEAR	6.5000	12.0000
25	6.5000	LINEAR	3.0000	12.0000

95% confidence interval for median time to response:

[6.5, 12] hours

In Sas: proc lifetest

Nonparametric confidence interval for Median survival time

Linear: $$-Z_{1-\alpha/2} \le \frac{\hat{S}(t)-0.5}{\sqrt{\hat{V}\left(\hat{S}(t)\right)}} \le Z_{1-\alpha/2}$$

Log-transformed:

$$-Z_{1-\alpha/2} \le \frac{\left[\ln\left\{-\ln\left(\hat{S}(t)\right)\right\}-\ln\left\{-\ln 0.5\right\}\right]\left[\hat{S}(t)\ln\left(\hat{S}(t)\right)\right]}{\sqrt{\hat{V}\left(\hat{S}(t)\right)}} \le Z_{1-\alpha/2}$$

Arcsine-square root transformed:

$$-Z_{1-\alpha/2} \le \frac{2\left\{\arcsin\sqrt{\hat{S}(t)}-\arcsin\sqrt{0.5}\right\}\sqrt{\hat{S}(t)\left(1-\hat{S}(t)\right)}}{\sqrt{\hat{V}\left(\hat{S}(t)\right)}} \le Z_{1-\alpha/2}$$

Assumptions for confidence intervals for median survival time

Assumption (in addition to the assumptions for the estimator of $S(t)$):

- large sample(s) (intervals based on asymptotic normality)

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Choice of transformation

With large amounts of data (events), all the alternatives produce acceptable intervals.

With small samples (n < approx. 40 with up to 30% cens.), the log-transformation produces the best results (good coverage probability and not extremely wide intervals) for the Kaplan-Meier estimator, and the linear transformation works best for the Nelson-Aalen estimator*.

* *Talsma (2023), Estimation of median survival time and its 95% confidence intervalusing SAS PROC LIFETEST, Journal of Biopharmaceutical Statistics, 34(3), 366–378.*

Example: Time to treatment response (h)

Quartile Estimates				
Percent	Point Estimate	95% Confidence Interval		
		Transform	[Lower	Upper)
75	12.0000	LINEAR	10.0000	.
50	12.0000	LINEAR	6.5000	12.0000
25	6.5000	LINEAR	3.0000	12.0000

Quartile Estimates				
Percent	Point Estimate	95% Confidence Interval		
		Transform	[Lower	Upper)
75	12.0000	LOGLOG	10.0000	.
50	12.0000	LOGLOG	6.5000	.
25	6.5000	LOGLOG	3.0000	12.0000

Quartile Estimates				
Percent	Point Estimate	95% Confidence Interval		
		Transform	[Lower	Upper)
75	12.0000	ASINSQRT	10.0000	
50	12.0000	ASINSQRT	6.5000	.
25	6.5000	ASINSQRT	3.0000	12.0000

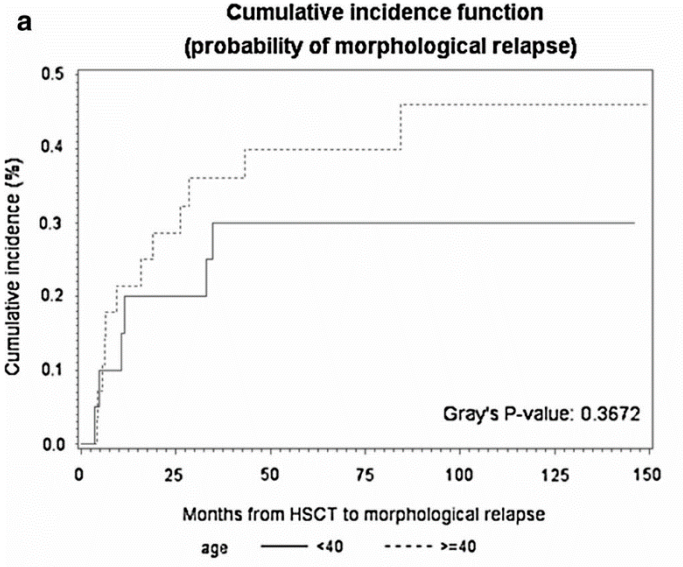
No upper limit defined

Either no events after median time included in interval, or no events with defined variance exist.

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Analysis of Survival Data

Lecture 4:
Summary curves for competing risks
Hypothesis testing



$$H_0 : h_1(t) = h_2(t) \quad \text{for all } t \le \tau$$

$$H_a : h_1(t) \neq h_2(t) \quad \text{for some } t \le \tau$$

Inger Persson

Program L4

- **Summary curves for competing risks**
  - Cumulative incidence function
- **Hypothesis testing**
  - K samples tests
    - Log-rank test
    - Gehan's test
    - Gray's test
  - Stratified tests
  - Differences at a fixed point in time

Competing risks

If a subject may fail due to one of several causes we have **competing risks**.

Example:
X = survival time for patients with a certain heart disease.

Competing risk:
Death from other causes (old age, other diseases, etc.).

If one of the competing risks occurs the event of interest cannot be observed.

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Competing risks vs censoring

Special cases of competing risks can be handled by random censoring.

E.g. when people in the study move abroad, or no longer want to participate.

If the competing events are independent, it is usually okay to denote these observations as censored. In these situations the censoring will be noninformative.

But if the competing events are dependent, the censoring is no longer noninformative.

Competing risks and Kaplan-Meier

The KM method assumes that all events are independent, and thus, censors events other than the event of interest.

If there are competing risks the KM estimates are biased.

Competing risks and Kaplan-Meier

Example: Studies of bone marrow transplantation and leukemia relapse.

Competing risk: death caused by transplantation complications.

Leukemia relapse and transplantation related mortality are not independent, since both events are likely related to immunologic mechanism following transplantation.

The K-M method is inappropriate for estimating the rate of relapse since it censors transplantation related mortality.

Summary curves for competing risks

In the case of competing risks the event times and censoring times may not be independent.

Three techniques for summarizing competing risks data:

1) The complement of the Kaplan-Meier estimator, $1-\hat{S}(t)$ Not recommended.
2) The cumulative incidence function
3) Conditional probability for the competing risk

Complement of the Kaplan-Meier estimator

Probability of experiencing the event before time t:

$$1-\hat{S}(t)$$

where occurrences of the competing event(s) are treated as censored observations.

This can be interpreted as the risk of experiencing the event by time t if all competing risks are removed.

This is rarely of clinical interest and this estimator is not recommended.

Cumulative incidence function

One of the competing risks occurs at K distinct times, $t_1 < t_2 < \cdots < t_K$

Y_i = number of individuals at risk at t_i (number of individuals who have not yet experienced any of the competing risks or experience one of them at t_i).

r_i = number of individuals who experience the event of interest at t_i.

d_i = number of individuals who experience any of the other events at t_i.

Cumulative incidence function

The **cumulative incidence function** is defined as

$$CI(t) = \begin{cases} 0 & \text{if } t < t_1 \\ \sum_{t_i \le t} \left(\prod_{j=1}^{i-1} 1 - \frac{d_j + r_j}{Y_j} \right) \frac{r_i}{Y_i} & \text{if } t_1 \le t \end{cases}$$

$$= \begin{cases} 0 & \text{if } t < t_1 \\ \sum_{t_i \le t} \hat{S}(t_{i-}) \frac{r_i}{Y_i} & \text{if } t_1 \le t \end{cases}$$

The Kaplan-Meier estimate just before t_i, treating any of the competing risks as an event.

Cumulative incidence function

The cumulative incidence function estimates the probability that the event of interest occurs at or before time *t* and that it occurs before any of the competing risks.

Example: Time to relapse

t_i	*Event*	Y_i	$\hat{S}_{KM}(t)$
10	R	10	
20+		9	
35	R	8	
40	T	7	
50+		6	
55	R	5	
70	T	4	
71	T	3	
80	R	2	
90+		1	

$$\hat{S}(t) = \begin{cases} 1 & \text{if } t < t_1 \\ \prod_{t_i \le t} \left(1 - \frac{d_i + r_i}{Y_i}\right) & \text{if } t_1 \le t \end{cases}$$

R = relapse

T = treatment-related death

Example: Time to relapse

$$\hat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i + r_i}{Y_i}\right) \qquad \hat{S}(10) = \left(1 - \frac{d_{10}}{Y_{10}}\right) = \left(1 - \frac{1}{10}\right) = 0.9$$

$$\hat{S}(35) = \prod_{t_i \le 35} \left(1 - \frac{d_i + r_i}{Y_i}\right) = 0.9\left(1 - \frac{1}{8}\right) = 0.7875$$

$$\hat{S}(40) = \prod_{t_i \le 40} \left(1 - \frac{d_i + r_i}{Y_i}\right) = 0.7875\left(1 - \frac{1}{7}\right) = 0.675$$

$$\hat{S}(55) = \prod_{t_i \le 55} \left(1 - \frac{d_i + r_i}{Y_i}\right) = 0.675\left(1 - \frac{1}{5}\right) = 0.54$$

Example: Time to relapse

$$CI(t) = \begin{cases} 0 & \text{if } t < t_1 \\ \sum_{t_i \le t} \hat{S}(t_{i-}) \frac{r_i}{Y_i} & \text{if } t_1 \le t \end{cases}$$

$$CI(10) = 1\frac{1}{10} = 0.10$$

$$CI(35) = 0.10 + 0.9\frac{1}{8} = 0.2125$$

Example: Time to relapse

$$CI(55) = 0.2125 + 0.675\frac{1}{5} = 0.3475$$

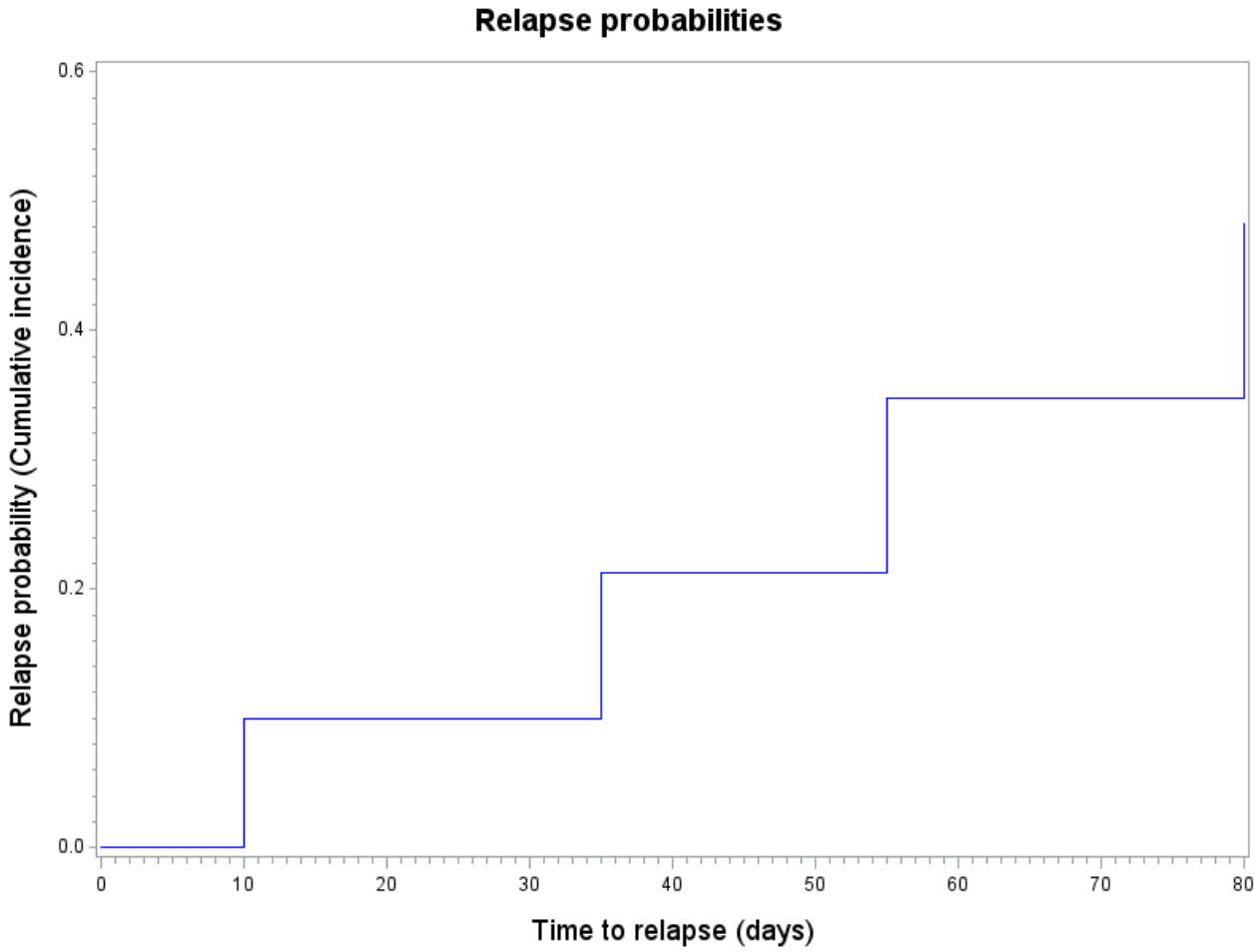
$$CI(80) = 0.3475 + 0.27\frac{1}{2} = 0.4825$$

Example: Time to relapse

t_i	*Event*	Y_i	$\hat{S}_{KM}(t)$	$CI(t)$
10	R	10	0.9	0.10
20+		9		
35	R	8	0.7875	0.2125
40	T	7	0.675	
50+		6		
55	R	5	0.54	0.3475
70	T	4	0.405	
71	T	3	0.27	
80	R	2	0.135	0.4825
90+		1		

R = relapse
T = treatment-related death

Example: Time to relapse



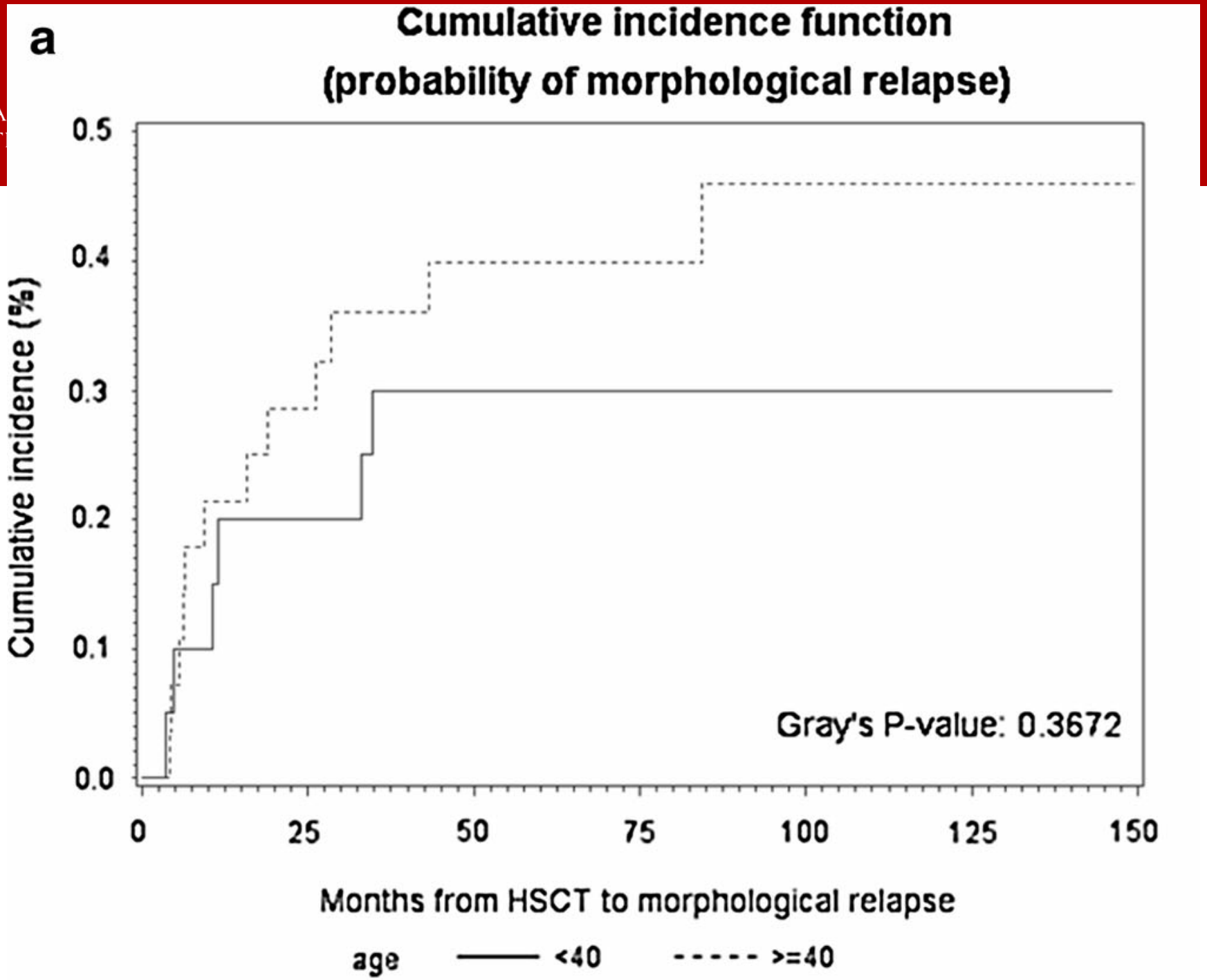
Example: ALL outcome after stem cell transplantation

Med Oncol (2014) 31:66
DOI 10.1007/s12032-014-0066-9

ORIGINAL PAPER

Outcome after HSCT in Philadelphia chromosome positive acute lymphoblastic leukemia in Sweden: a population-based study

E. Hulegårdh · H. Hägglund · L. Ahlberg · K. Karlsson · H. Karbach · A. Markuszewska · I. Persson · M. Åström · H. Hallböök


a
Cumulative incidence function
(probability of morphological relapse)
Cumulative incidence (%)
0.0
0.1
0.2
0.3
0.4
0.5
Gray's P-value: 0.3672
0
25
50
75
100
125
150
Months from HSCT to morphological relapse
age
<40
>=40

Assumptions

Assumptions for the Cumulative Incidence Function (same as for the Kaplan-Meier estimator*):

- random sample(s)
- independent samples (if >1 group)
- noninformative censoring
- right censored data

**Kaplan and Meier, Nonparametric Estimation from Incomplete Observations, JASA, 1958;*

Conditional probability function for the competing risk

The conditional probability is an estimate of the probability of the occurrence of event K by time t, given that none of the competing risks have occurred by that time.

Cumulative incidence for risk K

$$CP_K(t) = \frac{CI_K(t)}{1 - CI_{K^c}(t)}$$

Cumulative incidence for all other risks lumped together.

Program L4

- **Summary curves for competing risks**
  - Cumulative incidence function
- **Hypothesis testing**
  - K samples tests
    - Log-rank test
    - Gehan's test
    - Gray's test
  - Stratified tests
  - Differences at a fixed point in time

K sample tests (K ≥ 2)

Independent right censored samples (K samples/groups).

$$H_0 : h_1(t) = h_2(t) = \ldots h_k(t) \text{ for all } t \leq \tau$$

$$H_a : \text{At least one of the } h_j(t)'s \text{ different for some } t \leq \tau$$

τ = largest time at which all of the groups have at least one subject at risk

K sample tests (K ≥ 2)

Event times $t_1 < t_2 < \ldots < t_D$ in the pooled sample.

$$d_i = \sum_{j=1}^{K} d_{ij}$$

i: Time point; j: Group

= # patients that experience the event in the pooled sample

$$Y_i = \sum_{j=1}^{K} Y_{ij}$$

= # patients at risk in the pooled sample

Nelson-Aalen crude estimate of $h(t_i)$

Recap: An estimator of the cumulative hazard rate $H(t)$, is the **Nelson-Aalen estimator**.

$$\tilde{H}(t) = \begin{cases} 0 & \text{if } t < t_1 \\ \sum_{t_i \le t} \left(\frac{d_i}{Y_i} \right) & \text{if } t_1 \le t \end{cases}$$

A crude estimate of the hazard rate at an event time t_i is given by $\frac{d_i}{Y_i}$

K sample tests (K ≥ 2)

$$P(\text{experience the event at time } t_i) = \frac{d_i}{Y_i} \mid H_0 \text{ true}$$

Reasonable to compare $\frac{d_{ij}}{Y_{ij}}$ to $\frac{d_i}{Y_i}$

Proportion that experience the event in group j

Proportion that experience the event in all groups

K sample tests (K ≥ 2)

Test function:

$$Z_j(\tau) = \sum_{i=1}^{D} W_j(t_i)\left\{\frac{d_{ij}}{Y_{ij}} - \frac{d_i}{Y_i}\right\}$$

Positive weight function
$W_j(t)$ = zero when Y_{ij} is zero.

K sample tests (K ≥ 2)

All of the commonly used tests have

$W_j(t_i) = Y_{ij} W(t_i)$ where $W(t_i)$ is common for all groups

$$Z_j(\tau) = \sum_{i=1}^{D} W(t_i) \left\{ d_{ij} - Y_{ij} \frac{d_i}{Y_i} \right\}$$

Observed # events in group j

Expected # events in group j

$$\hat{V}\left(Z_j(\tau)\right) = \sum_{i=1}^{D} W(t_i)^2 \frac{Y_{ij}}{Y_i} \left(1 - \frac{Y_{ij}}{Y_i}\right) \left(\frac{Y_i - d_i}{Y_i - 1}\right) d_i$$

K sample tests (K ≥ 2)

$$\sum_{i=1}^{K} Z_j(\tau) = 0 \quad \Longrightarrow \quad Z_j\text{'s are linearly dependent}$$

The test statistic is constructed by selecting any K-1 of the Z_j's.

For two groups: $$\frac{Z_j(\tau)^2}{\hat{V}\left(Z_j(\tau)\right)} \sim \chi^2$$

j = 1 or 2, doesn't matter which group

Log-rank and Gehan's test

Two most common weight functions:

1) $W(t) = 1 \text{ for all } t$ gives the **Log-rank test** *(alias Mantel-Cox, or Generalized Savage test)*

2) $W(t_i) = Y_i$ gives **Gehan's test** *(alias Breslow-Gehan, or Wilcoxon test)*

Assumptions

Assumptions for the Log-rank and Gehan's tests*:

- random, independent samples
- noninformative censoring
- right censored data
- survival probabilities are the same for subjects recruited early and late in the study
- large samples (tests are based on large-sample approximations to the chi-square distribution)

**Log rank test described by Bland & Altman, BMJ. May 1, 2004; 328(7447): 1073.*

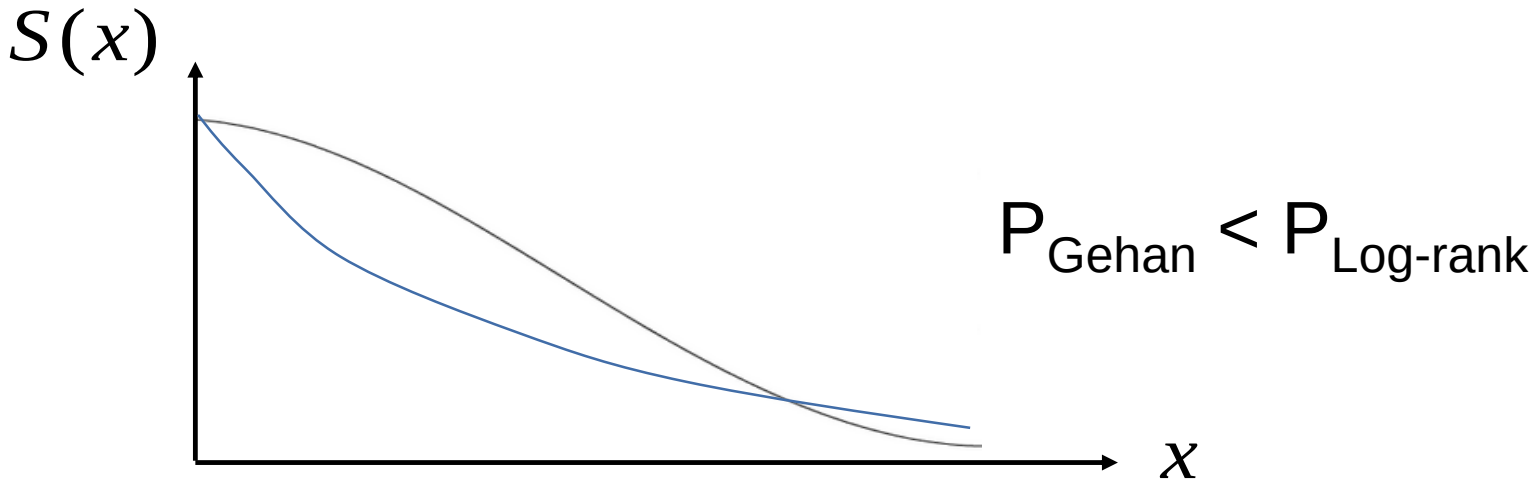
Log-rank test vs. Gehan’s test

The **Log-rank test** has optimum power to detect differences when the hazard rates in all groups are proportional to each other.

Gehan’s test puts larger weight to early differences.

Both test can have misleading results when censoring patterns are different in different groups (Gehan's test more sensitive).

Log-rank test vs. Gehan's test



Your hypothesis decides which test to use.

Example: Lifetime of refrigerators

Refrigerators with two different types of components.

$$H_0 : h_1(t) = h_2(t) \quad \text{for all } t \le \tau$$
$$H_a : h_1(t) \neq h_2(t) \quad \text{for some } t \le \tau$$

τ = largest time at which both groups of refrigerators have at least one subject at risk

Example: Lifetime of refrigerators

Significance level?

Always reflect upon the consequences of wrongly rejecting H_0.

Wrongly rejecting H_0 in this case means that we would claim that there is a difference in survival between the two types of components, when in fact there is no difference.

Example: Lifetime of refrigerators

Does this have any serious consequences?

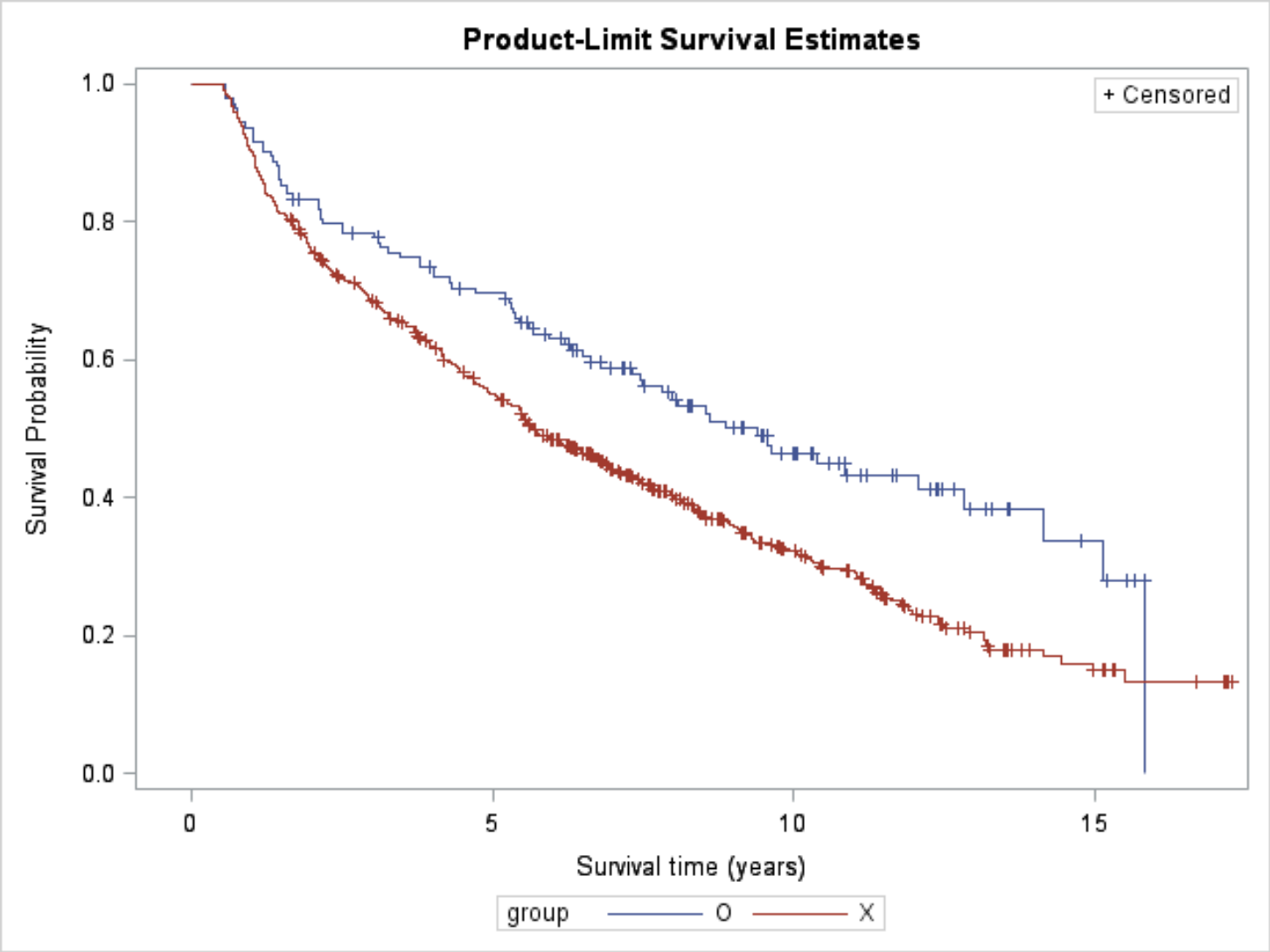
Depends on the circumstances.

One possible consequence of claiming that one component is better than the other, if it isn't, is that the trust/reputation of the company/product may go down.

Economical consequences could also be considered. If one of the components is extremely expensive, we might want to choose a lower significance level to get a "safer" result to base our investment decision on.

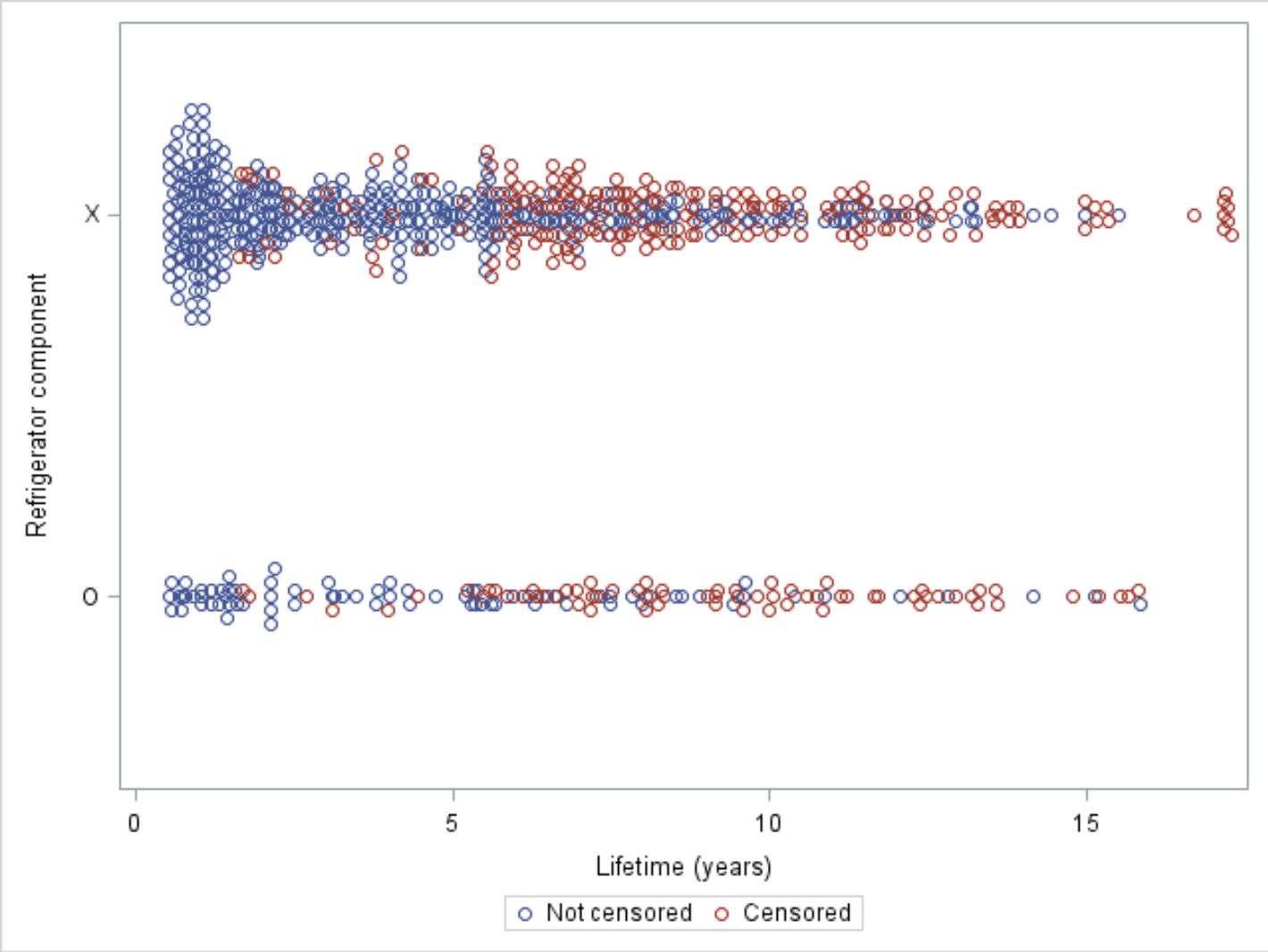
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Example: Lifetime of refrigerators



In Sas: proc lifetest, strata statement.

Example: Lifetime of refrigerators



Similar pattern in both groups, i.e., doesn't contradict noninformative censoring

Example: Lifetime of refrigerators

Summary of the Number of Censored and Uncensored Values					
Stratum	group	Total	Failed	Censored	Percent Censored
1	O	144	76	68	47.22
2	X	610	410	200	32.79
Total		754	486	268	35.54

No. events

In Sas: proc lifetest, strata statement.

Recap: Assumptions

Assumptions for the Log-rank and Gehan's tests*:

- random, independent samples ✓
- noninformative censoring - *To be discussed with field experts*
- right censored data ✓
- survival probabilities are the same for subjects recruited early and late in the study ✓ - *To be discussed with field experts (not an issue in this study since all refrigerators are manufactured at the same time)*
- large samples (tests are based on large-sample approximations to the chi-square distribution) ✓

**Log rank test described by Bland & Altman, BMJ. May 1, 2004; 328(7447): 1073.*

Example: Lifetime of refrigerators

The LIFETEST Procedure

Testing Homogeneity of Survival Curves for time over Strata

Rank Statistics		
group	Log-Rank	Wilcoxon
O	-31.719	-13736
X	31.719	13736

$= Z_j(\tau)$

Example: Lifetime of refrigerators

Covariance Matrix for the Wilcoxon Statistics		
group	O	X
O	20133277	-2.013E7
X	-2.013E7	20133277

$$= \hat{V}\left(Z_j(\tau)\right)$$

Example: Lifetime of refrigerators

Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	12.0543	1	0.0005
Wilcoxon	9.3714	1	0.0022
-2Log(LR)	13.0272	1	0.0003

Log-Rank test ---->

Gehan's test ----->

Likelihood ratio test. Based on the exponential distribution which is rarely applicable in a survival model. Can be ignored.

Other weight functions

$$W(t_i) = \sqrt{Y_i}$$

Tarone and Ware.
Gives more weight to differences at time points where there is most data.

$$W(t_i) = \tilde{S}(t_i) \quad \text{where} \quad \tilde{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{Y_i + 1}\right)$$

Peto and Peto, Kalbfleish and Prentice.

$$W(t_i) = \tilde{S}(t_i)\frac{Y_i}{Y_i + 1}$$

Andersen et al.

The weights depend on the combined survival experience in the pooled sample.

Other weight functions

$$W_{p,q}(t_i) = \hat{S}_{KM}(t_{i-1})^p \left(1 - \hat{S}_{KM}(t_{i-1})\right)^q, \quad p \geq 0,\ q \geq 0$$

Fleming and Harrington.
A very general class of tests.
E.g. p=q=0 gives the Log-rank test.

When q=0 and p>0 most weight is given to early departures,
and when p=0 and q>0 most weight is given to late departures

Gray's test for cumulative incidence

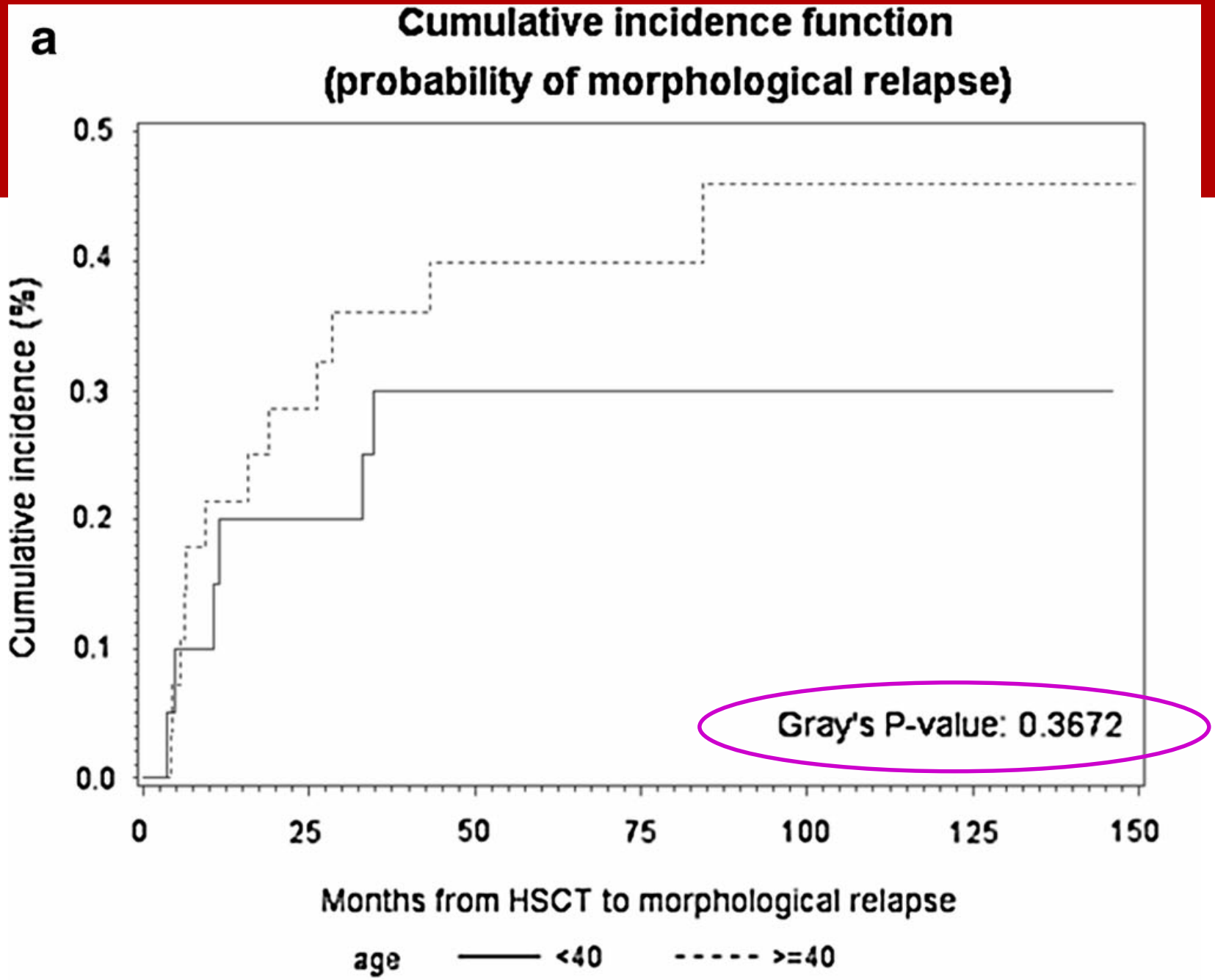
Independent right censored samples (K samples/groups).

$$H_0 : CI_1(t) = CI_2(t) = \ldots CI_k(t) \text{ for all } t \leq \tau$$

$$H_a : \text{At least one of the } CI_j(t)'s \text{ different for some } t \leq \tau$$

τ = largest time at which all of the groups have at least one subject at risk

Gray, R. (1988), A Class of K-Sample Tests for Comparing the Cumulative Incidence of a Competing Risk. The Annals of Statistics, 16, 1141–1154.


a
Cumulative incidence function
(probability of morphological relapse)
Cumulative incidence (%)
0.0
0.1
0.2
0.3
0.4
0.5
0
25
50
75
100
125
150
Months from HSCT to morphological relapse
Gray's P-value: 0.3672
age
<40
>=40

Assumptions

Assumptions for Gray's test:

- random, independent samples[1]
- noninformative censoring[1]
- right censored data[1]
- time to event and event type independent and identically distributed within each group[2]
- large samples (asymptotic results)[2]

[1] *Same as for the Kaplan-Meier estimator*

[2] *Gray, R. (1988), A Class of K-Sample Tests for Comparing the Cumulative Incidence of a Competing Risk. The Annals of Statistics, 16, 1141–1154.*

Survival vs. cumulative incidence

"..cause-specific hazards and cumulative incidence curves capture different aspects of the event histories in competing risks data, inference on these metrics may yield different results."

".. the log-rank test correctly detects differences in cause-specific hazards, and, unless there is strong dependence between failure times, is largely unaffected by between-group differences in hazards for other competing events.."

Dignam JJ, Kocherginsky MN (2008) Choice and interpretation of statistical tests used when competing risks are present. J Clin Oncol 26: 4027-4034.

Survival vs. cumulative incidence

"On the other hand, Gray's test correctly detects whether there is a difference in cumulative incidence between groups for a given event, whether that difference is caused by a difference in hazards between the groups for the event itself or by a difference in hazards for the competing events."

"Thus, when choosing and interpreting a statistical test, one must take these properties into account, as well as whether primary interest is in contrasting the relative rate of events or the absolute difference in incidence of events between groups."

Program L4

- **Summary curves for competing risks**
  - Cumulative incidence function
- **Hypothesis testing**
  - K samples tests
    - Log-rank test
    - Gehan's test
    - Gray's test
  - Stratified tests
  - Differences at a fixed point in time

Stratified tests

There might be some other covariates that affect the event rates in the K different populations.

Example: A breast cancer treatment is being tested.

Two groups are compared, A= new treatment, B= old treatment.

There are more smokers in group A than in group B. It would be wise to adjust for the variable smoking.

This can be done by stratifying a K sample test (e.g. the Log-rank test).

Stratified tests

Independent right censored samples (K samples/groups, M strata).

$$H_0 : h_{1s}(t) = h_{2s}(t) = \ldots h_{Ks}(t), \quad \text{for s} = 1,\ldots,M, \quad t < \tau$$

$$H_a : \text{At least one of the } h_{js}(t)\text{'s different for some } t < \tau$$

τ = largest time at which all of the groups have at least one subject at risk

Stratified tests

$$Z_{js}(\tau) = \sum_{i=1}^{D} W(t_i)\left\{ d_{ij} - Y_{ij}\frac{d_i}{Y_i} \right\} \text{ for stratum } s$$

$$Z_j(\tau) = \sum_{s=1}^{M} Z_{js}(\tau) \text{ for all strata}$$

$$\hat{V}\left(Z_j(\tau)\right) = \sum_{s=1}^{M} \hat{V}\left(Z_{js}(\tau)\right)$$

Power of stratified tests

Stratified tests have good power against alternative hypotheses that are in the same direction in each stratum.

When this is not the case, stratified tests may have very low power.

In such situations it is better to use separate tests for each stratum.

Example: Hodgkin's disease

A small study comparing the effectiveness of allogeneic transplants versus autogeneic transplants for Hodgkin's disease or non-Hodgkin's lymphoma is presented in section 1.10.

Allogeneic transplant: from a matching donor

Autogeneic transplant: you own bone marrow is cleansed and returned after a high dose of chemotherapy

Is there a difference in disease-free survival between allogeneic and autogeneic transplants?

Example: Hodgkin's disease

Variables:

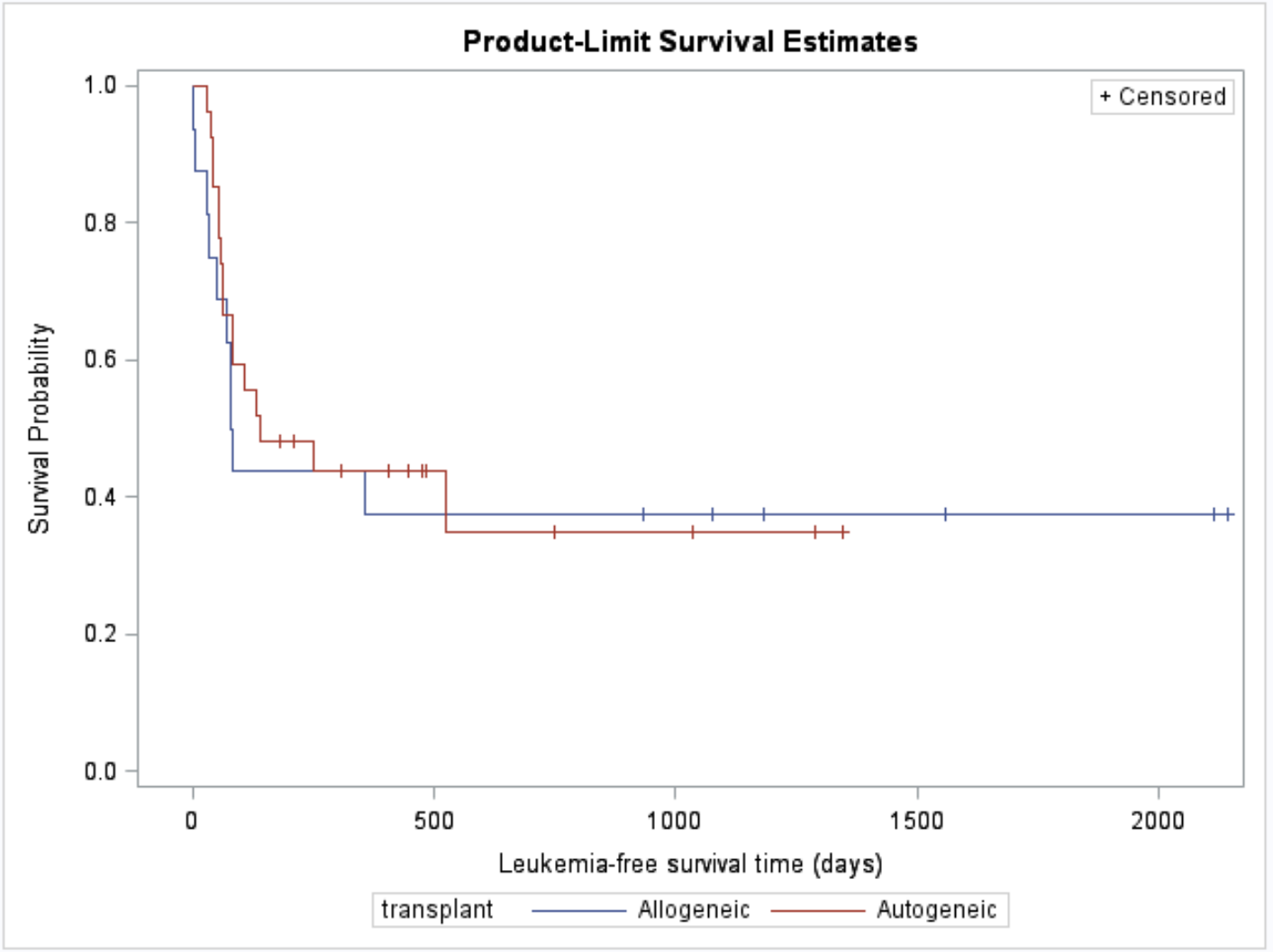
freetime = time to death or relapse (days)
transplant = type of transplant (0=allogeneic, 1=autogeneic)
event = event indicator (1=dead or relapse,
0=alive without relapse)
disease = disease type (0=non-Hodgkin's lymphoma,
1=Hodgkin's disease)

Example: Hodgkin's disease

To plot the Kaplan-Meier (Product-Limit) survival, use the **strata** command to compare the transplant groups.

```
proc lifetest data=hodgkins;
   time freetime*event(0);
   strata transplant;
run;
```

Example: Hodgkin's disease



Example: Hodgkin's disease

To stratify the test, use the **strata** command for the stratifying variable and the /**group=** option for the variable to be tested.

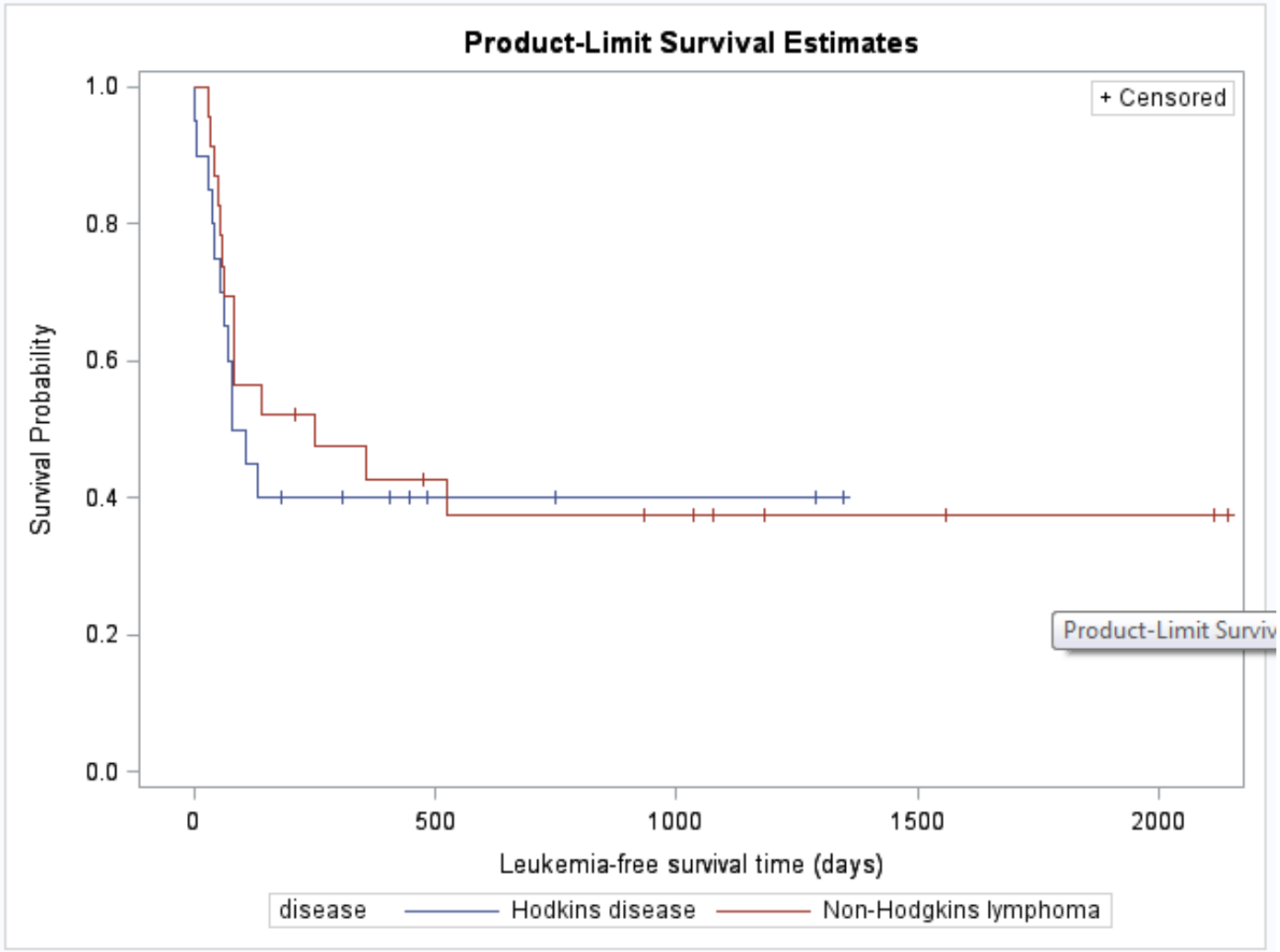
```
proc lifetest data=hodgkins;
   time freetime*event(0);
   strata disease / group=transplant;
run;
```

Stratifying variable

Variable containing groups to be compared

Example: Hodgkin's disease

Note: Survival curves by the stratification variable only (disease)



Example: Hodgkin's disease

Group = type of transplant

Stratified by disease

Stratified Test of Equality over Group			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	0.1202	1	0.7288
Wilcoxon	0.2942	1	0.5875

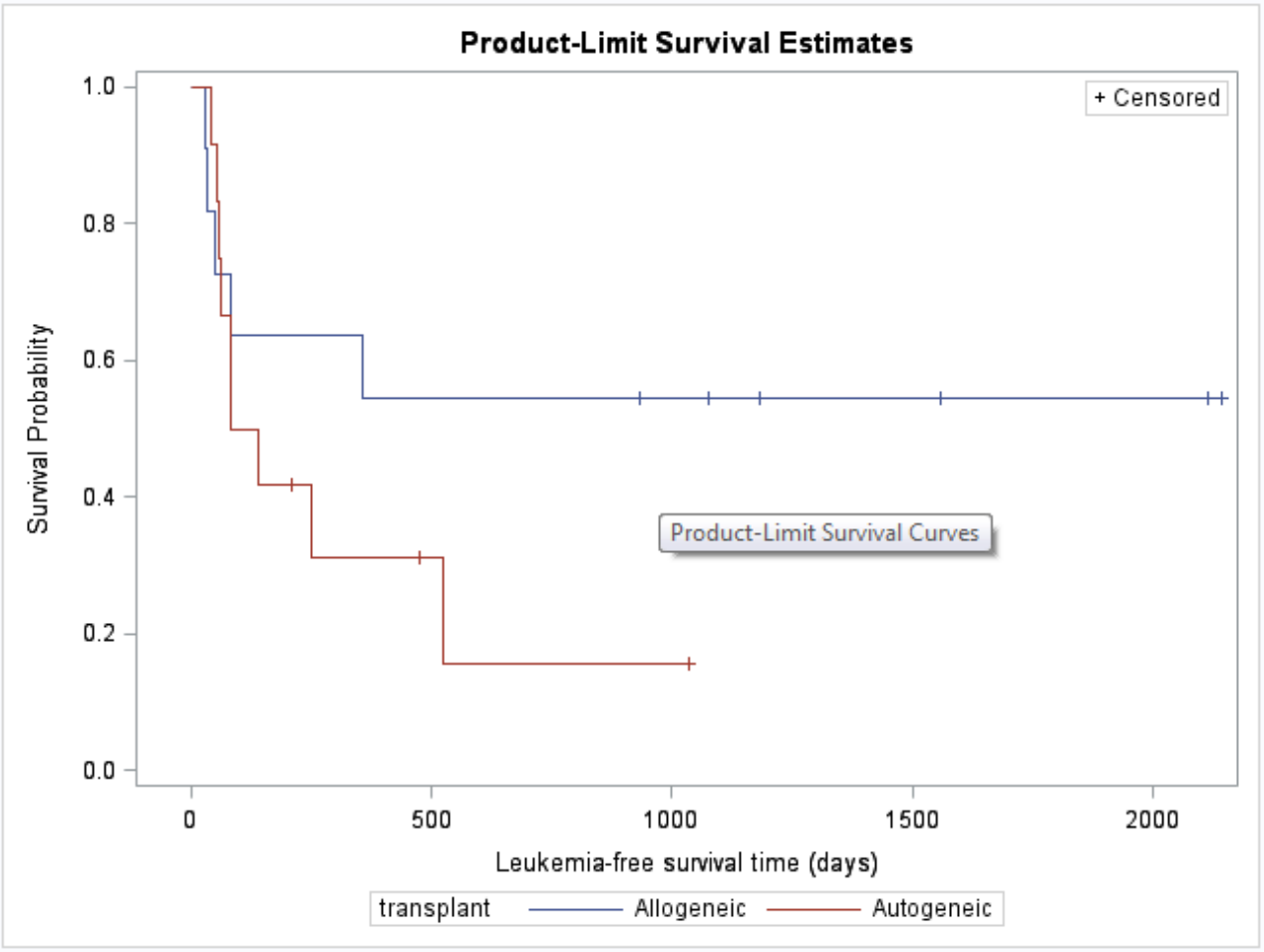
Example: Hodgkin's disease

Alternative to use if the alternative hypothesis is not in the same direction in the disease groups; testing the difference between types of transplant for each disease group separately:

```
proc lifetest data=hodgkins;
   time freetime*event(0);
   strata transplant;
   by disease;
run;
```

Example: Hodgkin's disease

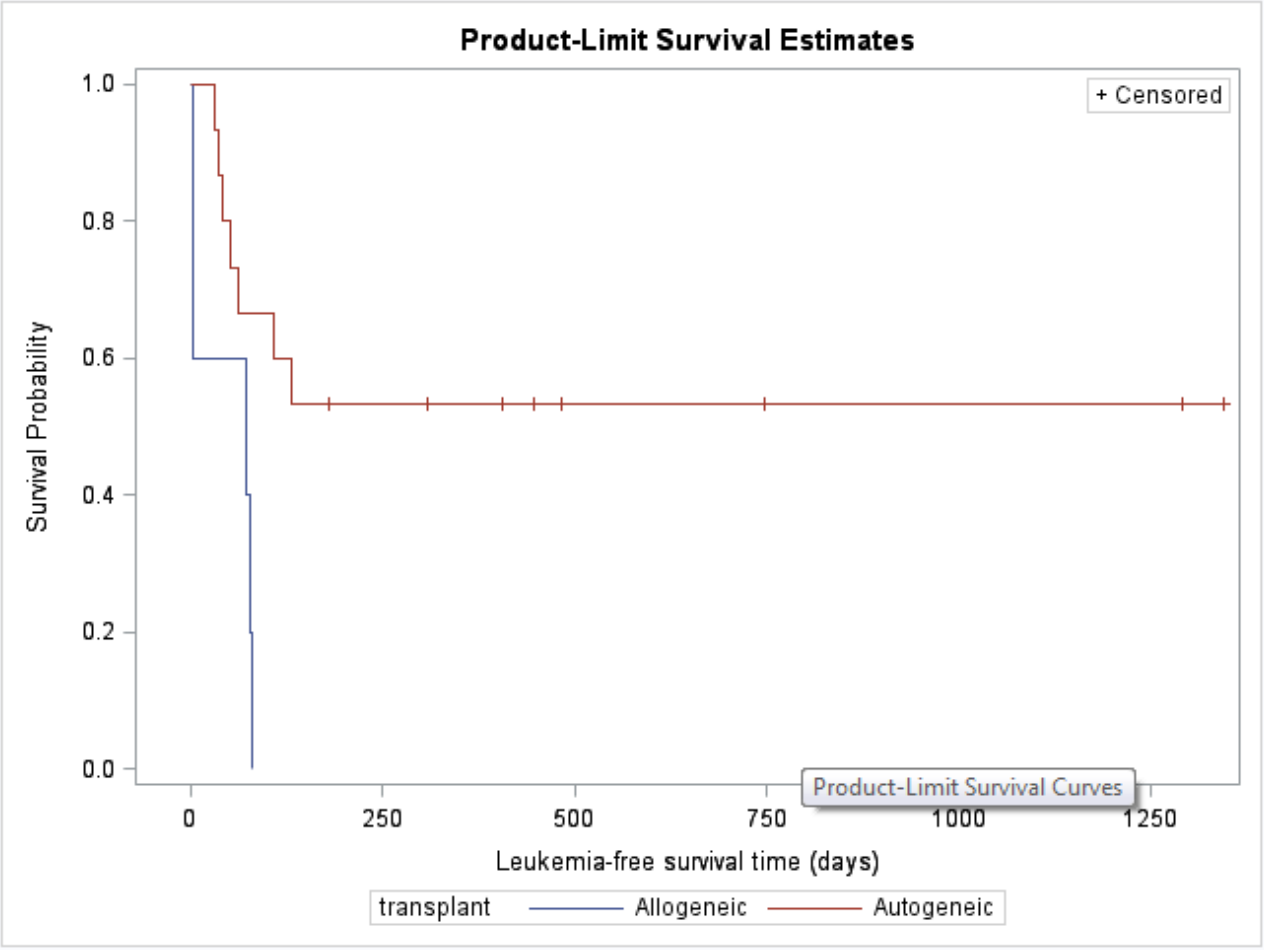
disease=Non-Hodgkins lymphoma



Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	1.6552	1	0.1983
Wilcoxon	0.6447	1	0.4220
-2Log(LR)	10.1967	1	0.0014

Example: Hodgkin's disease

disease=Hodkins disease



Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	6.3574	1	0.0117
Wilcoxon	5.1923	1	0.0227
-2Log(LR)	16.5414	1	<.0001

Program L4

- **Summary curves for competing risks**
  - Cumulative incidence function
- **Hypothesis testing**
  - K samples tests
    - Log-rank test
    - Gehan's test
    - Gray's test
  - Stratified tests
  - Differences at a fixed point in time

Differences in outcome at a fixed point in time

Sometimes you might want to compare K survival curves or cumulative incidence curves at a predetermined fixed point in time, t_0.

Important to select t_0 before examining the data.

Test statistics are special cases of tests for contrasts between a set of parameters.

Tests for differences in outcome at a fixed point in time

$$H_0 : S_1(t_0) = S_2(t_0) = \ldots = S_K(t_0)$$

H_a : At least one of the $S_j(t_0)$'s is different, for predetermined t_0

or

$$H_0 : CI_1(t_0) = CI_2(t_0) = \ldots = CI_K(t_0)$$

H_a : At least one of the $CI_j(t_0)$'s is different, for predetermined t_0

Tests for differences in outcome at a fixed point in time

Kaplan-Meier (Product-Limit) estimates of the survival function, or cumulative incidence estimates are used, together with the estimated variances of these statistics.

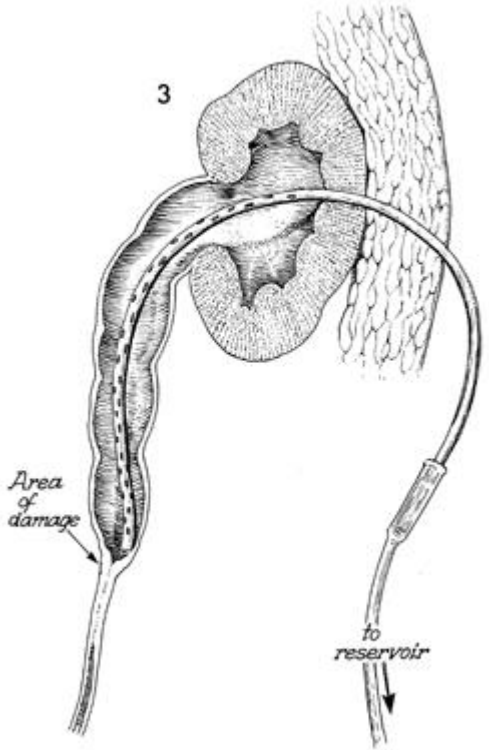
Test statistic when comparing the survival in two groups:

$$Z = \frac{\hat{S}_1(t_0) - \hat{S}_2(t_0)}{\sqrt{\hat{V}\left(\hat{S}_1(t_0)\right) + \hat{V}\left(\hat{S}_2(t_0)\right)}}$$

Example: Renal catheters

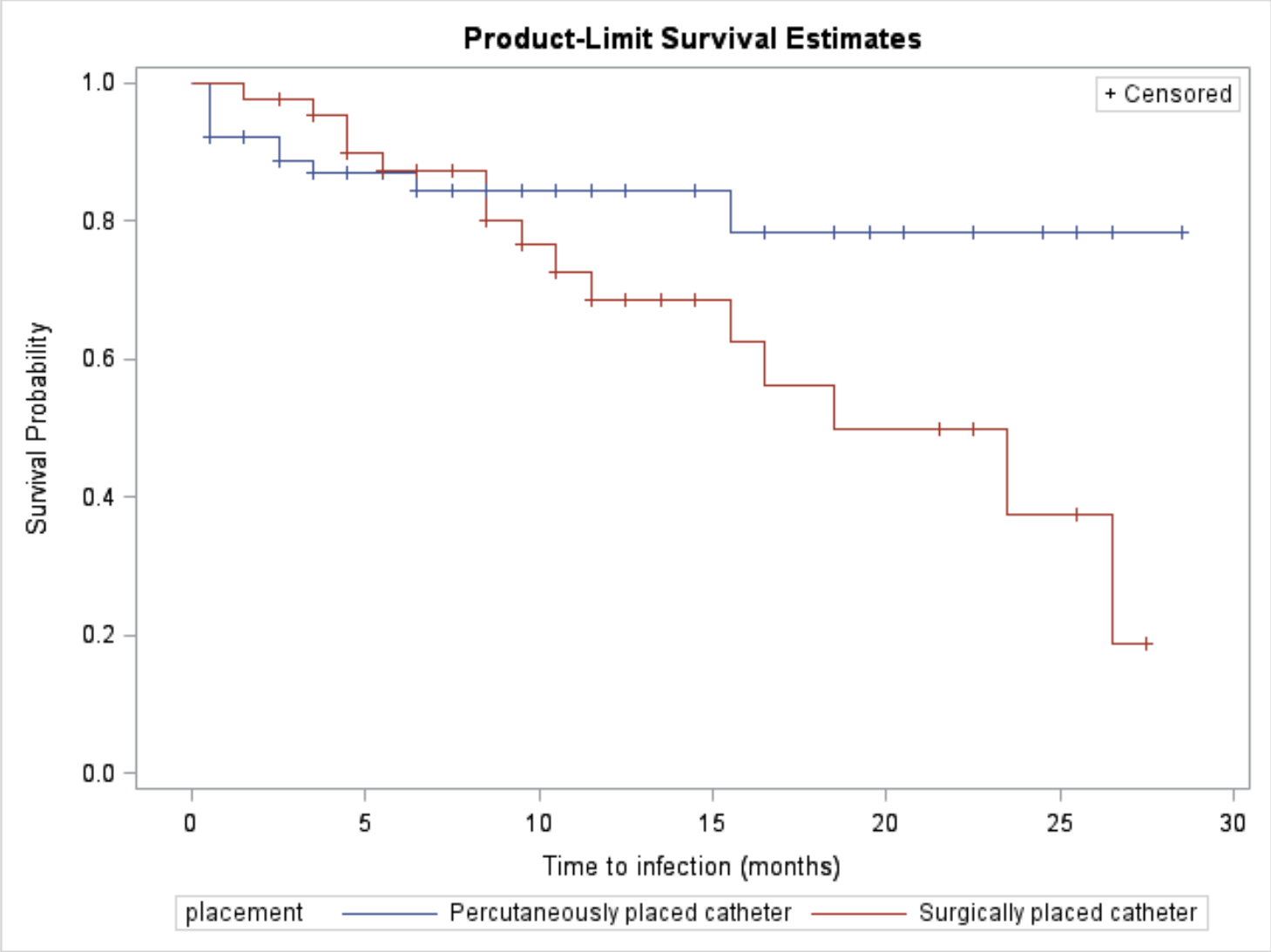
In section 1.4 the times to first exit-site infection (in months) of patients with renal insufficiency was reported.

In the study 43 patients had a surgically placed catheter (Group 1) and 76 patients had a percutaneous placement of their catheter (Group 0).



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Example: Renal catheters



Example: Renal catheters

To compare the survival functions at e.g. 3 months:

$$H_0 : S_1(3) = S_0(3)$$
$$H_a : S_1(3) \neq S_0(3)$$

Test statistic:

$$Z = \frac{\hat{S}_1(3) - \hat{S}_0(3)}{\sqrt{\hat{V}\left(\hat{S}_1(3)\right) + \hat{V}\left(\hat{S}_0(3)\right)}}$$

Example: Renal catheters

Stratum 1: placement = Percutaneously placed catheter

Product-Limit Survival Estimates

time		Survival	Failure	Survival Standard Error	Number Failed	Number Left
0.0000		1.0000	0	0	0	76
0.5000		.	.	.	1	75
0.6000					2	74
...						
2.5000		.	.	.	7	55
2.5000		0.8882	0.1118	0.0376	8	54
2.5000	*	.	.	.	8	53
2.5000	*	.	.	.	8	52
2.5000	*	.	.	.	8	51
2.5000	*	.	.	.	8	50
2.5000	*	.	.	.	8	49
3.5000		0.8700	0.1300	0.0409	9	48
3.5000	*	.	.	.	9	47

$$\hat{S}_0(3) = 0.8882$$

$$\hat{V}\left(\hat{S}_0(3)\right) = 0.0376^2$$

$$= 0.00141$$

Example: Renal catheters

Stratum 2: placement = Surgically placed catheter

Product-Limit Survival Estimates

time		Survival	Failure	Survival Standard Error	Number Failed	Number Left
0.0000		1.0000	0	0	0	43
1.5000		0.9767	0.0233	0.0230	1	42
2.5000	*	.	.	.	1	41
2.5000	*	.	.	.	1	40
3.5000		0.9523	0.0477	0.0329	2	39
3.5000	*				2	38

$$\hat{S}_1(3) = 0.9767$$

$$\hat{V}\left(\hat{S}_1(3)\right) = 0.0230^2$$

$$= 0.00053$$

Example: Renal catheters

To compare the survival functions at e.g. 3 months:

$$H_0 : S_1(3) = S_0(3)$$
$$H_a : S_1(3) \neq S_0(3)$$

Test statistic:

$$Z = \frac{\hat{S}_1(3) - \hat{S}_0(3)}{\sqrt{\hat{V}\left(\hat{S}_1(3)\right) + \hat{V}\left(\hat{S}_0(3)\right)}} = 2.01$$

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Analysis of Survival Data

Lecture 5:
Regression for survival data

$$h(t \mid \mathbf{Z}) = h_0(t)\exp(\boldsymbol{\beta}^t \mathbf{Z})$$

Cox, D.R. (1972), Regression Models and Life Tables, *Journal of the Royal Statistical Society,* B34

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{D} L_i = \prod_{i=1}^{D} \frac{\exp\left(\sum_{k=1}^{p} \beta_k Z_{(i)k}\right)}{\sum_{j \in R(t_i)} \exp\left(\sum_{k=1}^{p} \beta_k Z_{jk}\right)}$$

Partial likelihood for the *i*th event time

Risk set at time t_i

Inger Persson

Program L5

- **Regression for survival data**
  - Cox's proportional hazards regression
    - Partial maximum likelihood
    - Interpretation of estimated coefficients
    - Continuous vs categorical covariates
    - Ties
    - Local tests
    - Model building
    - Time-dependent covariates

Comparing two or more groups

When the groups to be compared are similar, except for the grouping variable, the K samples tests (Log-rank, Gehan's, Gray's test, etc.) can be used.

When there are other covariates that affect the event rates in the K different populations, stratified tests can be used. But stratified tests will not provide information about the size of the effect that the covariates might have on the outcome.

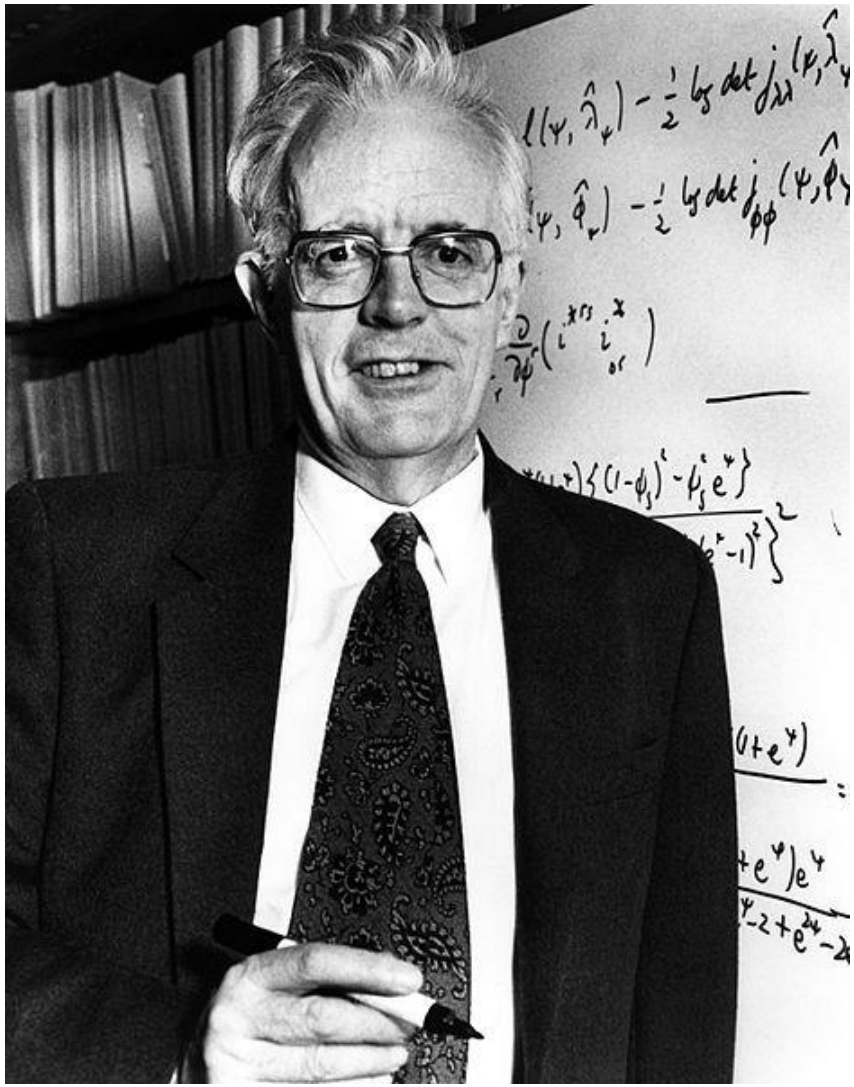
Alternative: regression

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Regression models

Continuous data	Linear regression
Dichotomous data	Logistic regression
No. of events	Poisson regression
Survival data	Cox regression

Sir David Cox



Cox, D.R. (1972), Regression Models and Life Tables, *Journal of the Royal Statistical Society,* B34

Still one of the most frequently cited journal articles in statistics and medicine.

Cox regression (a.k.a. proportional hazards regression, PH regression)

X = time to some event (e.g. survival time)

Data:

$(T_j, \delta_j, \mathbf{Z}_j(t))$ $\quad\quad$ $j = 1, \ldots, n$

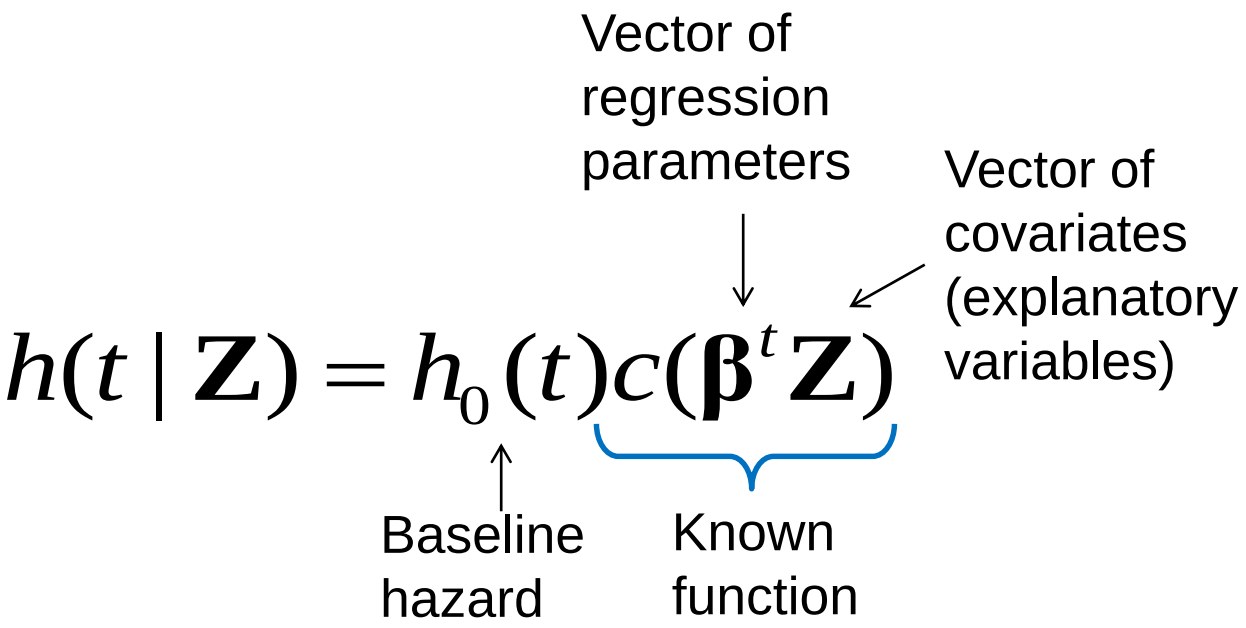
T = observed time (time to event or censoring)

δ = event indicator (1=event, 0=right censored obs.)

$\mathbf{Z}(t)$= vector of p covariates

We'll start focusing on fixed covariates (that do not depend on t)

Cox's proportional hazards model



The baseline hazard rate $h_0(t)$ is an unknown (arbitrary) function, giving the hazard function for the standard set of conditions $\mathbf{Z} = \mathbf{0}$.

A semi-parametric model

Making special assumptions about the baseline hazard $h_0(t)$ leads to parametric models, e.g. the exponential and Weibull distributions.

The advantage of Cox' model is the fact that such assumptions can be avoided.

Cox's approach is said to be **semi-parametric**.

Common choice of c

$$h(t \mid \mathbf{Z}) = h_0(t)c(\boldsymbol{\beta}^t \mathbf{Z})$$

$h(t \mid \mathbf{Z}) > 0$ ➡ c is chosen so it never can be negative

Common model for $c(\boldsymbol{\beta}^t \mathbf{Z})$: $\exp(\boldsymbol{\beta}^t \mathbf{Z})$

⬇

$$\boxed{h(t \mid \mathbf{Z}) = h_0(t)\exp(\boldsymbol{\beta}^t \mathbf{Z})}$$

Example: 3 explanatory variables

p = 3 covariates (explanatory variables)

$$\boldsymbol{\beta}^t = (\beta_1\ \beta_2\ \beta_3) \qquad \mathbf{Z}(t) = \begin{pmatrix} Z_1 \\ Z_2 \\ Z_3 \end{pmatrix}$$

$$\boldsymbol{\beta}^t\mathbf{Z} = \beta_1 Z_1 + \beta_2 Z_2 + \beta_3 Z_3 = \sum_{k=1}^{p} \beta_k Z_k$$

$$h(t \mid \mathbf{Z}) = h_0(t)\exp\left(\sum_{k=1}^{3} \beta_k Z_k\right)$$

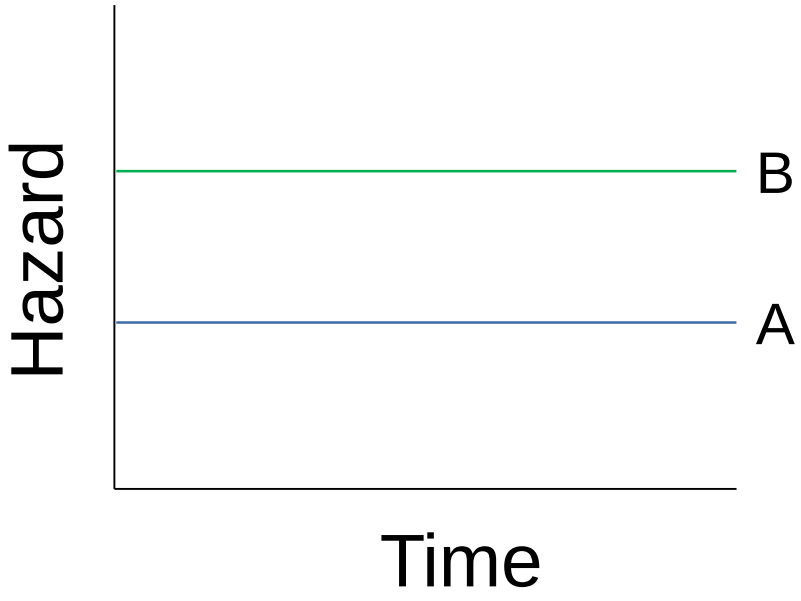
Comparing two individuals (proportional hazards)

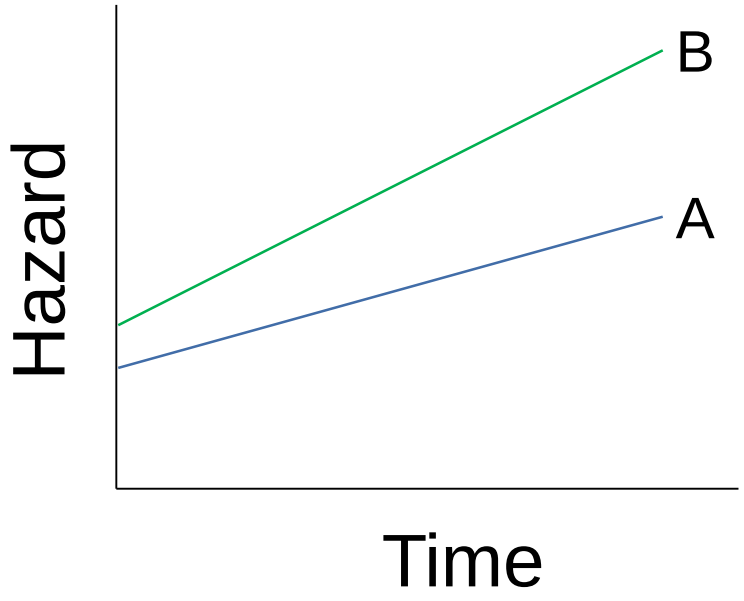
One way to compare two individuals with covariate values $\mathbf{Z}$ and $\mathbf{Z}^*$:

$$\frac{h(t \mid \mathbf{Z})}{h(t \mid \mathbf{Z}^*)} = \frac{\cancel{h_0(t)} \exp\left(\sum_{k=1}^{p} \beta_k Z_k\right)}{\cancel{h_0(t)} \exp\left(\sum_{k=1}^{p} \beta_k Z_k^*\right)} = \exp\left(\sum_{k=1}^{p} \beta_k (Z_k - Z_k^*)\right)$$

The ratio between two hazards is constant, independent of t. This means that the hazard rates are **proportional**.

Proportional hazards





The ratio between the two hazards is constant.

Relative risk

$$\exp\left(\sum_{k=1}^{p}\beta_k(Z_k - Z_k^*)\right)$$ is called the **relative risk** (hazard ratio)

Describes the risk of an individual with covariates $\mathbf{Z}$ experiencing the event, compared to an individual with covariates $\mathbf{Z}^*$.

Example: Relative risk for women compared to men

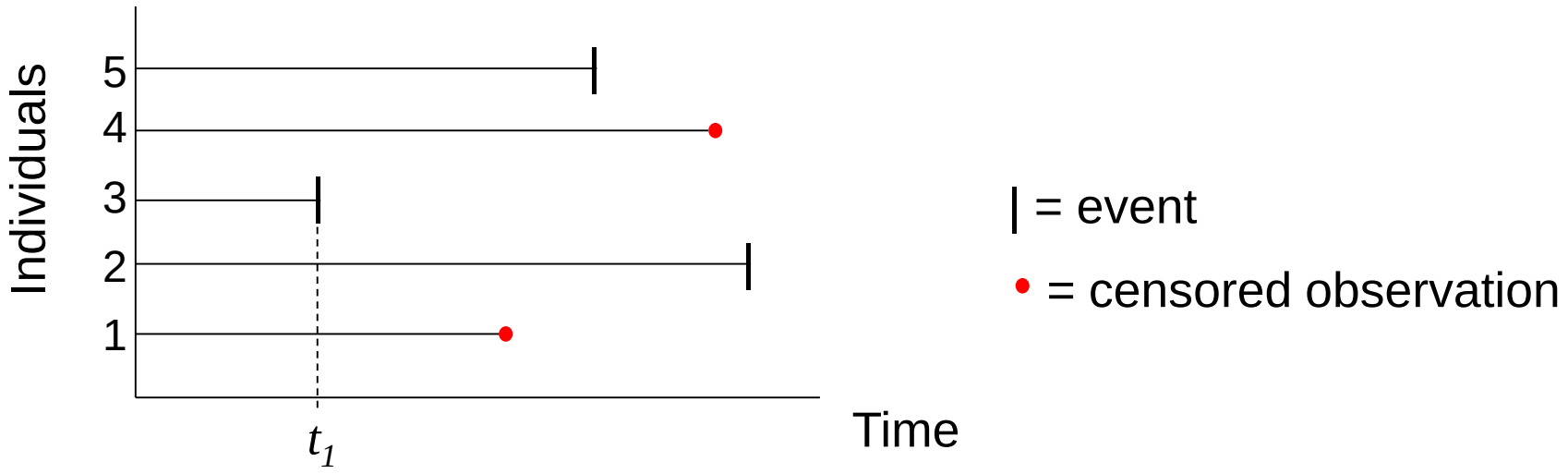
$$Z_1 = \begin{cases} 0 & \text{if man} \\ 1 & \text{if woman} \end{cases}$$

If all other covariates have the same value, the hazard ratio

$$\frac{h(t \mid \mathbf{Z})}{h(t \mid \mathbf{Z}^*)} = \exp(\beta_1)$$

describes the risk of experiencing the event for women compared to the risk for men

Example: estimating the parameters β



Given that a person experiences the event at t_1, what is the probability that it is individual no. 3?

We want a model that gives a large probability that it is individual no. 3.

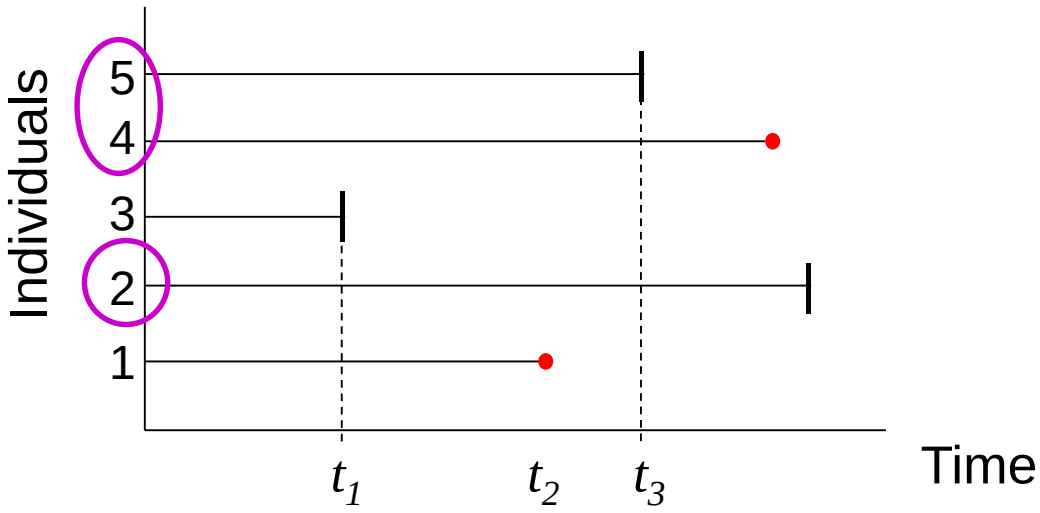
Example: estimating the parameters β

h_3 = the probability that individual no. 3 experiences the event at t_1, given that he/she has not experienced the event before that.

$$L_1 = \frac{h_3}{h_1 + h_2 + h_3 + h_4 + h_5} \leftarrow \text{The risk set}$$

= probability (likelihood) that no. 3 experiences the event, compared to all the individuals at risk

Example: estimating the parameters β



There is a term in the likelihood for each event, not for each individual.

Next event is at time t_3.

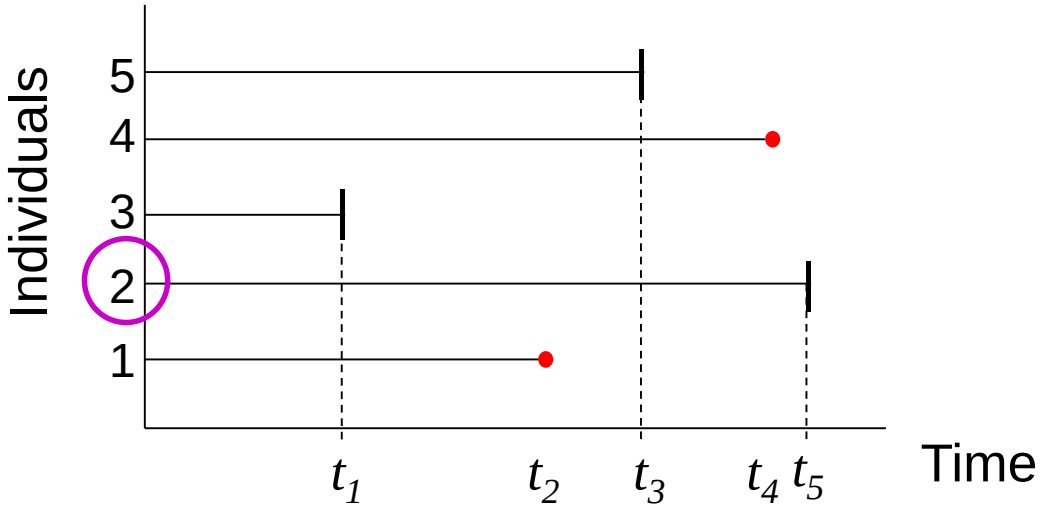
At that time there are three individuals still at risk.

Example: estimating the parameters β

$$L_2 = \frac{h_5}{h_2 + h_4 + h_5}$$

= probability (likelihood) that individual no. 5 experiences the event, compared to all the individuals still at risk

Example: estimating the parameters β



Next event is at time t_5.

At that time there is only one individual still at risk.

Example: estimating the parameters β

$$L_3 = \frac{h_2}{h_2} = 1$$

= probability (likelihood) that individual no. 2 experiences the event, compared to all the individuals still at risk

The “partial likelihood”

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{D} L_i = \prod_{i=1}^{D} \frac{\exp\left(\sum_{k=1}^{p} \beta_k Z_{(i)k}\right)}{\sum_{j \in R(t_i)} \exp\left(\sum_{k=1}^{p} \beta_k Z_{jk}\right)}$$

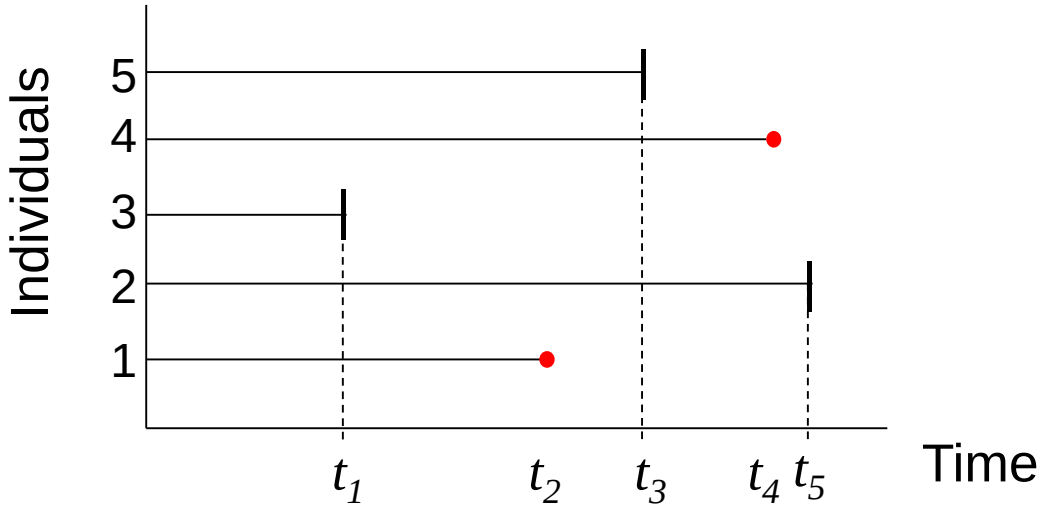
Partial likelihood for the *i*th event time

Risk set at time t_i

Inference for β is based on this **partial likelihood**.

This partial likelihood method estimates β only, not the hazard.

Example: partial maximum likelihood



Suppose $p = 1$

$$Z = \begin{cases} 0 & \text{if man} \\ 1 & \text{if woman} \end{cases}$$

Suppose individuals no. 1, 3, and 5 are women.

Example: partial maximum likelihood

First event time, individual 3: $h(t_1|Z) = h_0(t_1)e^{\beta Z_3} = h_0(t_1)e^{\beta}$

$$L_1 = \frac{h_3}{h_1 + h_2 + h_3 + h_4 + h_5} = \frac{e^{\beta}}{e^{\beta} + 1 + e^{\beta} + 1 + e^{\beta}} = \frac{e^{\beta}}{2 + 3e^{\beta}}$$

Second event time (t_3), individual 5:

$$L_2 = \frac{h_5}{h_2 + h_4 + h_5} = \frac{e^{\beta}}{1 + 1 + e^{\beta}} = \frac{e^{\beta}}{2 + e^{\beta}}$$

Estimating the parameters β: (partial) maximum likelihood

Estimates of β are found by maximizing the partial likelihood

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{D} \frac{\exp\left(\sum_{k=1}^{p} \beta_k Z_{(i)k}\right)}{\sum_{j \in R(t_i)} \exp\left(\sum_{k=1}^{p} \beta_k Z_{jk}\right)}$$

Even though this is not a likelihood in the traditional sense, it is treated as one, and inference is carried out by usual means.

Estimating the parameters β: (partial) maximum likelihood

The maximization of the partial likelihood (solving for the most likely values of β) cannot be done analytically, numerical methods must be employed.

The maximization can be done using a Newton-Raphson technique (or some other iterative method for optimization).

Estimating the parameters β: (partial) maximum likelihood

The partial likelihood does not depend upon the baseline hazard rate $h_0(t)$, which means that inference on the effects of explanatory variables (covariates) can be made without any knowledge about the baseline hazard.

In these analyses, the baseline hazard $h_0(t)$ is treated as a nuisance parameter function.

Characteristics of Cox proportional hazards regression

- Does not require that you choose some particular probability model to represent survival times, and is therefore more robust than parametric methods
- Semi-parametric (parametric assumptions can be avoided)
- Can accommodate both discrete and continuous measures of event times

Characteristics of Cox proportional hazards regression, cont'd

- Easy to incorporate time-dependent covariates (covariates that may change in value over the course of the observation period)
- Cox regression models the effect of covariates on the hazard rate but leaves the baseline hazard rate unspecified
- Estimates relative rather than absolute risk

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Assumptions

Assumptions of the Cox model:

- random sample(s) (for inference)
- independent observations
- noninformative censoring
- right censored or left truncated data
- large sample (common rule of thumb: ≥ 10 events/cov.)*
- proportional hazards

**NOTE: This is a recommendation, not a strict rule. See e.g.*

- *Peduzzi et al., Importance of Events Per Independent Variable in Proportional Hazards regression Analysis, J Clin Epidemiol, 1995; Vol 48, No. 12*
- *Vittinghoff & McCulloch, Relaxing the Rule of Ten Events per Variable in Logistic and Cox Regression, Am J Epidemiol, 2006; Vol. 165, No. 6*

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Proportional hazards assumption

The Cox regression model assumes that the hazard rates are proportional.

This assumption can be checked by a number of tests and graphical methods (we'll learn more later on).

Program L5

- **Regression for survival data**
  - Cox's proportional hazards regression
    - Partial maximum likelihood
    - Interpretation of estimated coefficients
    - Continuous vs categorical covariates
    - Ties
    - Local tests
    - Model building
    - Time-dependent covariates

Example: Hodgkin's disease

A small study comparing the effectiveness of allogeneic transplants versus autogeneic transplants for Hodgkin's disease or non-Hodgkin's lymphoma is presented in section 1.10.

Allogeneic transplant: from a matching donor

Autogeneic transplant: your own bone marrow is cleansed and returned after a high dose of chemotherapy

Is there a difference in disease-free survival between allogeneic and autogeneic transplants?

Example: Hodgkin's disease

Variables:

freetime = time to death or relapse (days)
transplant = type of transplant (0=allogeneic, 1=autogeneic)
event = event indicator (1=dead or relapse,
0=alive without relapse)
disease = disease type (0=non-Hodgkin's lymphoma,
1=Hodgkin's disease)
karnofsky = pretransplant Karnofsky score, 0-100
(higher score = less functional impairment)
waitingtime = waiting time from diagnosis to transplant
(months)

Example: Hodgkin's disease

To estimate the hazard ratios using the Cox proportional hazards model, use the **phreg** procedure.

```
proc phreg data=hodgkins;
  model freetime*event(0)=transplant disease karnofsky waitingtime;
run;
```

Time to event variable

Event/censoring variable (with censoring value)

Explanatory variables (covariates)

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Example: Hodgkin's disease

The PHREG Procedure

Model Information		
Data Set	WORK.HODGKINS	
Dependent Variable	freetime	Leukemia-free survival time (months)
Censoring Variable	event	
Censoring Value(s)	0	
Ties Handling	BRESLOW	

Number of Observations Read	43
Number of Observations Used	43

Summary of the Number of Event and Censored Values			
Total	Event	Censored	Percent Censored
43	26	17	39.53

Example: Hodgkin's disease

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	174.595	148.071
AIC	174.595	156.071
SBC	174.595	161.104

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	26.5239	4	<.0001
Score	31.2621	4	<.0001
Wald	24.3090	4	<.0001

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Convergence criterion

The iterative process of maximizing the (partial) likelihood is declared converged when the relative change in log likelihoods between successive steps is less than 0.0001.

Global tests

Three common tests of $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ (e.g. $H_0 : \boldsymbol{\beta} = 0$)

1) **Likelihood ratio test**, based on the likelihood ratio (how many times more likely the data are under the model with compared to without covariates)
2) **Wald's test**, based on asymptotic normality of the (partial) maximum likelihood estimates
3) **Scores test**, based on efficient scores (same scores used when finding the partial maximum likelihood estimates), asymptotically *p*-variate normal.

Global tests

The scores test is identical to the log-rank test if there are no ties between event times.

The Wald and Scores tests are both approximations of (and asymptotically equivalent to) the Likelihood ratio test.

With regards to size (α) and power, Li et al (1996) showed that the likelihood ratio test outperforms the Wald test especially for small samples. The scores test is not recommended, it tends to inflate the size of the test.

All three tests assume that the hazard rates are proportional.

Example: Hodgkin's disease

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
transplant	1	-0.24344	0.44299	0.3020	0.5826	0.784
disease	1	0.99262	0.52319	3.5995	0.0578	2.698
Karnofsky	1	-0.05555	0.01215	20.9069	<.0001	0.946
waitingtime	1	-0.00792	0.00790	1.0066	0.3157	0.992

Hazard ratio = 0.784

Interpretation:

The risk of dying or relapsing for autogeneic transplanted patients is 78.4% of the same risk for allogeneic transplanted, on average.

Hazard ratio = 0.946

Interpretation:

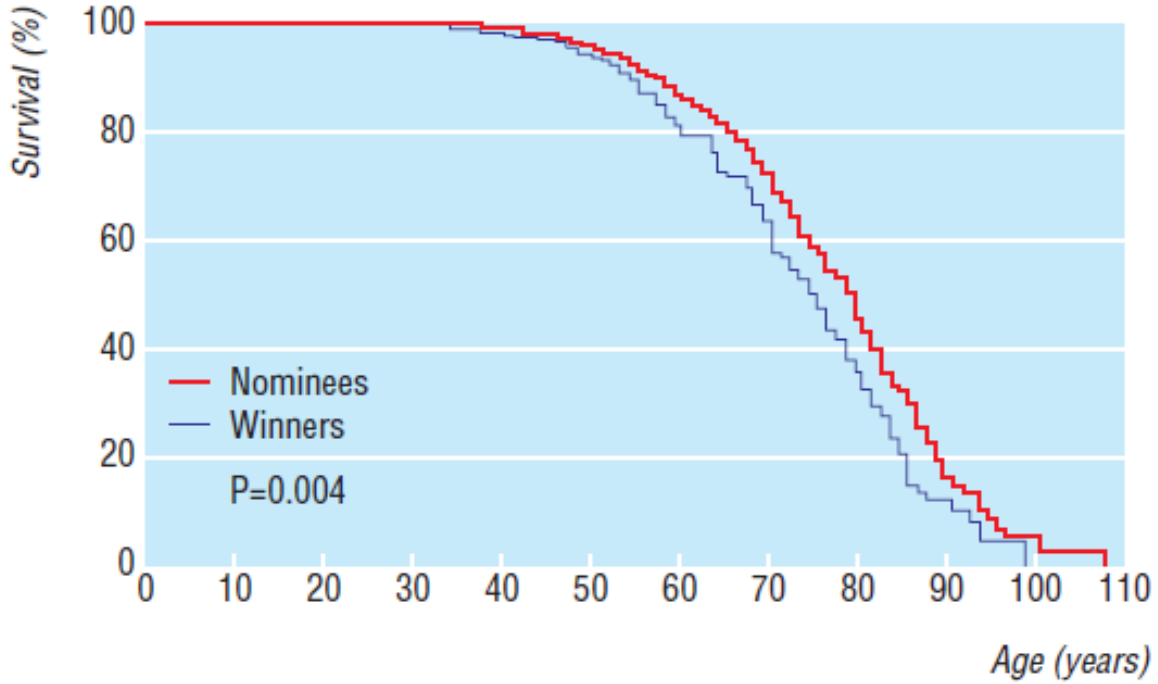
The risk of dying or relapsing decreases by 5.4% with each one-unit increase in the Karnofsky score, on average. Equivalently, the risk decreases by 24.2% with each 5-unit increase of Karnofsky score, on average ($0.946^5 = 0.758$).

Hazard ratio = 2.698

Interpretation:

The risk of dying or relapsing is 2.7 times higher for patients with Hodgkin's disease than for patients with Non-Hodkin's lymphoma, on average.

Example: Study of longevity (life length) in academy award winning screenwriters



Survival of winners and nominees of academy awards for screenwriting. The graph shows the percentage of each group alive, plotted by using the Kaplan-Meier technique. Primary statistical analysis is based on a log rank test comparing winners to nominees (n=185, deaths=112 and n=610, deaths=316, respectively)

p. 1494

Ref: D.A. Redelmeier, S.M. Singh (2001). *Longevity of screenwriters who win an academy award: longitudinal study.* BMJ 2001; 323: 1491-6

Table 2 Death rates for screenwriters who have won an academy award.* Values are percentages (95% confidence intervals) and are adjusted for the factor indicated

Factor	Relative increase in death rate for winners
Basic analysis	37 (10 to 70)
Adjusted analysis	
Demographic:	
Year of birth	32 (6 to 64)
Sex	36 (10 to 69)
Documented education	39 (12 to 73)
All three factors	33 (7 to 65)
Professional:	
Film genre	37 (10 to 70)
Total films	39 (12 to 73)
Total four star films	40 (13 to 75)
Total nominations	43 (14 to 79)
Age at first film	36 (9 to 68)
Age at first nomination	32 (6 to 64)
All six factors	40 (11 to 76)
All nine factors	35 (7 to 70)

*Results from Cox regression model with hazard ratios reported as relative increases.

Hazard ratio = 1.37

Interpretation:

The risk of dying is on average 37% higher for winners compared to nominees

Hazard ratio = 1.43

Interpretation:

The risk of dying is on average 43% higher for every extra nomination

Hazard ratio confidence intervals

Confidence intervals can of course be calculated around hazard ratios estimated by Cox proportional hazards regression.

Two methods available:

1) **Wald** (standard confidence intervals). May work poorly for maximum-likelihood estimation.

2) **Profile-likelihood**, inverts a likelihood-ratio test to obtain a CI for the parameter in question. Applicable to all likelihood-based statistical analyses.

Example: Hodgkin's disease

Use proc phreg and the **risklimit** option.

```
proc phreg data=hodgkins;
  model freetime*event(0)=transplant disease karnofsky
            waitingtime/risklimit=pl;
run;
```

Produces confidence intervals for hazard ratios

Profile-likelihood intervals

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Example: Hodgkin's disease

Analysis of Maximum Likelihood Estimates								
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Profile Likelihood Confidence Limits	
transplant	1	-0.23317	0.44299	0.2771	0.5986	0.792	0.332	1.928
disease	1	0.98058	0.52264	3.5201	0.0606	2.666	0.948	7.495
Karnofsky	1	-0.05584	0.01216	21.1007	<.0001	0.946	0.922	0.968
waitingtime	1	-0.00786	0.00788	0.9936	0.3189	0.992	0.975	1.006

Program L5

- **Regression for survival data**
  - Cox's proportional hazards regression
    - Partial maximum likelihood
    - Interpretation of estimated coefficients
    - Continuous vs categorical covariates
    - Ties
    - Local tests
    - Model building
    - Time-dependent covariates

Continuous vs. categorical variables

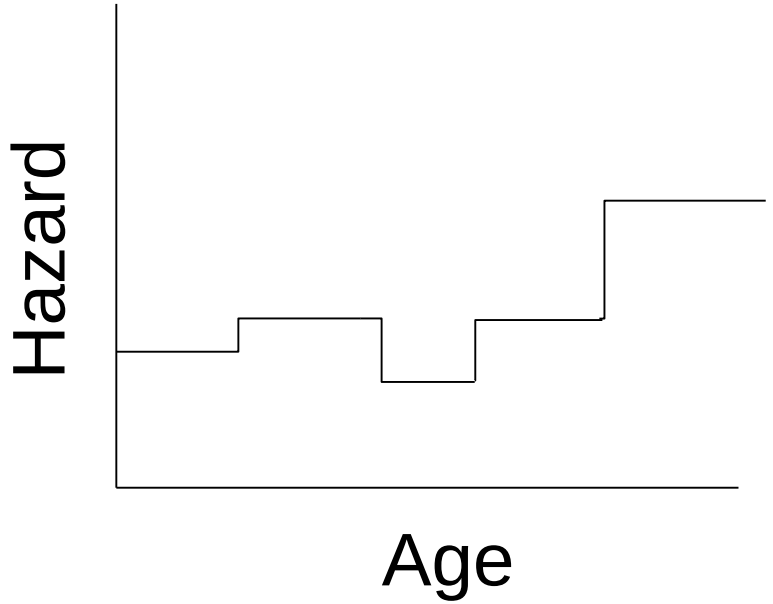
Continuous variables can be used as they are in the regression model.

The regression estimate of β will then describe the effect of a one-unit increase of the explanatory variable.

If the hazard does not increase/decrease continuously it might be better to categorize the covariate.

Example: Age covariate





Continuously increasing hazard

Not continuously increasing hazard

Dummy variables and interaction effects

Dummy variables (for categorical variables) and interaction variables can be used with Cox's proportional hazards model, just as for linear models.

Categorization of covariates

There are different ways of categorizing a continuous variable.

One way is to divide into groups of equal size (the same number of observations in each group).

The optimal strategy is to determine cut points based on scientific reasoning.

Avoid dichotomization of covariates

Avoid dichotomization (using only two categories)!

Dichotomizing is a way of effectively losing a great deal of information, with a serious loss of power to detect real relationships.

Dichotomizing may also increase the probability of false positive results.

Further reading: P. Royston, D.G. Altman, W. Sauerbrei (2006). Dichotomizing continuous preditors in multiple regression: a bad idea. *Statistics in medicine*, 2006; 25: 127-141.

Transformation of variables

Keep the continuous variable continuous, if possible, to avoid loss of information.

Variables can be transformed (using logarithms, squares, etc) to find a stronger relationship between explanatory and dependent variables.

Ties

Events occur at D times, $t_1 < t_2 < \cdots < t_D$

At time t_i there are d_i events (there can be ties between event times).

Common methods of constructing the partial likelihood when ties are present:

1) Exact
2) Breslow (approximation)
3) Efron (approximation)
4) Cox's discrete

“Exact” method of handling ties

Time is treated as a continuous variable, and ties are assumed being a result of imprecise measurements of time.

Assumes there is a true unknown order of events in time.

Calculates the exact probability of all possible orderings of events.

Complex computations, but usually no noticeable extra computer time.

Kalbfleisch, J. D. and Prentice, R. L. (1980), *The Statistical Analysis of Failure Time Data*, New York: John Wiley & Sons

Breslow's method of handling ties

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{D} \frac{\exp\left(\boldsymbol{\beta}^t \mathbf{s}_i\right)}{\left(\sum_{j \in R_i} \exp\left(\boldsymbol{\beta}^t \mathbf{Z}_j\right)\right)^{d_i}}$$

Set of all individuals at risk just prior to time t_i

$$\mathbf{s}_i = \sum_{j \in D_i} \mathbf{Z}_j$$

Set of all individuals who experience the event at time t_i

Works well when there are few ties.

The default method in SAS.

Efron's method of handling ties

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{D} \frac{\exp\left(\boldsymbol{\beta}^t \mathbf{s}_i\right)}{\prod_{j=1}^{d_i}\left(\sum_{k\in R_i}\exp\left(\boldsymbol{\beta}^t \mathbf{Z}_k\right) - \frac{j-1}{d_i}\sum_{k\in \mathrm{D}_i}\exp\left(\boldsymbol{\beta}^t \mathbf{Z}_k\right)\right)}$$

Set of all individuals at risk just prior to time t_i

$$\mathbf{s}_i = \sum_{j\in \mathrm{D}_i} \mathbf{Z}_j$$

Set of all individuals who experience the event at time t_i

Closer to the correct partial likelihood based on a discrete hazard model than Breslow's likelihood.

Similar to Breslow's likelihood when there are few ties.

Cox's discrete method of handling ties

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{D} \frac{\exp\left(\boldsymbol{\beta}^t \mathbf{s}_i\right)}{\sum_{q \in Q_i} \exp\left(\boldsymbol{\beta}^t \mathbf{s}_q^*\right)}$$

Set of all subsets of d_i individuals who could be selected from the risk set R_i

$$\mathbf{s}_i = \sum_{j \in \mathrm{D}_i} \mathbf{Z}_j$$

Set of all individuals who experience the event at time t_i

$$\mathbf{s}_q^* = \sum_{j=1}^{d_j} \mathbf{Z}_{qj}$$

Assumes time is discrete (not very common in reality).

Gives exact estimates, no approximations are used.

A logistic model is assumed for the hazard rate, models proportional odds (not hazards).

Choice of ties handling method

When there are no ties, all methods give exactly the same results.

When there are few ties, the choice of method has a very small impact on the results.

When there are many ties, the Breslow and Efron approximations give poor results (coefficients are biased towards 0)

Choice of ties handling method, cont'd

Base the choice of method on substantive grounds – are the tied events truly tied, or are they a result of imprecise measurement?

Prefer discrete or exact method over Breslow and Efron approximations.

If you have to use approximations: Prefer Efron's method over Breslow's.

Example: Hodgkin's disease

Variables:

freetime = time to death or relapse (days)
transplant = type of transplant (0=allogeneic, 1=autogeneic)
event = event indicator (1=dead or relapse,
0=alive without relapse)
disease = disease type (0=non-Hodgkin's lymphoma,
1=Hodgkin's disease)
karnofsky = pretransplant Karnofsky score, 0-100
(higher score = less functional impairment)
waitingtime = waiting time from diagnosis to transplant
(months)

Example: Hodgkin's disease

To choose the tie handling method, use the **ties** option.

```
proc phreg data=hodgkins;
  model freetime*event(0)=transplant disease karnofsky waitingtime
   /ties=exact;
run;
```

Choice of method

```
/ties=exact;

/ties=discrete;

/ties=efron;
```

Example: Hodgkin's disease

Breslow

Parameter	Parameter Estimate	Standard Error	Hazard Ratio
transplant	-0.24344	0.44299	0.784
disease	0.99262	0.52319	2.698
Karnofsky	-0.05555	0.01215	0.946
waitingtime	-0.00792	0.00790	0.992

Efron

Parameter	Parameter Estimate	Standard Error	Hazard Ratio
transplant	-0.23317	0.44299	0.785
disease	0.98058	0.52264	2.719
Karnofsky	-0.05584	0.01216	0.946
waitingtime	-0.00786	0.00788	0.992

Discrete (Cox)

Parameter	Parameter Estimate	Standard Error	Hazard Ratio
transplant	-0.24187	0.44396	0.785
disease	1.00025	0.52674	2.719
Karnofsky	-0.05581	0.01218	0.946
waitingtime	-0.00801	0.00792	0.992

Exact

Parameter	Parameter Estimate	Standard Error	Hazard Ratio
transplant	-0.23317	0.44299	0.792
disease	0.98058	0.52264	2.666
Karnofsky	-0.05584	0.01216	0.946
waitingtime	-0.00786	0.00788	0.992

The FREQ Procedure

Leukemia-free survival time (months)				
freetime	Frequency	Percent	Cumulative Frequency	Cumulative Percent
2	1	2.33	1	2.33
4	1	2.33	2	4.65
28	1	2.33	3	6.98
30	1	2.33	4	9.30
32	1	2.33	5	11.63
36	1	2.33	6	13.95
...				
63	1	2.33	14	32.56
72	1	2.33	15	34.88
77	1	2.33	16	37.21
79	1	2.33	17	39.53
81	2	4.65	19	44.19
84	1	2.33	20	46.51
108	1	2.33	21	48.84
133	1	2.33	22	51.16

Program L5

- **Regression for survival data**
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Local tests

Local hypotheses can be tested, e.g.

$$H_0 : \beta_1 = 0 \, , \beta_2 = 0, \text{ etc.}$$

$$H_0 : \beta_1 = \beta_3$$

$$H_0 : \beta_1 = \beta_3 = \beta_5$$

The Wald test is used in SAS (OK for large samples).

The Likelihood ratio test (and the scores test) can be calculated (SAS code in document *SAS examples.pdf*)

Example: Hodgkin's disease

To test the local hypothesis that the effect of type of transplant is the same as the effect of disease type:

$$H_0 : \beta_{transplant} = \beta_{disease}$$

Use proc phreg and the **test** statement.

```
proc phreg data=hodgkins;
   model freetime*event(0)=transplant disease
                karnofsky waitingtime/ties=exact;
   test transplant=disease;
run;
```

Example: Hodgkin's disease

Linear Hypotheses Testing Results				
	Label	Wald Chi-Square	DF	Pr > ChiSq
	Test 1	2.3073	1	0.1288

Relative risks that don't appear directly in the result table

You might be interested in relative risks not directly presented in the regression results.

Example:
three treatments are being compared; A, B, and C.

Treatment A is being used as the reference category, and dummy variables are created for treatments B and C.

Relative risks that don't appear directly in the result table

You are interested in the risk of experiencing the event for patients receiving treatment B relative to the risk of experiencing the event for patients receiving treatment C. This will not appear directly with the regression results.

$$\frac{Risk_B}{Risk_C} = \frac{e^{\beta_B}}{e^{\beta_C}} = e^{(\beta_B - \beta_C)}$$

Relative risks that don't appear directly in the result table

To find the confidence interval for this relative risk you need the standard error of b_B-b_C.

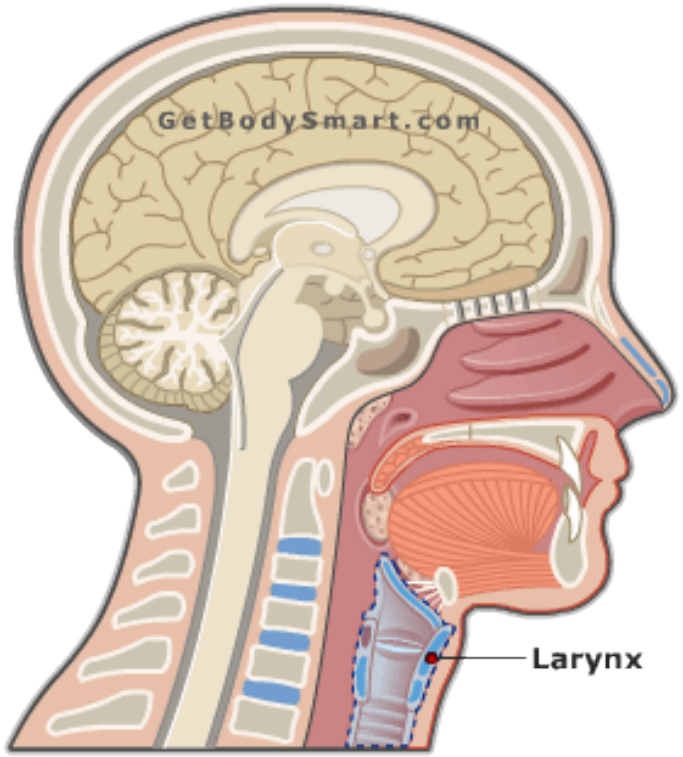
The variance-covariance matrix of the b_i's can be estimated in SAS by using the **covb** option.

A confidence interval for $(\beta_B - \beta_C)$ can be constructed by using $V(b_B - b_C) = V(b_B) + V(b_C) - 2\,Cov(b_B, b_C)$

And a confidence interval for the relative risk $e^{(\beta_B - \beta_C)}$ is then obtained by exponentiating the lower and upper limits.

Example: larynx cancer

A study of 90 males with cancer of the larynx is described in section 1.8.



Example: larynx cancer

X = survival time from first treatment (years)

There are four stages of the disease, stage I – stage IV, ordered from least serious to most serious.

```
proc phreg data=larynx;
  model time*death(0)= age stage2-stage4 /ties=exact covb;
run;
```

Dummy variables for stages II, III and IV

Produces the variance-covariance matrix

Example: larynx cancer

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
age	1	0.01903	0.01426	1.7815	0.1820	1.019
stage2	1	0.13992	0.46254	0.0915	0.7623	1.150
stage3	1	0.64232	0.35618	3.2521	0.0713	1.901
stage4	1	1.70693	0.42201	16.3600	<.0001	5.512

Estimated Covariance Matrix				
Parameter	age	stage2	stage3	stage4
age	0.0002033325	0.0008248711	0.0003274704	-.0003914790
stage2	0.0008248711	0.2139472417	0.0683891516	0.0689194854
stage3	0.0003274704	0.0683891516	0.1268657657	0.0680920074
stage4	-.0003914790	0.0689194854	0.0680920074	0.1780937953

$$\frac{Risk_{IV}}{Risk_{III}} = \frac{5.512}{1.901}$$

$$= 2.9$$

Example: larynx cancer

95% CI for (β_{IV}- β_{III}) :

b_{IV}-b_{III} ± 1.96 sqrt(V(b_{IV}-b_{III}))

V(b_{IV}-b_{III}) = V(b_{IV}) + V(b_{III}) – 2 Cov(b_{IV}, b_{III}) =

= 0.17809 + 0.12687 – 2 × 0.06809 =

= 0.16878

1.70693 - 0.64232 ± 1.96 sqrt(0.16878)

1.06461 ± 0.80522

[0.2594; 1.8698]

Example: larynx cancer

95% CI for the relative risk $e^{(\beta_{IV}-\beta_{III})}$:

[exp(0.2594); exp(1.8698)]

[1.296; 6.487]

There is a significant difference in the risk of dying for stage IV compared to stage III (the relative risk is significantly different from 1)

Program L5

- **Regression for survival data**
  - Cox's proportional hazards regression
    - Partial maximum likelihood
    - Interpretation of estimated coefficients
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    - Model building
    - Time-dependent covariates

Model building

Different ways of building a model:

1) Hypothesis based modelling. The time distribution is predicted by explanatory variables selected to fit a specific hypothesis.
2) Predicting the distribution of time by selecting from a number of explanatory variables with no particular prior hypothesis in mind.

Hypothesis based modelling

When a particular hypothesis is in mind, the explanatory variables fitting that hypothesis are included in the model.

Other explanatory variables can also be added, variables that can be seen as adjusters or confounders (variables that might affect the relationship between the hypothesis based variables and the outcome).

E.g. demographic variables (age, sex, etc.) can be confounders, or the severity of a patient's illness, the size of a tumour, etc.

Hypothesis based modelling – forward selection approach

1) Fit the model including only the explanatory variables fitting the hypothesis.
2) Add one of the possible confounders to the model, to analyze the relationship between that variable and survival (adjusting for all hypothesis based variables).
3) Repeat 2) for each possible confounder, one at a time.
4) Include the confounder with the strongest significant relationship to survival to the model.
5) Repeat steps 2-4, with the “basic” model now including the confounder added in step 4). Stop when no more significant confounders are found.

P-value and information criteria approaches

Different ways of deciding which variable is most related to survival:

1) **p-value approach**: choose the variable with the lowest *p*-value.

2) **Information criterion approach**: choose the variable which yields the model with the lowest information criteria value. Information criteria are based on the likelihood function, their value increase when added variables are unnecessary.

Can be combined: add significant variables with not increasing information criteria values.

Common information criteria

Akaike information criterion (AIC)

$$AIC = -2\log L + 2k$$

k = no. of parameters in model

Bayesian information criterion (BIC)

$$BIC = -2\log L + k\log n$$

Akaike information criterion corrected (AIC_C)

$$AIC_C = -2\log L + 2k + \frac{2k(k+1)}{n-k-1}$$

$$AIC_C = AIC + 2k(k+1)/(n-k-1)$$

Which to choose?

AIC/AIC_C preferred over BIC, shown by Burnham and Anderson*.

They also recommend AIC_C over AIC, especially for small *n* or large *k*.

*Burnham, K. P.; Anderson, D. R. (2002), Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach (2nd ed.), Springer-Verlag

Example: infection in burns

In section 1.6 a study is described to evaluate a protocol change in disinfectant practice in a large midwestern university medical center.

Control of infection is the primary concern for the 154 patients entered into the burn unit with varying degrees of burns.

The outcome variable is the time until infection from admission to the unit. Censoring variables are discharge from the hospital without an infection or death without an infection.

84 patients were in a group which had a body-cleansing method (disinfectant: chlorhexidine) and 70 patients received the routine bathing care method (disinfectant: povidone-iodine).

Example: infection in burns

Variables:

Trt = treatment (0=routine bathing, 1=body cleansing)

TimeStaph = Time to staphylococcus infection (days)

Staph = Staphylococcus indicator (1=infection, 0=no inf.)

Possible confounders*:*

Area (percentage burned, % of total surface area)

BurnSite (head, buttock, trunk, etc. – 6 indicators)

BurnType (1=chemical, 2=scald, 3=electric, 4=flame)

Example: infection in burns

Akaike information criterion
Decreases as variables are added to the model. If it increases, the added variable is unnecessary.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	390.415	386.686
AIC	390.415	388.686
SBC	390.415	390.557

$$AIC_C = AIC + 2k(k+1)/(n-k-1) = 388.686+2\cdot1\cdot2/(154-1-1) = 388.7123$$

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
Trt	1	-0.56139	0.29336	3.6621	0.0557	0.570

Example: infection in burns

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	390.415	385.718
AIC	390.415	389.718
SBC	390.415	393.460

$AIC_C = AIC + 2k(k+1)/(n-k-1) =$
$= 389.718+2\cdot2\cdot3/(154-2-1)=$
$= 389.7975$

AIC_C increases compared to the model with trt only (388.7123)

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
Trt	1	-0.52479	0.29578	3.1481	0.0760	0.592
Area	1	0.00725	0.00715	1.0294	0.3103	1.007

Percentage of area burned not significant

Example: infection in burns

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	390.415	382.220
AIC	390.415	396.220
SBC	390.415	409.319

$$AIC_C = AIC + 2k(k+1)/(n-k-1) =$$
$$= 396.220+2\cdot7\cdot8/(154-7-1)=$$
$$= 396.9871$$

AIC_C increases compared to the model with trt only (388.7123)

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
Trt	1	-0.67096	0.30791	4.7485	0.0293	0.511
Head	1	0.10177	0.33099	0.0945	0.7585	1.107
Buttock	1	0.78846	0.40234	3.8404	0.0500	2.200
Trunk	1	0.12888	0.47906	0.0724	0.7879	1.138
LegUpper	1	-0.45655	0.37107	1.5138	0.2186	0.633
LegLower	1	-0.15951	0.35843	0.1980	0.6563	0.853
RespTract	1	0.04712	0.31967	0.0217	0.8828	1.048

Buttock is the only burn site which has a significant relationship with survival

Example: infection in burns

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Trt	1	4.0318	0.0446
BurnType	3	9.3473	0.0250

The variable Burn type has a significant relationship with survival

Analysis of Maximum Likelihood Estimates								
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	Label
Trt		1	-0.59591	0.29677	4.0318	0.0446	0.551	
BurnType	chemical	1	-0.98876	1.01601	0.9471	0.3305	0.372	BurnType chemical
BurnType	electric	1	1.27781	0.45222	7.9843	0.0047	3.589	BurnType electric
BurnType	scald	1	0.14402	0.44561	0.1045	0.7465	1.155	BurnType scald

Electric burn is the only burn type which has a significant relationship with survival

Example: infection in burns

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	390.415	378.874
AIC	390.415	386.874
SBC	390.415	394.359

$$AIC_C = AIC + 2k(k+1)/(n-k-1) =$$
$$= 386.874+2\cdot4\cdot5/(154-4-1)=$$
$$= 387.1425$$

AIC_C decreases compared to the model with trt only (388.7123)

This is a better model than using treatment as a single covariate

Example: infection in burns

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	390.415	377.464
AIC	390.415	387.464
SBC	390.415	396.820

$$AIC_C = AIC + 2k(k+1)/(n-k-1) =$$
$$= 387.464+2\cdot5\cdot6/(154-5-1)=$$
$$= 387.8694$$

AIC_C increases compared to the model with trt and burn type (387.1425)

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Trt	1	3.1978	0.0737
BurnType	3	10.1912	0.0170
Area	1	1.5145	0.2184

Percentage of area burned does not have a significant relationship with survival

Example: infection in burns

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	390.415	374.004
AIC	390.415	394.004
SBC	390.415	412.716

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Trt	1	4.4731	0.0344
BurnType	3	9.4849	0.0235
Head	1	0.0290	0.8647
Buttock	1	4.0774	0.0435
Trunk	1	0.0070	0.9332
LegUpper	1	0.5962	0.4400
LegLower	1	0.5099	0.4752
RespTract	1	0.3183	0.5726

$$AIC_C = AIC + 2k(k+1)/(n-k-1) =$$
$$= 394.004+2\cdot10\cdot11/(154-10-1)=$$
$$= 395.5425$$

AIC_C increases compared to the model with trt and burn type (387.1425)

Buttock is again the only burn site which has a significant relationship with survival

Example: infection in burns

The “best” model

Analysis of Maximum Likelihood Estimates								
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits	
Trt	1	-0.59591	0.29677	4.0318	0.0446	0.551	0.308	0.986
chemical	1	-0.98876	1.01601	0.9471	0.3305	0.372	0.051	2.725
scald	1	0.14402	0.44561	0.1045	0.7465	1.155	0.482	2.766
electric	1	1.27781	0.45222	7.9843	0.0047	3.589	1.479	8.707

Modelling without prior hypothesis – forward selection approach

1) Fit the model including one of the possible explanatory variables.
2) Repeat 1) for each explanatory variable, one at a time.
3) Choose the variable with the strongest significant relationship to survival.
4) Repeat steps 1-3, with the “basic” model now including the variable chosen in step 3). Stop when no more significant explanatory variables are found.

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Backward selection approach

1) Fit the model including all possible explanatory variables (covariates).
2) Remove the least significant covariate from the model.
3) Repeat 2) for each covariate, until no insignificant covariates are left.

Stepwise selection approach

The stepwise selection approach combines forward selection and backward selection, adding and deleting variables in an iterative manner.

Forward, backward, and stepwise selection are all available in SAS, based on the *p*-value approach.

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Best subset selection approach

A specified number of best models are found containing one, two, or three variables, and so on, up to the single model containing all of the explanatory variables.

The criterion used to determine the "best" subset is based on the global score chi-square statistic (the higher the value, the “better” the model – for that number of explanatory variables).

All possible model selection approach

The “all possible model” selection methodology is based on a combination of stepwise regression, Akaike information criteria, and the best subset selection.

All possible models, from the null model to the full model including all the explanatory variables are determined.

The models will be ordered by minimizing the AIC value at every step.

http://www2.sas.com/proceedings/forum2008/375-2008.pdf

Which approach to choose?

In different literature there are different suggestions on how to build statistical models.

Some consensuses:

1) Avoid blindfolded use of automatic selection procedures
2) Scientific knowledge (e.g. medical) plays an important role.

Program L5

- **Regression for survival data**
  - Cox's proportional hazards regression
    - Partial maximum likelihood
    - Interpretation of estimated coefficients
    - Continuous vs categorical covariates
    - Ties
    - Local tests
    - Model building
    - Time-dependent covariates

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Time-dependent covariates

Explanatory variables (covariates) that don't have fixed values (recorded at the start of the study) are said to be **time-dependent**.

The values of time-dependent covariates may change during the course of the study.

If a covariate Z is time-dependent we denote it $Z(t)$.

Example: Ex-cons



X = time to crime relapse for ex-convicts (months)

The risk of committing another crime decreases with time from prison release.

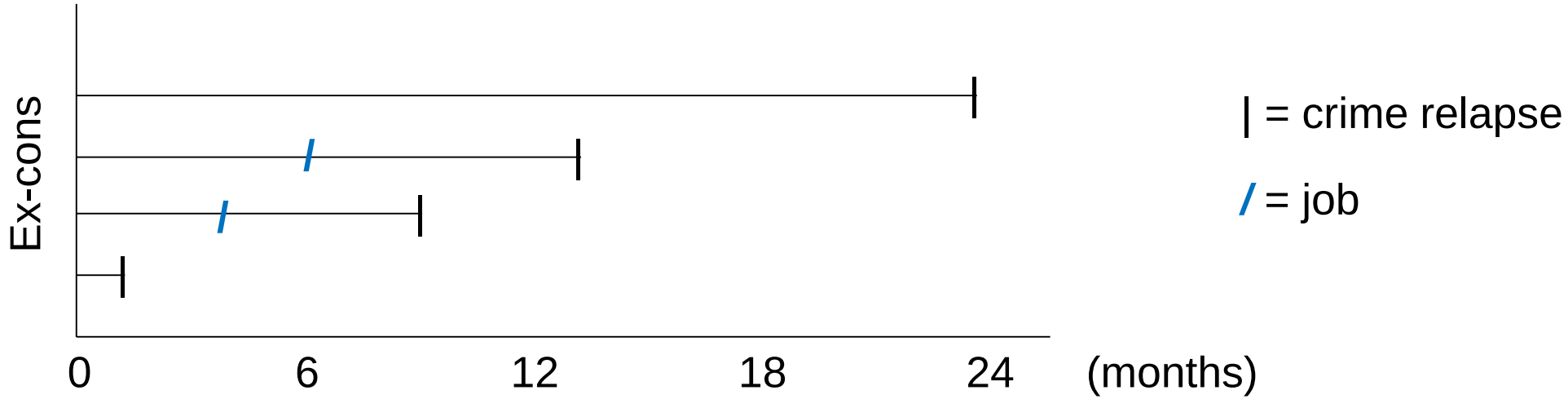
The risk of crime relapse also decreases if the ex-con gets a job (not known at the time of prison release).

Example: Ex-cons

t_i (*months*)	d_i	*Job* (*months*)	$Z(t)$ $t =$ 1	8	13	24
1	1	-	0	-	-	-
8	1	4	0	1	-	-
13	1	6	0	1	1	-
24	1	-	0	0	0	0

$Z(t)$ changes with time

Example: Ex-cons



The “partial likelihood” with time-dependent covariates

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{D} L_i = \prod_{i=1}^{D} \frac{\exp\left(\sum_{h=1}^{p} \beta_h Z_{(i)h}(t_i)\right)}{\sum_{j \in R(t_i)} \exp\left(\sum_{h=1}^{p} \beta_h Z_{jh}(t_i)\right)}$$

Partial likelihood for the *i*th event time

Risk set at time t_i

Individual hazards with time-dependent covariates

The hazard for an individual experiencing the event at time t_1:

$$h(t_1 \mid Z(t_i) = h_0(t)e^{\beta Z(t_i)}$$

Event time t_i

$= h_0(t)e^{\beta}$ if $Z(t)$=1 at this time for this individual

$= h_0(t)$ if $Z(t)$=0 at this time for this individual

Example: Ex-cons

t_i (months)	d_i	Job (months)	$Z(t)$ $t =$ 1	8	13	24
1	1	-	0	-	-	-
8	1	4	0	1	-	-
13	1	6	0	1	1	-
24	1	-	0	0	0	0

$$L_1 = \frac{e^{\beta \cdot 0}}{e^{\beta \cdot 0} + e^{\beta \cdot 0} + e^{\beta \cdot 0} + e^{\beta \cdot 0}} = \frac{1}{4}$$

= probability (likelihood) that no. 1 experiences the event, compared to all the individuals at risk

Example: Ex-cons

t_i (months)	d_i	Job (months)	$Z(t)$ $t=$ 1	8	13	24
1	1	-	0	-	-	-
8	1	4	0	1	-	-
13	1	6	0	1	1	-
24	1	-	0	0	0	0

$$L_2 = \frac{e^{\beta \cdot 1}}{e^{\beta \cdot 1} + e^{\beta \cdot 1} + e^{\beta \cdot 0}}$$

$$L_3 = \frac{e^{\beta \cdot 1}}{e^{\beta \cdot 1} + e^{\beta \cdot 0}}$$

$$L_4 = \frac{e^{\beta \cdot 0}}{e^{\beta \cdot 0}} = 1$$

Time-dependent covariates

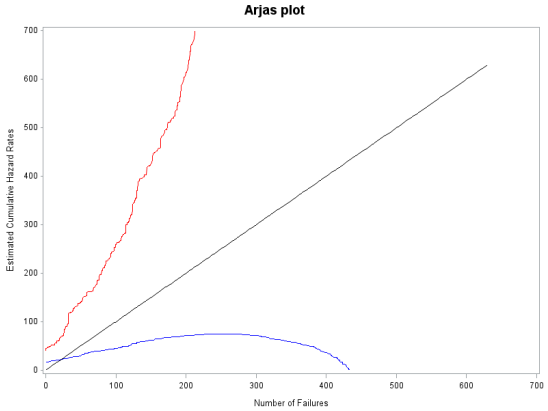
Read more (Studium module Articles):

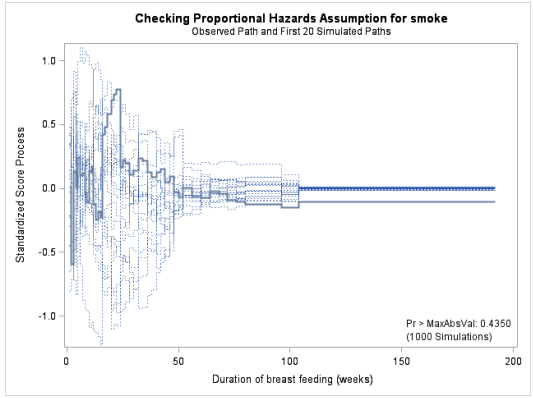
Therneau et al. (2018), Using Time Dependent Covariates and Time Dependent Coefficients in the Cox model.

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Analysis of Survival Data

Lecture 6: Regression diagnostics



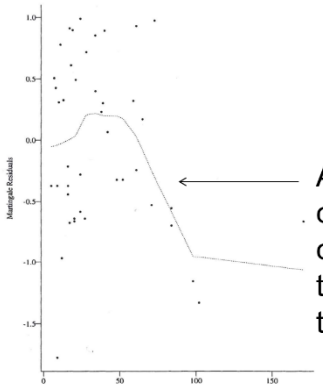


Fixed-time covariate (for which we want to test the PH assumption)

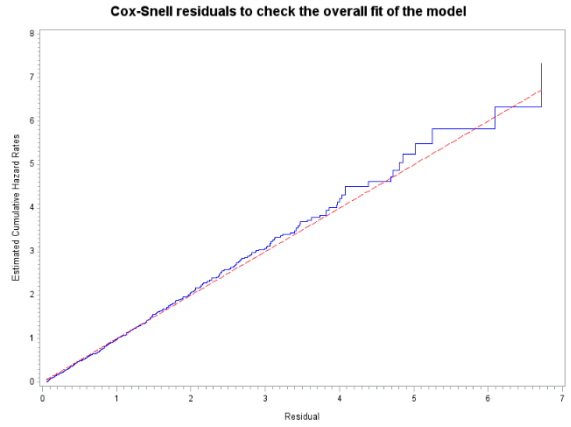
$g(t)$ is a known function of time

$$h(t \mid \mathbf{Z}) = h_0(t) \exp\left\{\beta_1 Z_1 + \beta_2 \left(Z_1 g(t)\right)\right\}$$

Artificially created time-dependent covariate



A drop, suggesting that this covariate could be categorized to better explain the relationship with the time variable



Inger Persson

Program L6

- **Regression for survival data**
  - Cox's proportional hazards regression, cont'd
    - Testing the PH assumption
    - Graphical methods to investigate the PH assumption
    - What to do if the proportional hazards assumption is violated
    - Regression diagnostics

Tests of PH assumption

A number of tests of the PH assumption available.

Most common:
Time-dependent covariate test (Cox).

Testing the assumption of proportional hazards

Time-dependent covariate methodology can be used to test the proportional hazards assumption.

One common way:

Fixed-time covariate (for which we want to test the PH assumption)

$g(t)$ is a known function of time

$$h(t \mid \mathbf{Z}) = h_0(t) \exp\{\beta_1 Z_1 + \beta_2 (Z_1 g(t))\}$$

Artificially created time-dependent covariate

Usually: $g(t) = \ln t$

Comparing two individuals (proportional hazards)

Comparing two individuals with distinct values of Z_1:

$$\frac{h(t \mid Z_1)}{h(t \mid Z_1^*)} = \exp\left(\beta_1(Z_1 - Z_1^*) + \beta_2 g(t)(Z_1 - Z_1^*)\right)$$

If $\beta_2 = 0$ this function does not depend on t.

Testing H_0: $\beta_2 = 0$ is thus a test for the PH assumption.

Note: PH ass. not to be checked for time-dep. covariates.

Example: infection in burns (recap.)

In section 1.6 a study is described to evaluate a protocol change in disinfectant practice in a large midwestern university medical center.

Control of infection is the primary concern for the 154 patients entered into the burn unit with varying degrees of burns.

The outcome variable is the time until infection from admission to the unit. Censoring variables are discharge from the hospital without an infection or death without an infection.

84 patients were in a group which had a body-cleansing method (disinfectant: chlorhexidine) and 70 patients received the routine bathing care method (disinfectant: povidone-iodine).

Example: infection in burns

Variables:

Trt = treatment (0=routine bathing, 1=body cleansing)

TimeStaph = Time to staphylococcus infection (days)

Staph = Staphylococcus indicator (1=infection, 0=no inf.)

Additional explanatory variables (found at previous lecture)*:*

BurnType (1=chemical, 2=scald, 3=electric, 4=flame)

Race (0=nonwhite, 1=white)

Example: infection in burns (recap.)

The “best” model

Analysis of Maximum Likelihood Estimates									
Parameter		DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Profile Likelihood Confidence Limits	
Trt		1	-0.60100	0.29787	4.0708	0.0436	0.548	0.302	0.978
BurnType	chemical	1	-1.00348	1.01626	0.9750	0.3234	0.367	0.021	1.706
BurnType	electric	1	1.16885	0.45143	6.7040	0.0096	3.218	1.200	7.274
BurnType	scald	1	0.57509	0.45294	1.6121	0.2042	1.777	0.660	4.018
Race		1	2.28344	1.02559	4.9571	0.0260	9.810	2.050	176.405

Example: infection in burns
Testing the PH assumption

```
proc phreg data=burns;
   model TimeStaph*Staph(0)=trt lntTrt /ties=exact;
   lntTrt=log(TimeStaph)*trt;
run;
```

Z $Z\,g(t)$ $Z\,g(t) = Z\ln(t)$

Example: infection in burns
Testing the PH assumption

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
Trt	1	-0.23378	0.74550	0.0983	0.7538	0.792
IntTrt	1	-0.15863	0.33385	0.2258	0.6347	0.853

OK for trt

(null hypothesis of $\beta = 0$, i.e. no time dependency, is not rejected)

Example: infection in burns
Testing the PH assumption

```
proc phreg data=burns;
   model TimeStaph*Staph(0)=chemical scald electric
         lntChemical lntScald lntElectric/ties=exact;
   lntChemical=log(timestaph)*chemical;
   lntScald=log(timestaph)*Scald;
   lntElectric=log(timestaph)*electric;
   test lntChemical=lntScald=lntElectric=0;
run;
```

Note: Dummies must be created for categorical variables (“class” statement cannot be used for this)

Example: infection in burns
Testing the PH assumption

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
chemical	1	-8.53847	8.16805	1.0928	0.2959	0.000
scald	1	2.90012	1.22275	5.6254	0.0177	18.176
electric	1	3.59420	1.30819	7.5485	0.0060	36.387
IntChemical	1	2.90466	2.81679	1.0634	0.3024	18.259
IntScald	1	-1.45124	0.70469	4.2411	0.0395	0.234
IntElectric	1	-1.45821	0.81732	3.1831	0.0744	0.233

Linear Hypotheses Testing Results				
	Label	Wald Chi-Square	DF	Pr > ChiSq
	Test 1	6.5349	3	0.0883

OK for burntype

Example: infection in burns
Testing the PH assumption

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
Race	1	15.97393	14.61024	1.1954	0.2742	8657459
IntRace	1	-4.28022	3.91857	1.1931	0.2747	0.014

OK for race

Example: infection in burns
Testing the PH assumption

But the power of these tests may be low due to small number of events in each group (which is extra important since we “want” a non-significant result)!

Frequency

Table of Staph by Trt

Staph	Trt 0	Trt 1	Total
0	42	64	106
1	28	20	48
Total	70	84	154

Table of Staph by BurnType

Staph	BurnType 1	2	3	4	Total
0	8	12	5	81	106
1	1	6	6	35	48
Total	9	18	11	116	154

Table of Staph by Race

Staph	Race 0	Race 1	Total
0	18	88	106
1	1	47	48
Total	19	135	154

Program L6

- **Regression for survival data**
  - Cox's proportional hazards regression, cont'd
    - Testing the PH assumption
    - Graphical methods to investigate the PH assumption
    - What to do if the proportional hazards assumption is violated
    - Regression diagnostics

Graphical methods to assess the assumption of proportional hazards

There are a number of graphical methods available to check the proportional hazards assumption.

1) Plots based on the log cumulative baseline hazard
2) Andersen plot, based on the cum. baseline hazard
3) Arjas plot, based on the expected number of events calculated from the estimated cumulative hazard rate and the observed number of events up to time t
4) Plots based on score residuals
5) Plots based on scaled Schoenfeld residuals

Plots of log cumulative baseline hazard rates

$$h_1(t) = ch_2(t)$$
$$H_1(t) = cH_2(t)$$

Same assumption of proportionality

$$\ln H_1(t) = \ln c + \ln H_2(t)$$

$$\ln H_1(t) - \ln H_2(t) = \ln c$$

A constant

This difference can be plotted to see if it is constant over time.

Plots of log cumulative baseline hazard rates

$$H(t) = \int_0^t h(u)du$$

$$h_1(t) = ch_2(t)$$

$$\int_0^t h_1(u)du = \int_0^t ch_2(u)du$$

$$\int_0^t h_1(u)du = c\int_0^t h_2(u)du \qquad H_1(t) = cH_2(t)$$

Plots of log cumulative baseline hazard rates

Procedure:

1) The covariate Z_1 for which we want to check the PH assumption has to have K possible values.

   Continuous covariates need to be stratified into K disjoint strata to be able to use this approach.

2) Fit a Cox model, stratified on the discrete values of Z_1.

3) Estimate the cumulative baseline hazard rate in each of the K strata, $\hat{H}_{go}(t)$ for g = 1, … K.

Plots of log cumulative baseline hazard rates

4) Plot $ln\widehat{H}_{go}(t) - ln\widehat{H}_{1o}(t)$ vs. t for g=2,….K.

5) If the proportional hazards assumption holds, each curve should be roughly constant over time.

NOTE! This plot should be interpreted with some care, especially in the right-hand tail of the plot, since the variances of the curves are not constant over time.

Example: Duration of breastfeeding

Section 1.14 describes a study with the aim to evaluate the relationship between duration of breastfeeding and a number of explanatory variables.

Example: Duration of breastfeeding

Variables:

time = duration of breastfeeding (weeks)

complete = indicator of completed breastfeeding (1=yes, 0=no)

poverty = mother in poverty (1=yes, 0=no)

smoke = mother smoked at birth (1=yes, 0=no)

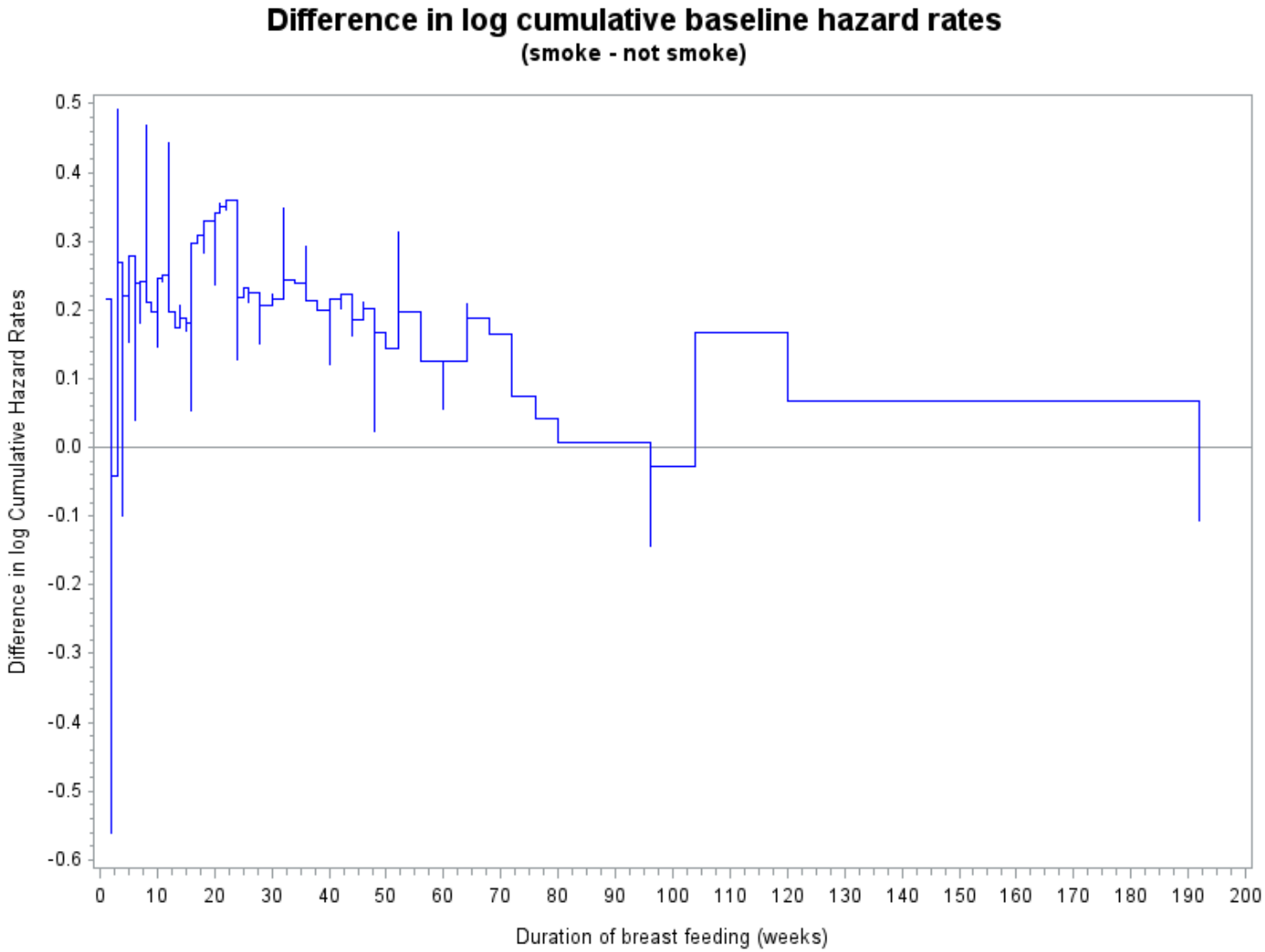
birthyear = birthyear of mother

education = mother's education (years of school)

Example: Duration of breastfeeding

Analysis of Maximum Likelihood Estimates								
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Profile Likelihood Confidence Limits	
poverty	1	-0.17178	0.09230	3.4634	0.0627	0.842	0.701	1.007
smoke	1	0.19272	0.07588	6.4503	0.0111	1.213	1.044	1.405
birthyear	1	0.07179	0.01786	16.1582	<.0001	1.074	1.038	1.113
education	1	-0.07349	0.01993	13.5932	0.0002	0.929	0.894	0.966

Example: Duration of breastfeeding



Roughly constant over time?

Not that easy to interpret.

Andersen plot (based on the cumulative baseline hazard)

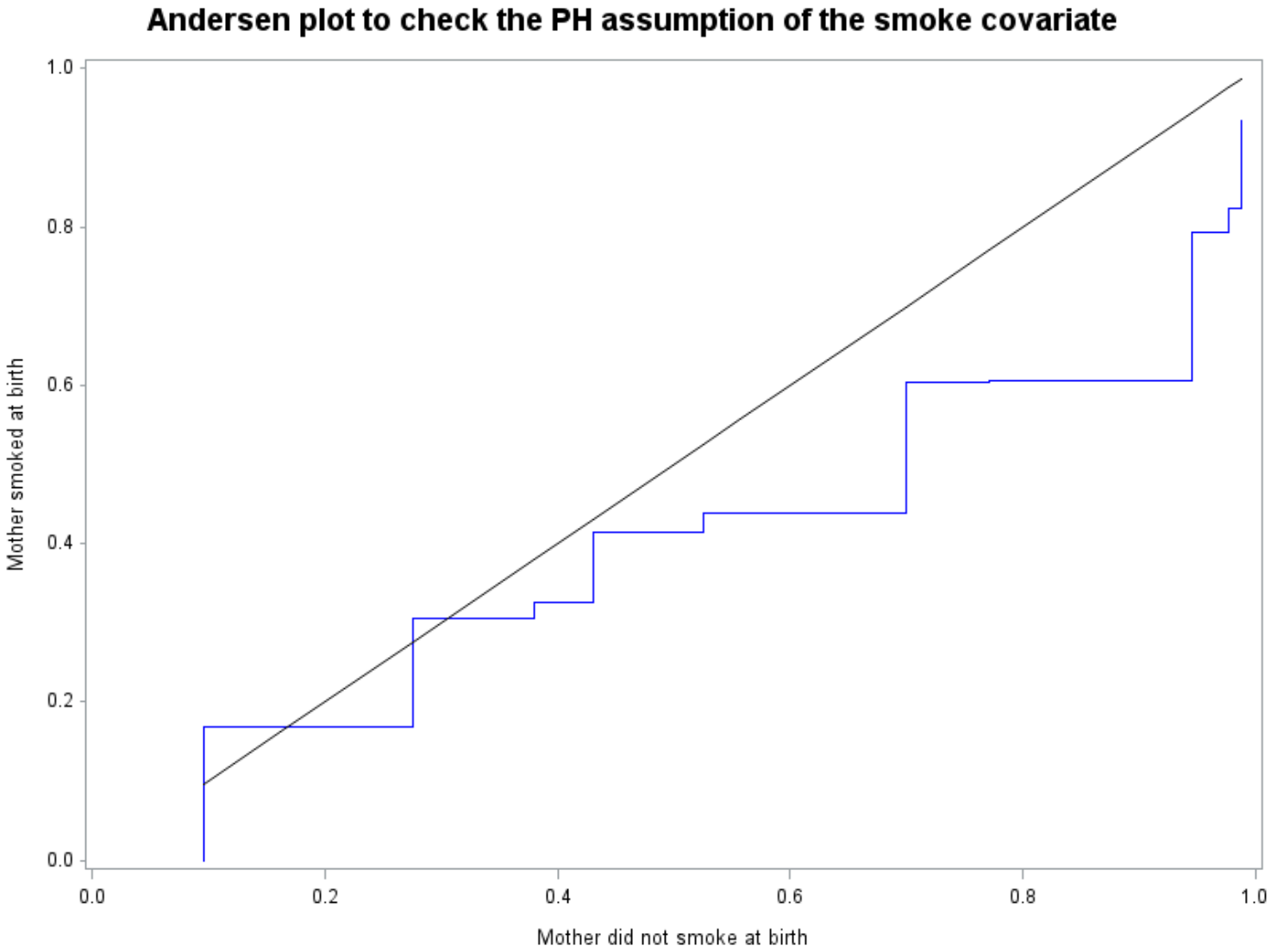
Procedure:

1) Plot, for all t (or a subset of t), $\widehat{H}_{1o}(t)$ versus $\widehat{H}_{go}(t)$ for g = 2, … K.

2) If the proportional hazard assumption holds, these curves should be straight lines through the origin.

NOTE! This plot should also be interpreted with some care, especially in the right-hand tail of the plot, again since the variances of the curves are not constant over time.

Example: Duration of breastfeeding



Roughly a straight line through the origin?

What does "roughly" mean?

Arjas plot

Arjas plots can be used both to check the proportional hazards assumption, and to check the overall fit of the Cox model.

Z_1 = a covariate for which we want to check the PH assumption, or check if the covariate should be included in the model.

Arjas plot

Procedure:

1) Fit the Cox model without the covariate Z_1.

2) For each event time and for each level of Z_1 calculate the expected number of events and the observed no. of events up to this time.

Arjas plot

Expected number of events up to time t_i:
(referred to as the "total time on test" in the book)

$$E_TOT_g(t_i) = \sum_{Z_{1j=g}} \hat{H}(\min(t_i, T_j)|\mathbf{Z}_j^*)$$

Stratum g

Sum over all individuals in stratum g

Cumulative hazard up to time t_i (or the last time for the individual)

based on a model of all covariates except Z_1.

Observed number of events up to time t_i:

$$N_g(t_i) = \sum_{Z_{1\,j=g}} \delta_j(T_j \le t_i)$$

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Arjas plot

Procedure:

1) Fit the Cox model without the covariate Z_1.

2) For each event time and for each level of Z_1 calculate the expected number of events and the observed no. of events up to this time.

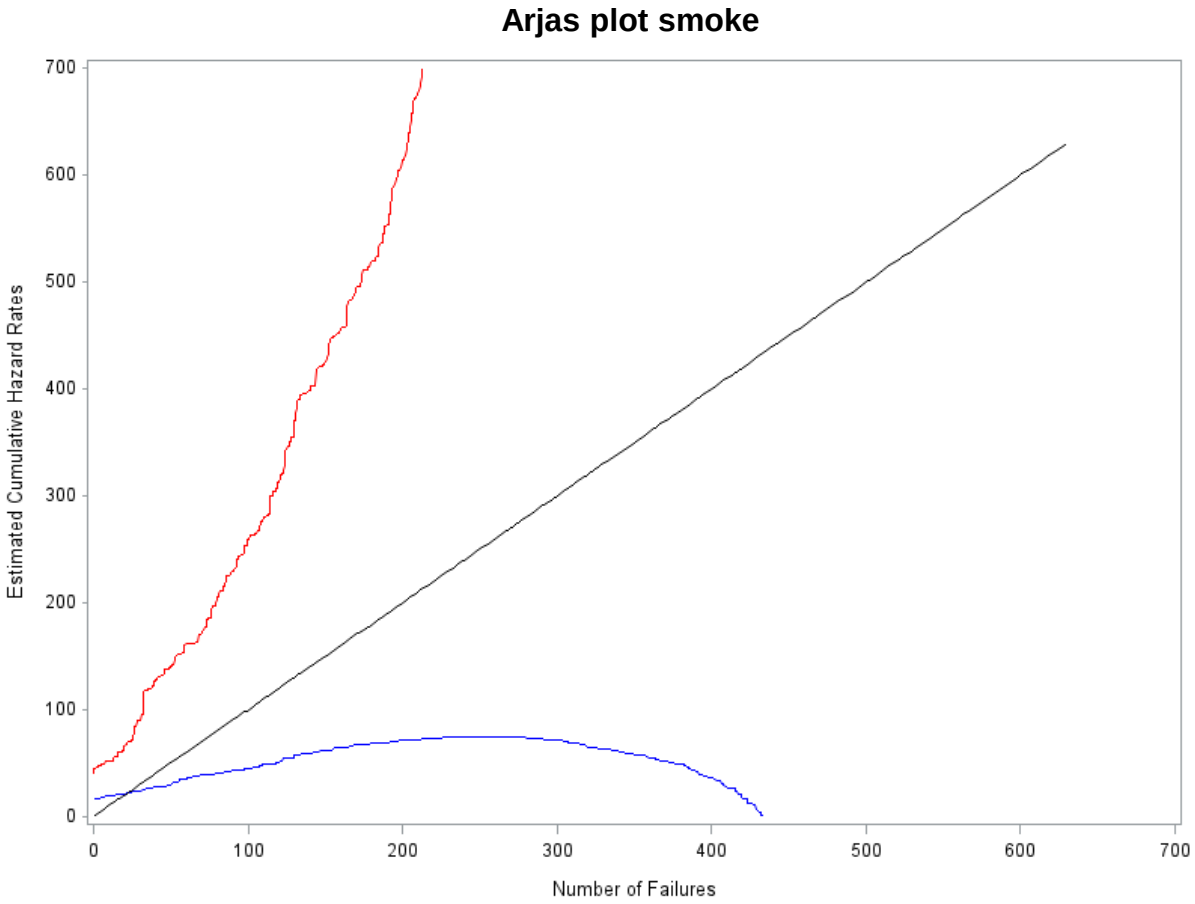
3) If the covariate Z_1 has a nonproportional hazards effect on the hazard rate, a plot of $N_g(t_i)$ versus $E_TOT_g(t_i)$ provides curves that differ nonlinearly from a 45 degree line through the origin.

Arjas plot

4) If the covariate Z_1 is redundant (does not need to be included in the model), the plot of $N_g(t_i)$ versus $E_TOT_g(t_i)$ provides curves that should roughly follow a 45 degree line through the origin.

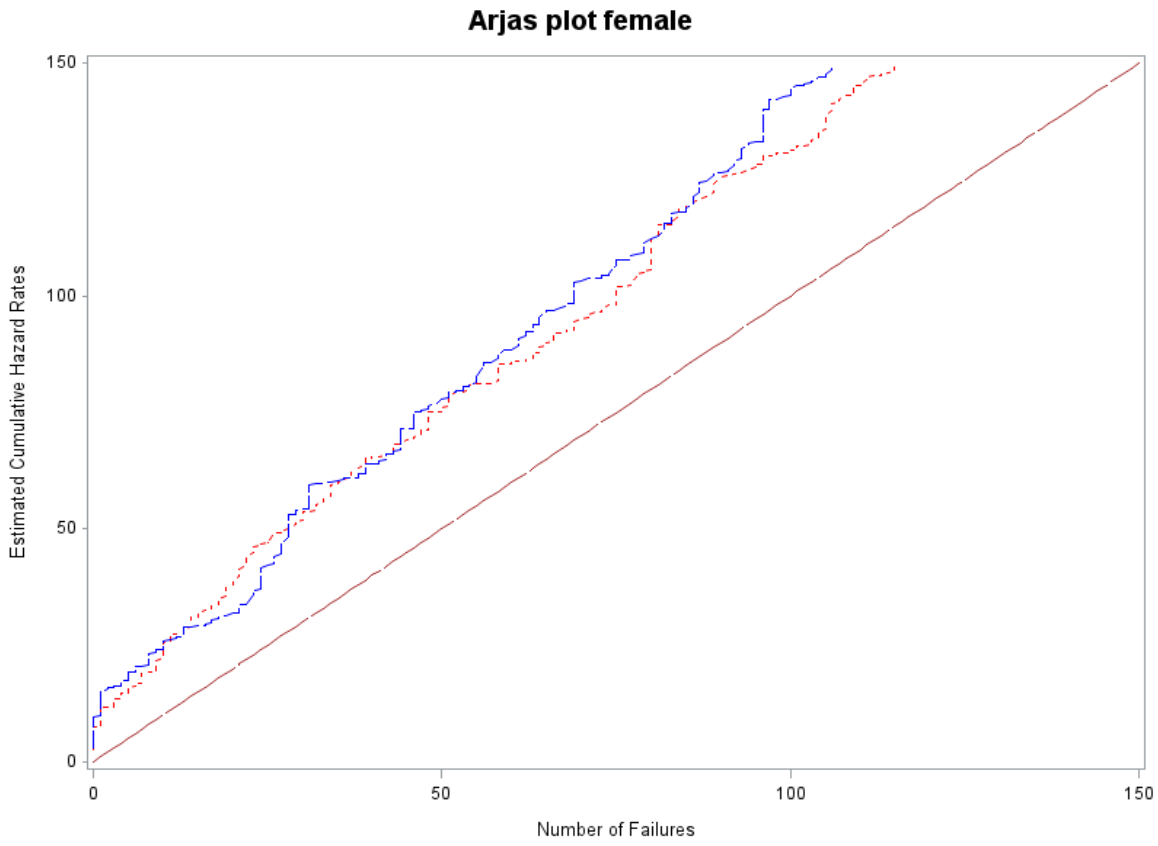
   If the covariate Z_1 should be included in the model, the plot provides curves that differ linearly from a 45 degree line through the origin (with slopes different from 1).

Example: Duration of breastfeeding



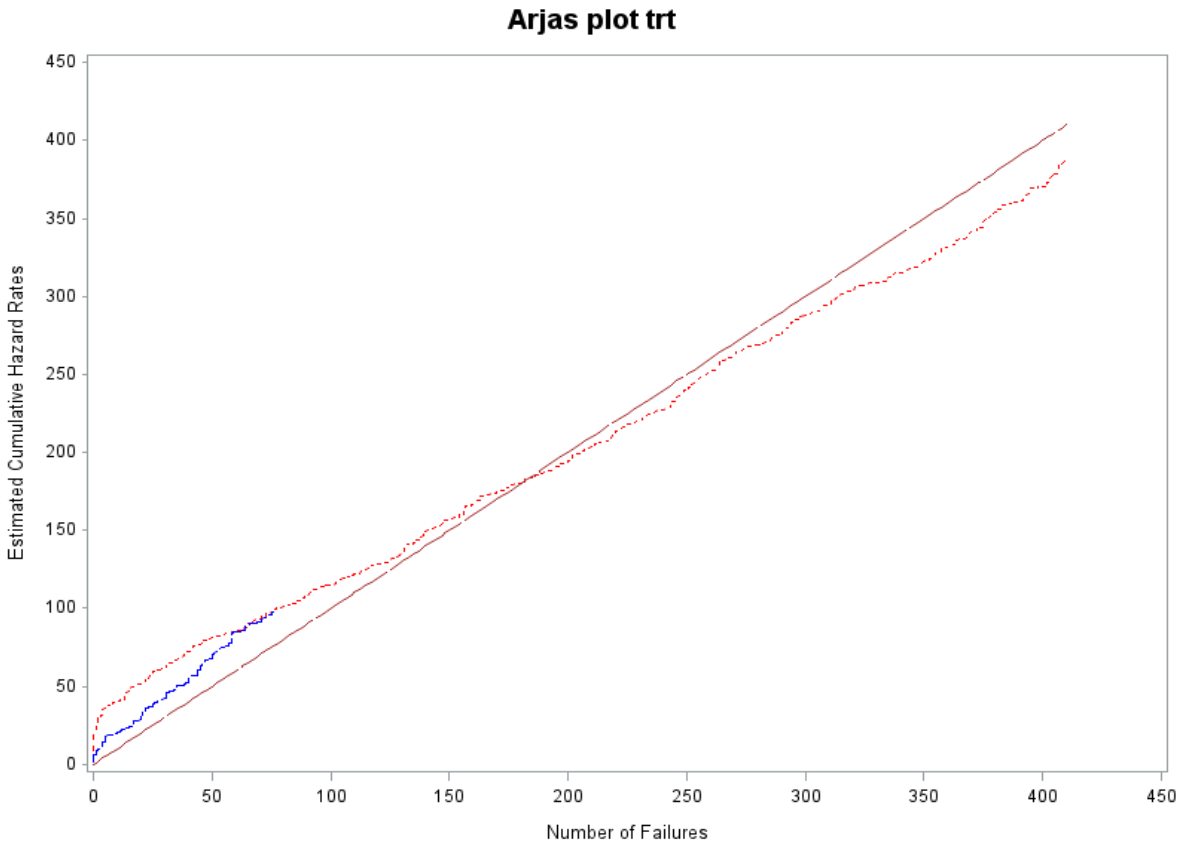
The curves (especially the blue one) differ non-linearly from the line, indicating nonproportional hazards

Example: Exam 2019-01-18 Task 3



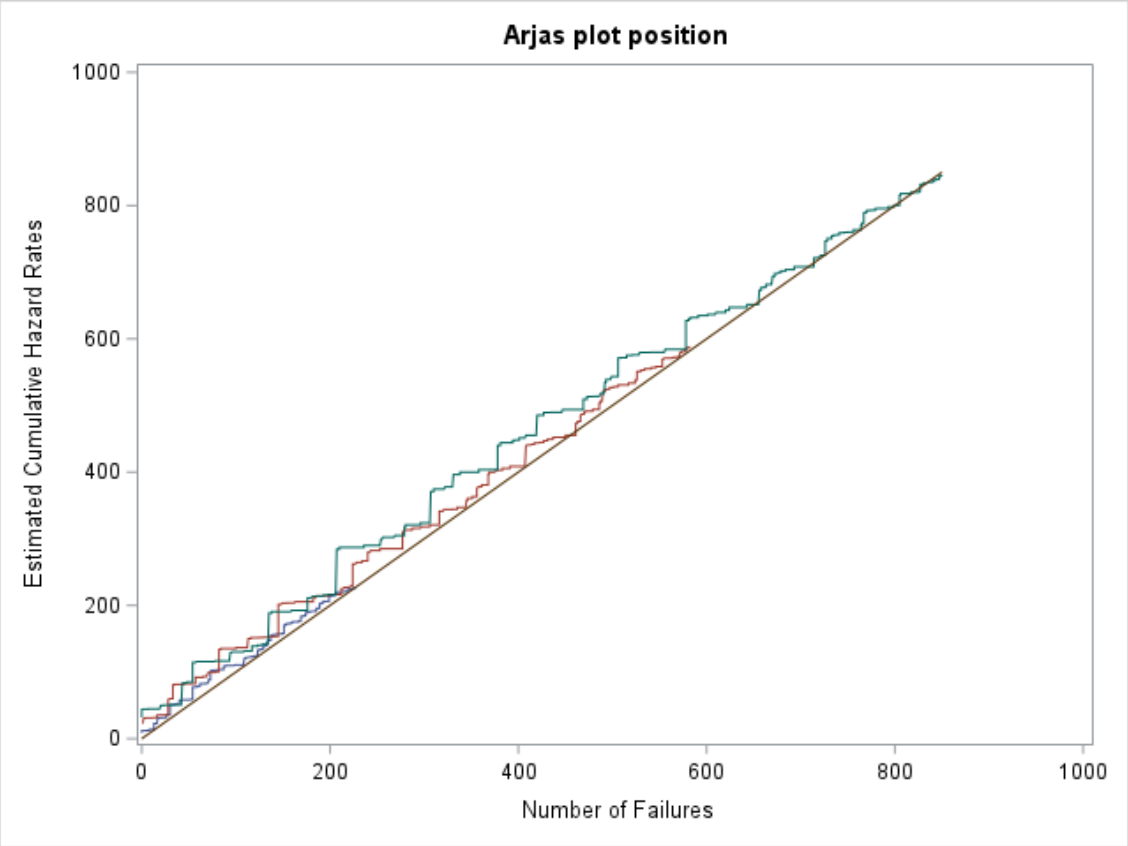
The curves differ linearly from the line, not contradicting proportional hazards

Example: Exam 2024-01-11 Task 3



The red curve differs non-linearly from the line, indicating nonproportional hazards

Example: Exam 2025-01-17 Task 3C



The curves are quite linear, not contradicting proportional hazards

But they roughly follow the line, which means that this covariate doesn't contribute much to the model.

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Arjas plot

One advantage of the Arjas plot is that you can change the way the investigated covariate is stratified without refitting the basic Cox model.

One disadvantage is that continuous covariates cannot be investigated as they are but have to be categorized.

Using Score residuals for graphical checks of the PH assumption

The score residuals are the first partial derivative with respect to β_k of the contribution to the log likelihood for the fitted Cox model, using only information of an individual j accumulated up to time t.

Score residuals are useful in assessing the influence of each subject on individual parameter estimates.

Interpreted as a weighted difference between the value of a given covariate for a given subject and the average value of this covariate in a risk set.

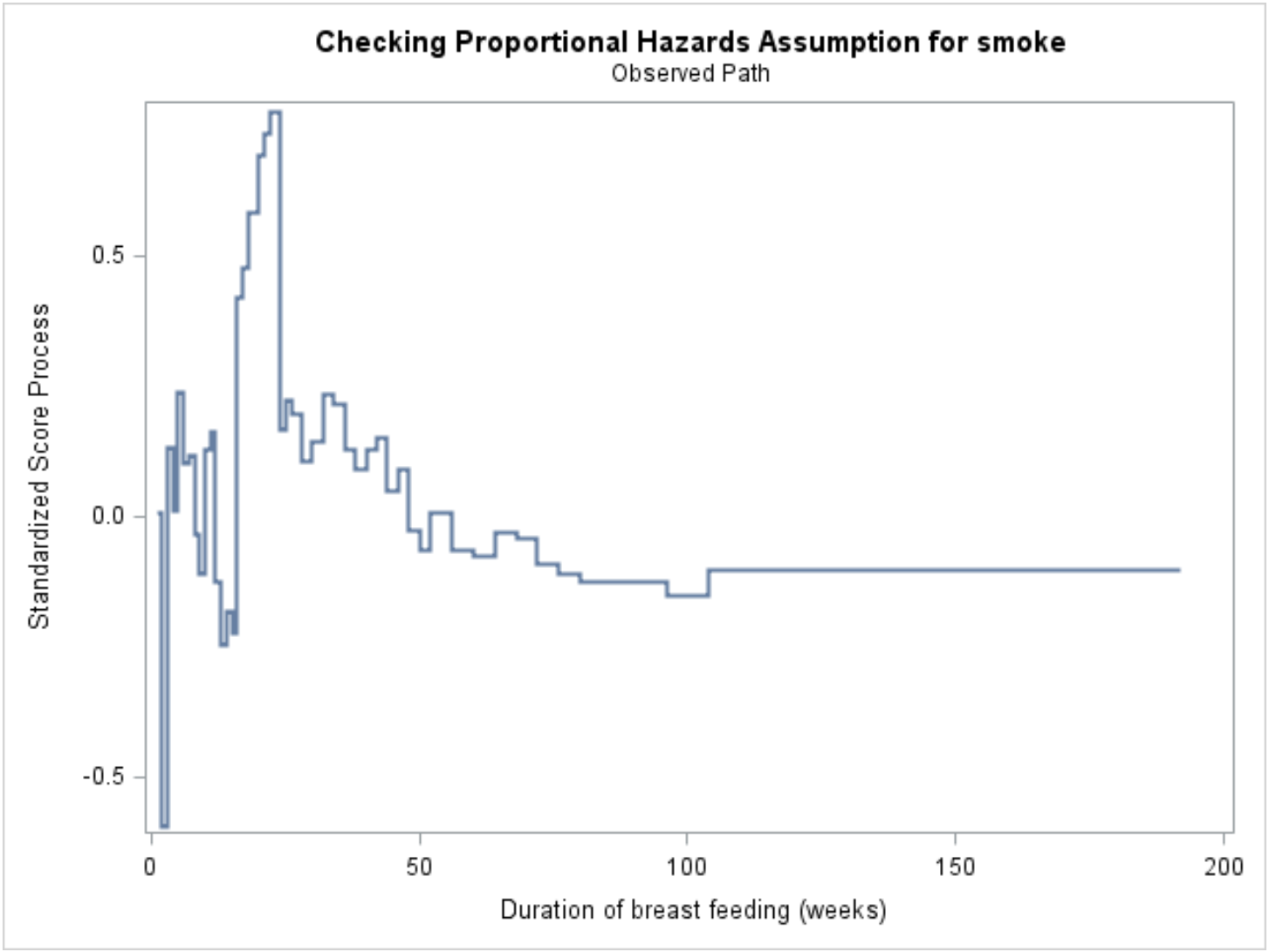
Using Score residuals for graphical checks of the PH assumption

Standardized score residuals can be plotted against time to check for nonproportional hazards.

If the model fits properly, the standardized residuals should behave like a Brownian Bridge. If the residuals get unusually large at any time point, we don't have PH.

One advantage of the score process is that a continuous covariate doesn't have to be discretized to make the plot.

Example: Duration of breastfeeding



No evidence of nonproportional hazards here

Not that easy to interpret

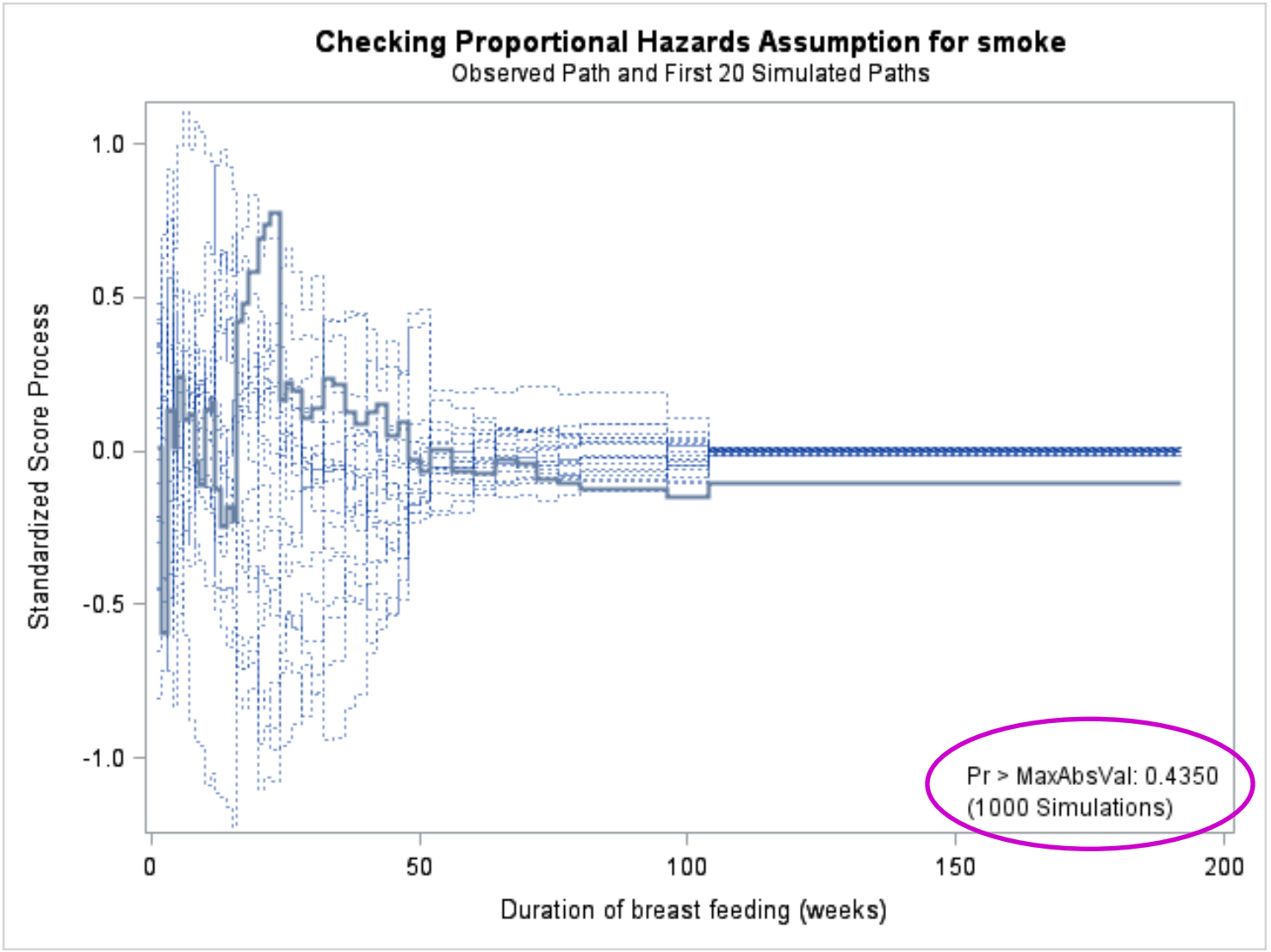
Using Score residuals for graphical checks of the PH assumption

The standardized score residuals can be used to provide a p-value of testing the PH assumption, based on simulations*.

This is included in proc phreg, option **assess ph**.

*Lin, Wei, Ying (1993). Checking the Cox Model with Cumulative Sums of Martingale-Based Residuals. *Biometrika*, 80, 3, pages 557-572

Example: Duration of breastfeeding



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Score residuals plot

One advantage of the score residuals plot is that continuous covariates can be investigated as they are (no need for categorization).

One disadvantage is that this method is computationally demanding and often cannot be used with the free online software SAS OnDemand for Academics.

Schoenfeld's partial residuals

Schoenfeld's partial residuals are a part of the score residuals and defined as

$$s_{jk} = \delta_j \{ Z_{jk} - \bar{Z}_k(T_j) \}$$

Defined for events only

Covariate value for individual j

Expected Covariate value for individual j

This describes the difference between a covariate value and the expected value of this covariate at this time.

Scaled Schoenfeld residuals

Grambsch and Therneau (1994) showed that a scaled version of the Schoenfeld residual at time *T* for a particular covariate will approximate the change in the regression coefficient at this time point:

$$E(s_{jk}^{*}) \approx \hat{\beta}_k + \beta_j\,(T_j)$$

Scaled Schoenfeld residual for individual *j*, covariate *k.*

$s_{jk}^{*} = s_{jk} / \hat{V}(\hat{\beta}_k)$

Time-invariant coefficient

Time-dependent coefficient

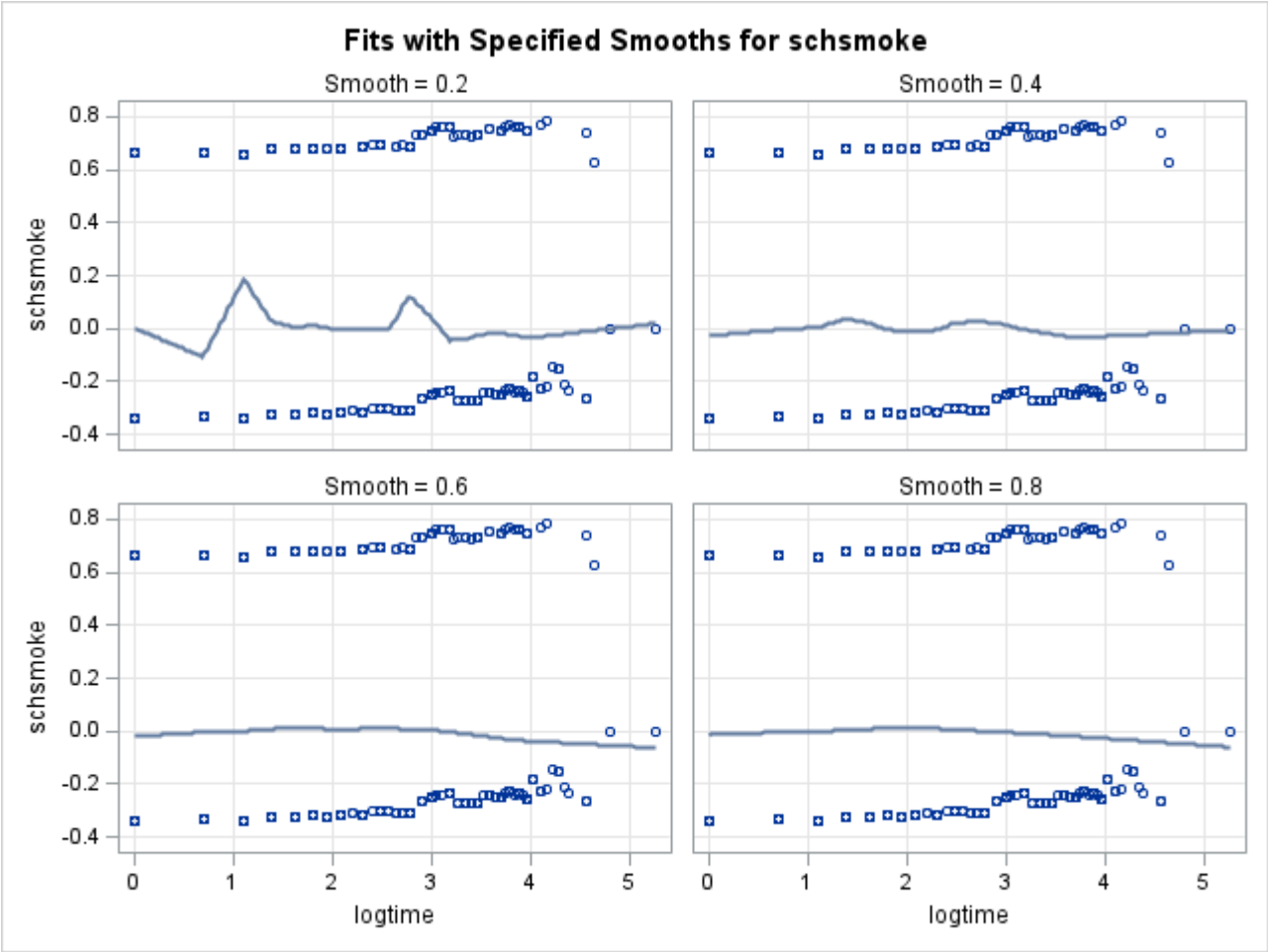
Grambsch and Therneau (1994). Proportional hazards tests and diagnostics based on weighted residuals. *Biometrika*, Vol. 81, No. 3, 1994, pp. 515-526

Plots based on scaled Schoenfeld residuals

Smoothed scatterplots are used to explore the scaled Schoenfeld residuals' relationship with time (preferably log(time) or rank(time) to find non-linear relationships)

If the average of the scaled Schoenfeld residuals is zero across time (flat smoothed lines at 0), this suggests that the coefficient does not vary over time (the assumption of proportional hazards holds for this covariate).

Example: Duration of breastfeeding



No evidence of nonproportional hazards here

Choice of method to check the assumption of proportional hazards

Tests:

A number of statistical tests, including the Cox model with time-dependent covariates (with different functions of time) have been evaluated by Persson and Khamis (2008).

The standard test, including a time-dependent covariate in the Cox model ($g(t) = \ln t$), is shown to work well in most situations (not good power if <~50 events/group).

Inger Persson, Harry Khamis (2008). A Comparison of Statistical Tests for Assessing the Proportional Hazards Assumption in the Cox Model. *Journal of Statistics and Applications*, Vol. 3, No. 1-2, 2008, pages 135-154

Choice of method to check the assumption of proportional hazards

Graphical methods:
A number of graphical methods have been evaluated by Persson and Khamis (2007).

The Arjas plot is shown to work well in most situations, and it is the easiest graphical method to evaluate visually.

Also, the method based on standardized score residuals (phreg option **assess ph**) seems to have good properties (not included in the study by Persson and Khamis).

Inger Persson, Harry Khamis (2007). A comparison of graphical methods for assessing the proportional hazards assumption in the Cox model. *Journal of Statistics and Applications*, Vol 2, No. 1-2, 2007, 1-32.

Choice of method to check the assumption of proportional hazards

Recommendations:

Combine test+graphical method.

If any of the chosen methods finds nonproportionality, reject the null hypothesis of proportional hazards (the power of any of these methods is not very high).

Example: Duration of breastfeeding Testing the PH assumption

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
smoke	1	0.25264	0.15330	2.7160	0.0993	1.287
IntSmoke	1	-0.01172	0.06365	0.0339	0.8538	0.988

OK for smoke

Time-dependent covariate test and scaled Schoenfeld residual plot show no deviation from proportionality. But the Arjas plot suggests non-proportionality, which means that the assumption of proportional hazards is violated.

Program L6

- **Regression for survival data**
  - Cox's proportional hazards regression, cont'd
    - Testing the PH assumption
    - Graphical methods to investigate the PH assumption
    - What to do if the proportional hazards assumption is violated
    - Regression diagnostics

If the PH assumption doesn't hold

In many instances the model diagnostics reveal only a small to moderate deviation from the proportional hazards assumption.

In such cases the hazard ratio estimated from the Cox model is quite close to a geometric average of the hazard ratio*. This does not exactly describe the truth, since the hazard rate changes over time when the hazards are not proportional. It can however provide an approximate estimate.

*Inger Persson, Harry Khamis (2005). Bias of the Cox Model Hazard Ratio. *Journal of Modern Applied Statistical Methods*, May, 2005, Vol. 4, No. 1, 90-99

“Rescuing” a violated proportional hazards assumption

Two possible solutions:

1) Stratify the model on the covariate that violates the PH assumption (might lead to the hazards being proportional within each stratum, for the other covariates).

2) Include the time-dependent covariate(s) in the model (may be harder to interpret, especially for continuous covariates)

Stratified proportional hazards models

With a stratified proportional hazards model, the regression coefficients are assumed to be the same in each stratum, but the baseline hazard function might differ between strata.

$$h_j(t \mid \mathbf{Z}) = h_{0j}(t)\exp(\boldsymbol{\beta}^t \mathbf{Z})$$

↑
Stratum j

Stratified proportional hazards models

Assumption of stratified models: covariates are acting similarly on the baseline hazard function in each stratum.

This can be tested, e.g. using the Likelihood ratio test.

$$H_0 : \text{All } \beta\text{'s are the same for all } s \text{ strata}$$

$$H_a : \text{At least one of the } \beta\text{'s is/are different}$$

Likelihood ratio test for the assumption of stratified models

1) Fit the stratified model (with common β's in each stratum) and obtain the log partial likelihood $LL(\mathbf{b})$.

2) Use only data from the jth stratum, and fit a Cox model to estimate β_j and corresponding $LL_j(\mathbf{b}_j)$. Do this for all strata.

3) Calculate the log likelihood under the model with distinct covariates for each of the s strata:

$$\sum_{j=1}^{s} LL_j(\mathbf{b}_j)$$

Likelihood ratio test for the assumption of stratified models

4) Calculate the likelihood ratio chi square test for

H_0: All β's are the same in the different strata

$$-2\left[LL(\mathbf{b}) - \sum_{j=1}^{s} LL_j(\mathbf{b}_j)\right] \sim \chi^2_{(s-1)p}$$

s = no. of strata

p = no. of covariates

Equivalently: $$\left[-2LL(\mathbf{b}) - \sum_{j=1}^{s} -2LL_j(\mathbf{b}_j)\right]$$

Example: Duration of breastfeeding

The **strata** statement is used to estimate a stratified proportional hazards model

```
/* Stratify on smoke */
proc phreg data=bf;
  model time*complete(0)=poverty education birthyear
   /ties=exact;
  strata smoke;
run;
```

Example: Duration of breastfeeding

Stratified by smoke (same parameter estimates, different baseline hazards)

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	5320.374	5296.306
AIC	5320.374	5302.306
SBC	5320.374	5316.686

← -2LogL = 5296.306

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
poverty	1	-0.16198	0.09260	3.0603	0.0802	0.850
education	1	-0.07309	0.01997	13.3963	0.0003	0.930
birthyear	1	0.07257	0.01794	16.3595	<.0001	1.075

Example: Duration of breastfeeding

The **by** statement is used to estimate separate models for each stratum

```
/* estimate model for each stratum separately */
proc phreg data=bf;
  model time*complete(0)= poverty education birthyear
   /ties=exact;
  by smoke;
run;
```

Example: Duration of breastfeeding

Smoked at birth

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	1498.413	1495.547
AIC	1498.413	1501.547
SBC	1498.413	1512.264

Did not smoke

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	3821.962	3797.824
AIC	3821.962	3803.824
SBC	3821.962	3817.157

Example: Duration of breastfeeding

$$\sum_{j=1}^{2} -2LL_j(\mathbf{b}_j) = 1495.547 + 3797.824$$

$$= 5293.371$$

$$-2\left[LL(\mathbf{b}) - \sum_{j=1}^{2} LL_j(\mathbf{b}_j)\right] = 5296.306 - 5293.371]$$

$$= 2.9 \sim \chi^2_{(2-1)3=3\,df}$$

Example: Duration of breastfeeding

The corresponding *p*-value is 0.407 (table C.2 only shows that it is larger than 0.1)

This means that the effects of the covariates are not significantly different for all strata.

A stratified model is thus appropriate in this case.

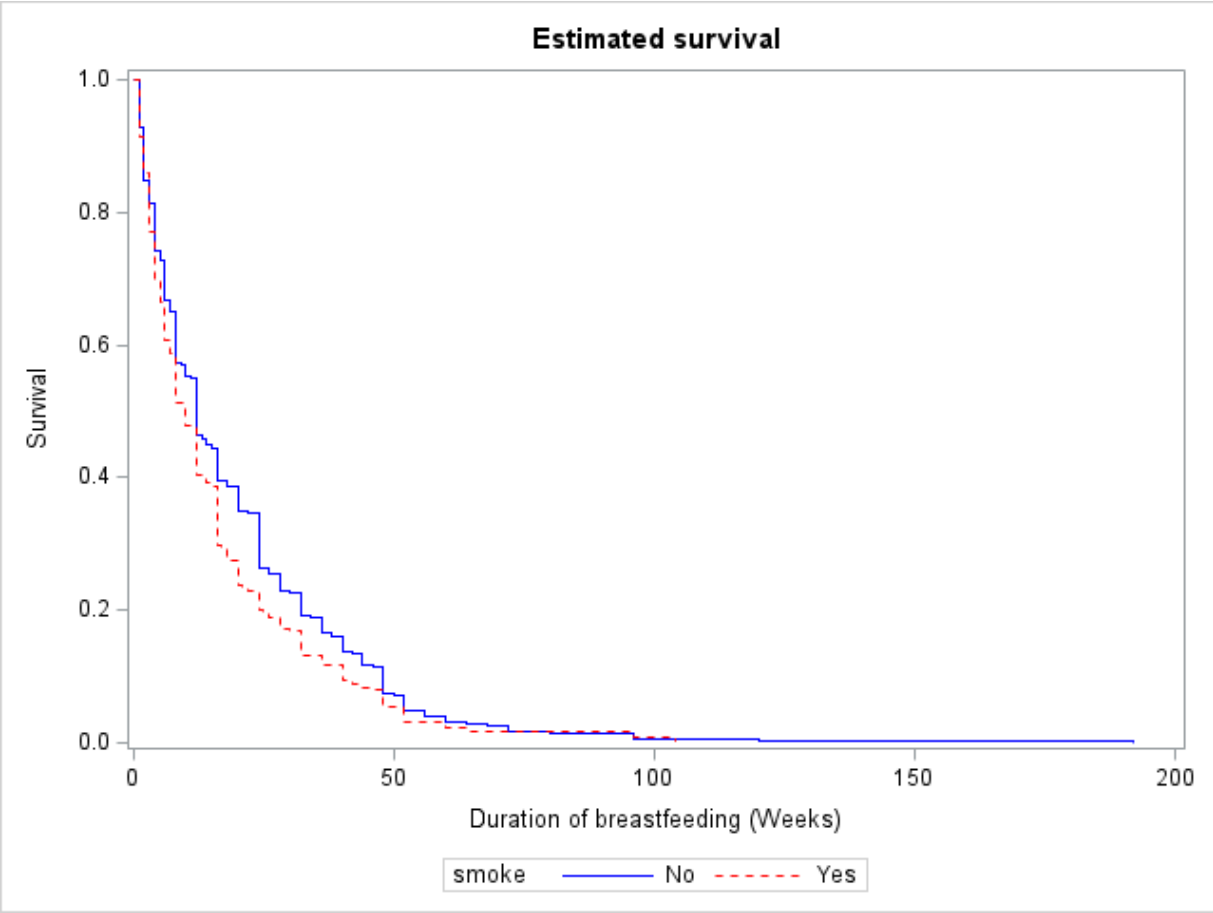
Example: Duration of breastfeeding

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
poverty	1	-0.16198	0.09260	3.0603	0.0802	0.850
education	1	-0.07309	0.01997	13.3963	0.0003	0.930
birthyear	1	0.07257	0.01794	16.3595	<.0001	1.075

The stratified model doesn't contain any information on the stratification covariate.

Information on this covariate can be provided by plots of the estimated survival for the different strata, for some values of the other covariates.

Example: Duration of breastfeeding



Survivor functions can be estimated by the Cox model for different values of the stratification variable, for selected values of the other covariates (most common values chosen here)

“Rescuing” a violated proportional hazards assumption

Two possible solutions:

1) Stratify the model on the covariate that violates the PH assumption (might lead to the hazards being proportional within each stratum, for the other covariates).

2) Include the time-dependent covariate(s) in the model (may be harder to interpret, especially for continuous covariates)

Example: Duration of breastfeeding

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
poverty	1	-0.13921	0.09225	2.2771	0.1313	0.870
education	1	-0.17740	0.03843	21.3128	<.0001	0.837
IntEducation	1	0.04855	0.01536	9.9834	0.0016	1.050
birthyear	1	0.07415	0.01794	17.0928	<.0001	1.077

The hazard ratio for education is not constant over time, needs to be interpreted separately for different time points.

Example: Duration of breastfeeding

HR for education:

$$\exp\{b_1 \cdot education + b_2 \cdot \ln(t) \cdot education\}$$

$$= \exp\{(b_1 + b_2 \cdot \ln(t))education\}$$

HR for a 1 unit's increase in education:

$$\exp\{b_1 + b_2 \cdot \ln(t)\}$$

Example: Duration of breastfeeding

HR for education at 3 months (13 weeks):

$$\exp\{-0.17740 + 0.04855 \cdot \ln(13)\}$$

$$= 0.949$$

At 6 months (26 weeks) the HR = 0.981

At 1 year (52 weeks) the HR = 1.01

Program L6

- **Regression for survival data**
  - Cox's proportional hazards regression, cont'd
    - Testing the PH assumption
    - Graphical methods to investigate the PH assumption
    - What to do if the proportional hazards assumption is violated
    - Regression diagnostics

Regression diagnostics

Four aspects of the Cox proportional hazards regression model are of interest to examine:

1) Which is the best functional form of a covariate to explain the effect on survival, adjusting for other covariates (Z, ln Z, etc)?
2) Does the assumption of proportional hazards hold?
3) What is the accuracy of the model for predicting the survival of a given subject?
4) What influence does each individual have on the model fit?

Residuals and Cox regression

	Observed value, y	Estimated value, $\hat{y}$	Residual $y - \hat{y}$
Linear regression	132	128	4
Cox regression	t	$h(t)$	?

In Cox regression there is no residual in the model (as there is for linear regression)

Residuals and Cox regression

There are a number of different residuals for the Cox model:

- **Cox-Snell residuals**, used to check the overall fit of a Cox model
- **Martingale residuals**, used to determine the functional form of a covariate to be included in a Cox model
- **Score residuals**, used for graphical checks of the proportional hazards assumption
- **Deviance residuals**, used to examine model accuracy for each individual
- **Schoenfeld residuals**, used to determine the influence of individual observations

Cox-Snell residuals

True hazard for an individual j:

$$H_j(t) = -\ln S_j(t) = H_0(t)\exp(\boldsymbol{\beta}^t \mathbf{Z}_j)$$

Baseline cumulative hazard

adjusted for values of $\mathbf{Z}$

If the model is correct, the estimated cumulative hazard for each individual at the time of their event or censoring should be like a censored sample from a unit exponential distribution*.

One cumulative hazard can be estimated for each individual j, and this is used as a residual.

Cox, D. R., and E. J. Snell (1968). A General Definition of Residuals. Journal of the Royal Statistical Society. Series B, Vol. 30, No. 2, pp. 248-275

Cox-Snell residuals

Cox-Snell residual:

$$r_j = \hat{H}_0(t)\exp(\mathbf{b}^t\mathbf{Z}_j) \quad = -\ln(\hat{S}_j(t))$$

↑

estimated by
the Cox model

This can be thought of as the expected number of events for individual j.

If the model is correct, and the estimated β's are close to the true β's, then the r_j's should behave as they come from a unit exponential distribution.

Using the Cox-Snell residuals to check the overall fit of the model

Procedure:

1) Calculate the Nelson-Aalen estimator of the cumulative hazard *of the residuals.*
2) Plot the estimated cumulative hazard vs. r_j
3) If the model fits, this plot should be a straight line through the origin with a 45 degree slope.

Using the Cox-Snell residuals to check the overall fit of the model

Drawbacks/warnings:

- If this plot is not linear, this method does not indicate the type of departure from the model
- This method is not appropriate for small samples
- The exponential distribution for the residuals holds only for the true parameter values. When using estimates (as we always do), departures from the exponential distribution may partly be due to the uncertainty in estimating β and H. This uncertainty is largest in the right of the plot and for small samples.

Example: Duration of breastfeeding

```
proc phreg data = bf noprint;
  model time*complete(0)= poverty smoke birthyear/ties=exact;
  output out = coxsnell LOGSURV = h;
run;
```

-logsurv gives the Cox-Snell residuals

```
data coxsnell; set coxsnell;
  r=-h;
run;
```

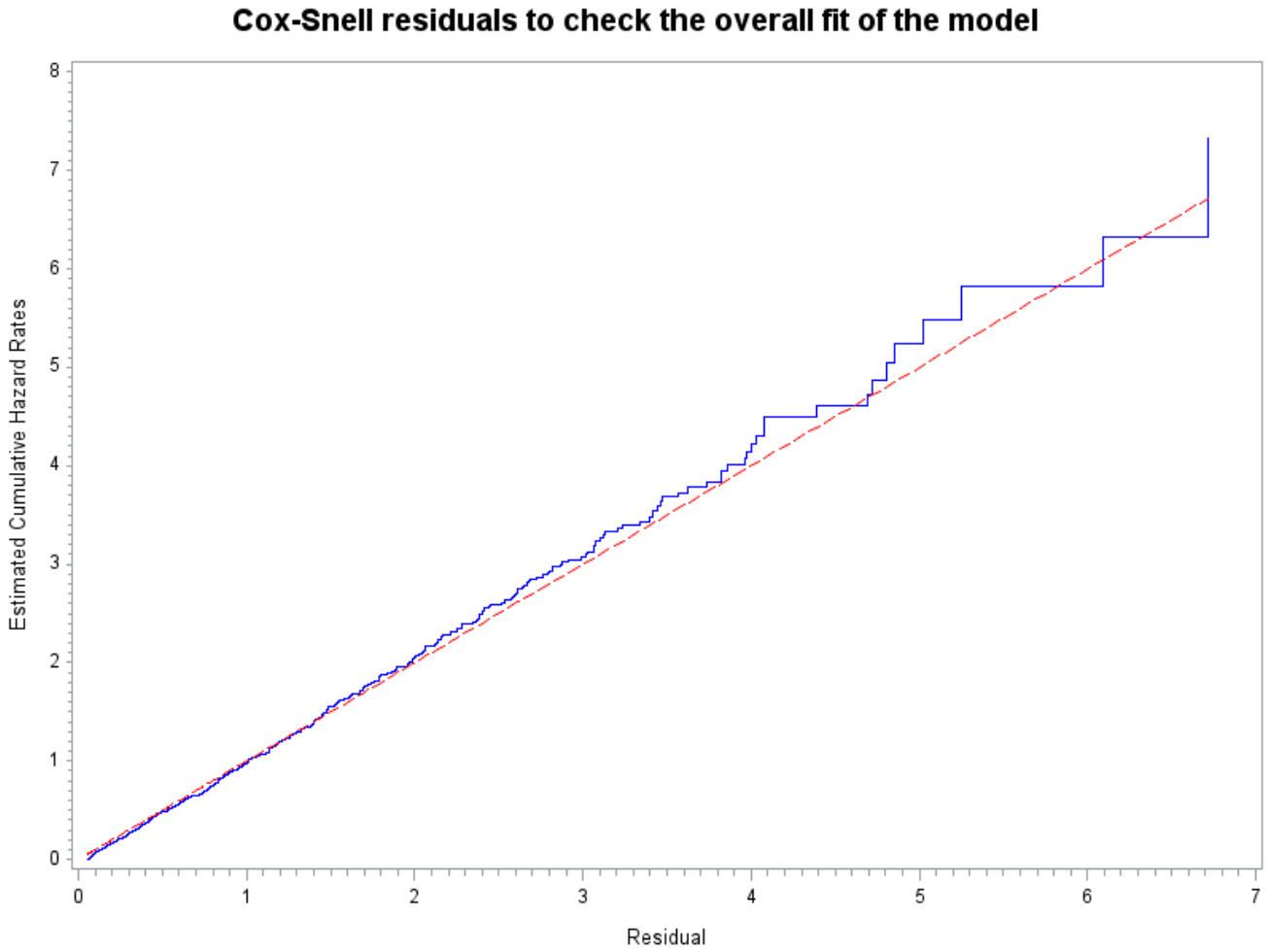
Example: Duration of breastfeeding

```
ods output ProductLimitEstimates=figure;
proc lifetest data=coxsnell nelson plots=none;
 time r*complete(0);
run;
```

Nelson-Aalen estimates

Based on the Cox-Snell residuals

Example: Duration of breastfeeding



The model fits the data pretty well

Generalized R^2 (Cox and Snell 1989)

$$R^2 = 1 - e^{-(LRT/n)}$$

where LRT= Likelihood ratio test statistic:

$$LRT = \text{-2logL(0)} - \text{[-2logL(p)]}$$

Likelihood of model without covariates

Likelihood of the fitted model with p covariates

Cox, D. R., and E. J. Snell, 1989. *The Analysis of Binary Data, 2nd ed.* London: Chapman and Hall.

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Generalized R^2 (Cox and Snell 1989)

The generalized R^2 is a statistic between 0 and 1 that is larger when the covariates are more strongly associated with the dependent variable.

NOT "proportion of variation explained by the model“.

Sensitive to the degree of censoring.

Example: Duration of breastfeeding

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	5485.411	5451.970
AIC	5485.411	5459.970
SBC	5485.411	5479.143

Summary of the Number of Event and Censored Values			
Total	Event	Censored	Percent Censored
927	892	35	3.78

$$LRT = 5485.411 - 5451.970 = 33.441$$

$$R^2 = 1 - e^{-(33.441/927)} = 0.037$$

There is a week association between the covariates and the duration of breastfeeding

Residuals and Cox regression

There are a number of different residuals for the Cox model:

- **Cox-Snell residuals**, used to check the overall fit of a Cox model
- **Martingale residuals**, used to determine the functional form of a covariate to be included in a Cox model
- **Score residuals**, used for graphical checks of the proportional hazards assumption
- **Deviance residuals**, used to examine model accuracy for each individual
- **Schoenfeld residuals**, used to determine the influence of individual observations

Martingale residuals

Martingale residual for fixed Z (not time-dependent):

$$M_j = \delta_j - r_j$$

Event indicator for individual *j* (0 or 1)

Cox-Snell residual for individual *j* (estimated cumulative hazard)

This can be thought of as the expected number of events for individual *j*.

Can be interpreted as the difference between observed and expected number of events over time, i.e. an estimate of the excess number of events seen in the data but not caught by the model.

Using Martingale residuals to determine the functional form of a covariate

Z_1 = a covariate for which we are uncertain of what functional form of the covariate we should use.

$f(Z_1)$ = best function of Z_1 to explain its effect on the time to event.

Using Martingale residuals to determine the functional form of a covariate

Procedure:

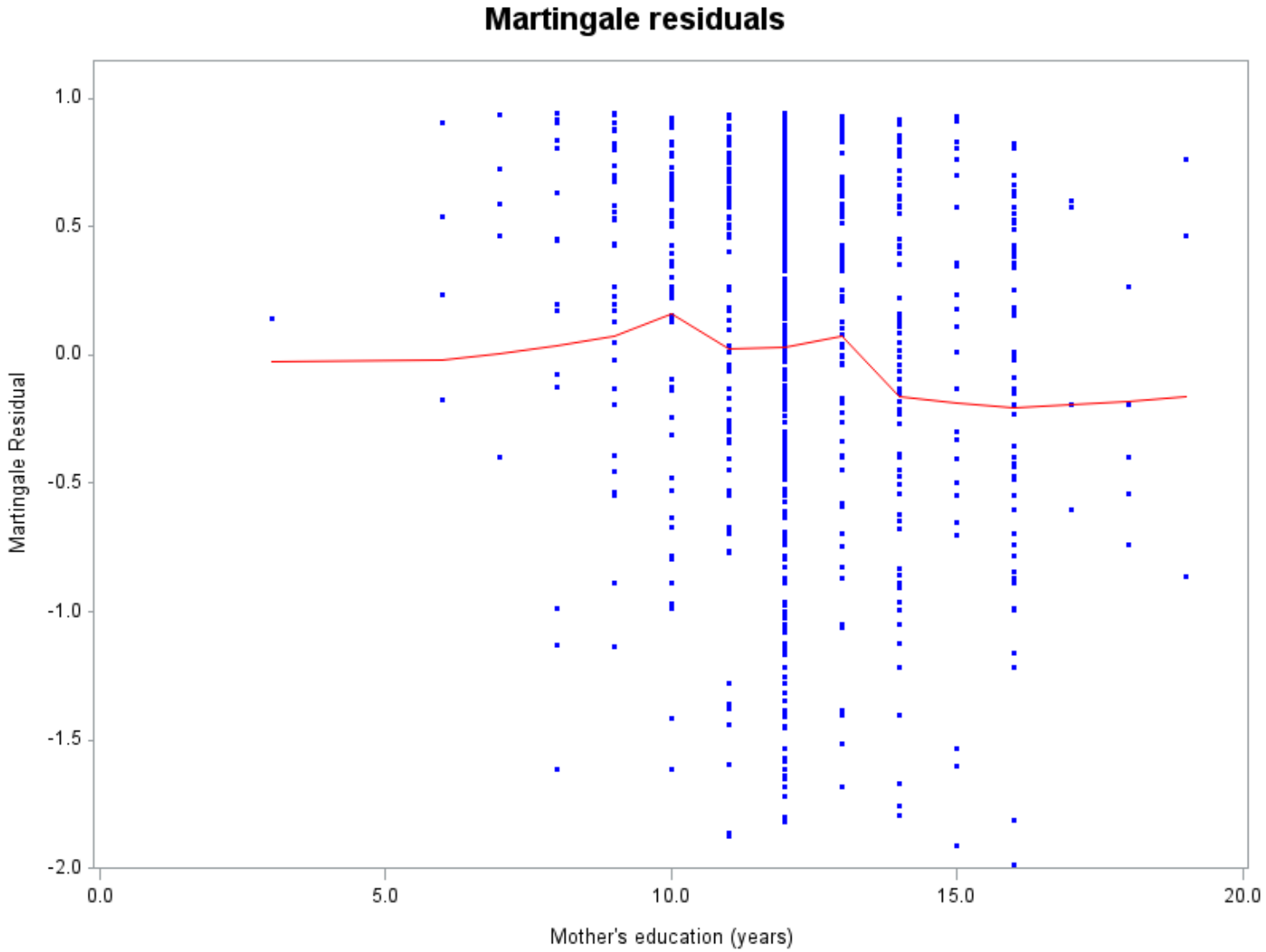
1) Fit a Cox model excluding the covariate which we want to investigate (Z_1), and calculate the Martingale residuals.
2) Plot the residuals vs. the value of Z_1 using a lowess smoothing function (lowess=LOcally Weighted Scatterplot Smooting).
3) If the plot is linear, no transformation of Z_1 is needed. If there is a threshold, a discretized version of Z_1 is indicated.

Example: Duration of breastfeeding

```
proc phreg data = bf noprint;
  model time*complete(0)= smoke poverty birthyear/ties=exact;
  output out = martingale resmart=mgale;
run;
```

Martingale residuals

Example: Duration of breastfeeding



Roughly linear, no obvious threshold.

Example of non-linear relationship

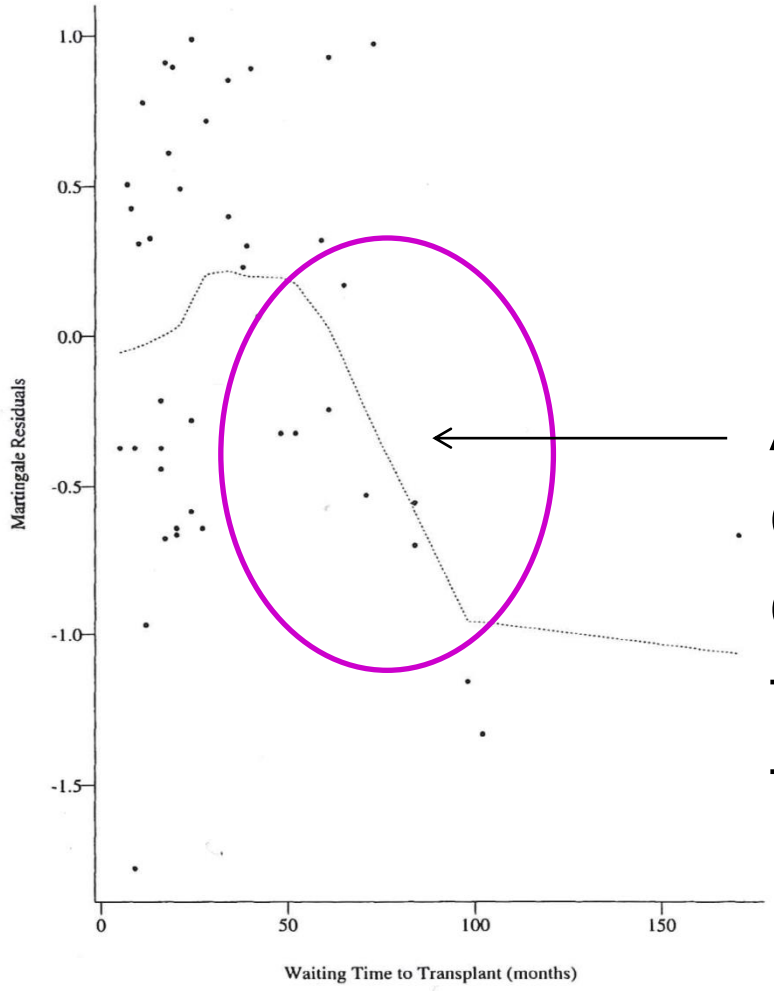


Figure 11.4 *Plot of martingale residual verus waiting time to transplant and LOWESS smooth*

Not linear.

A drop, suggesting that this covariate could be categorized to better explain the relationship with the time variable

Residuals and Cox regression

There are a number of different residuals for the Cox model:

- **Cox-Snell residuals**, used to check the overall fit of a Cox model
- **Martingale residuals**, used to determine the functional form of a covariate to be included in a Cox model
- **Score residuals**, used for graphical checks of the proportional hazards assumption
- **Deviance residuals**, used to examine model accuracy for each individual
- **Schoenfeld residuals**, used to determine the influence of individual observations

Searching for outliers

It is of interest to see if any of the observations has a response not well predicted by the fitted model, i.e. to examine the model for any outliers.

The martingale residual could be used, being a measure of the difference between the observed number of events and the expected number of events the individual would experience according to the model.

But: Martingale residuals are highly skewed, can take values from $-\infty$ to +1.

Deviance residuals

The deviance residuals are a transform of the martingale residuals, more symmetrically distributed around zero.

Deviance residuals have a distribution closer to the Normal distribution (asymptotically ND).

Using Deviance residuals to assess the effect of each individual (outlier check)

Procedure:

1) Plot the individual deviance residuals D_j versus the risk scores $\sum_{k=1}^{p} b_k Z_{jk}$

2) Under light to moderate censoring, D_j should look like a sample of Normally distributed noise.

3) Potential outliers will have deviance residuals with large absolute values that don't fit the Normal distribution.

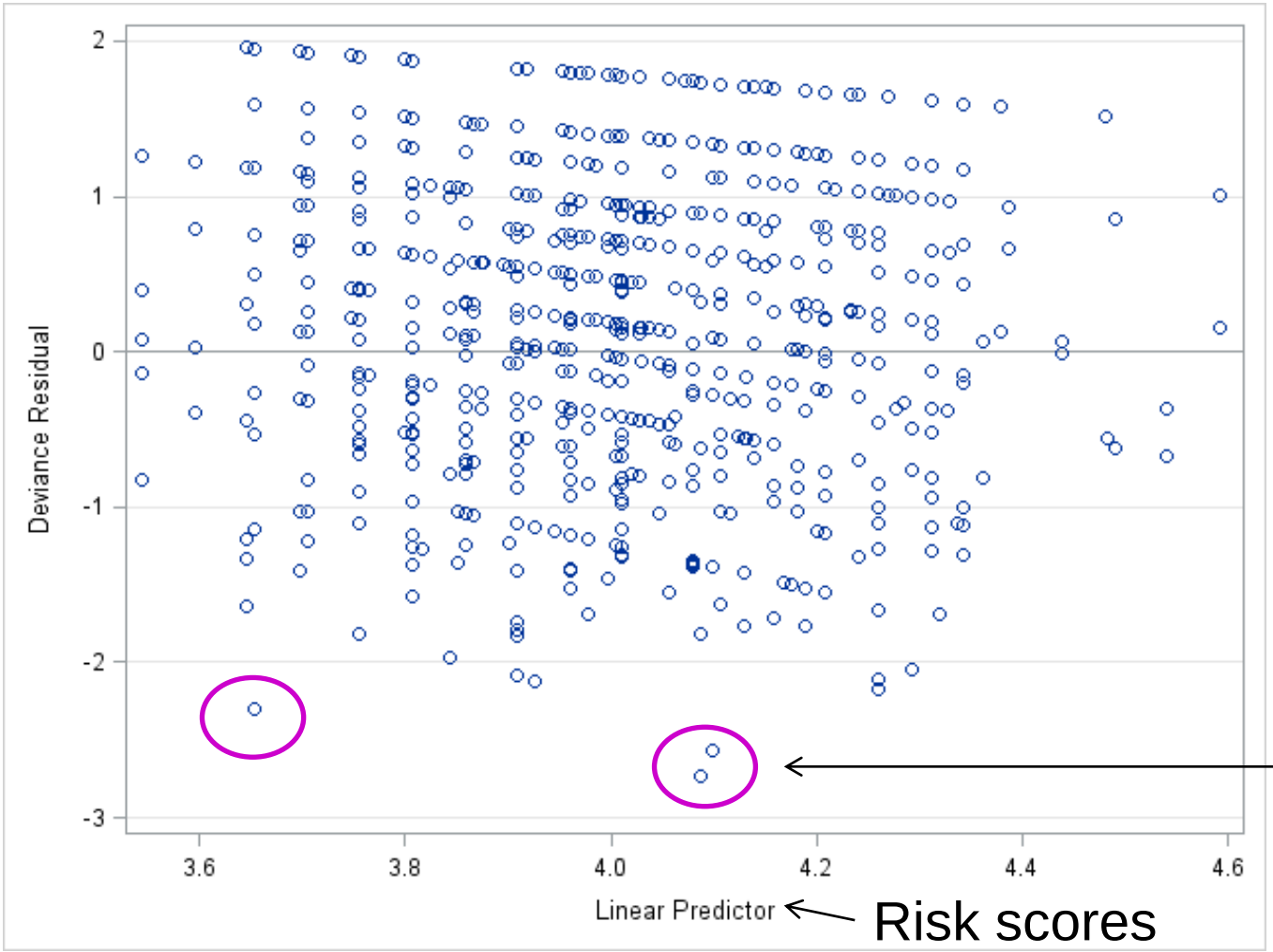
Risk score vs. the estimated risk of experiencing the event

Risk score: $\sum_{k=1}^{p} b_k Z_{jk}$

Estimated risk of experiencing the event for an individual:

$$\exp \sum_{k=1}^{p} b_k Z_{jk}$$

Example: Duration of breastfeeding



Potential outliers

Risk scores

Example: Duration of breastfeeding

	Duration of breast feeding (weeks)	Linear Predictor	Deviance Residual
1	120	4.0877810086	-2.737429889
2	104	4.0989420776	-2.566037036
3	192	3.6535222193	-2.294796572
4	64	4.2599687974	-2.174277282
5	96	3.9267542888	-2.118338552
6	60	4.2599687974	-2.110382689

The potential outliers

Based on their risk scores they should have relatively short/medium breast-feeding duration time, but they have in fact the longest times in the study.

Checking the influence of possible outliers

The influence of possible outliers on the estimation process can be checked by the following procedure:

1) Fit the Cox model including all observations to obtain **b**, estimates of β
2) Fit the Cox model excluding one of the possible outliers, obtaining $\mathbf{b}_j$ (excluding individual j).
3) If $\mathbf{b}-\mathbf{b}_j$ is close to 0, the jth observation has little influence on the estimate.

This is feasible for a relatively small number of outliers.

Retention or deletion of the outlier

Retain possible outliers unless demonstrable proof indicates that they are truly deviant and not representative of any observations in the population.

If they do represent any part of the population, no matter how uncommon they are, they should be retained to ensure generalizability to the entire population.

When deleting outliers, the analysis may improve, but the generalizability is limited. You have to redefine the population!

Residuals and Cox regression

There are a number of different residuals for the Cox model:

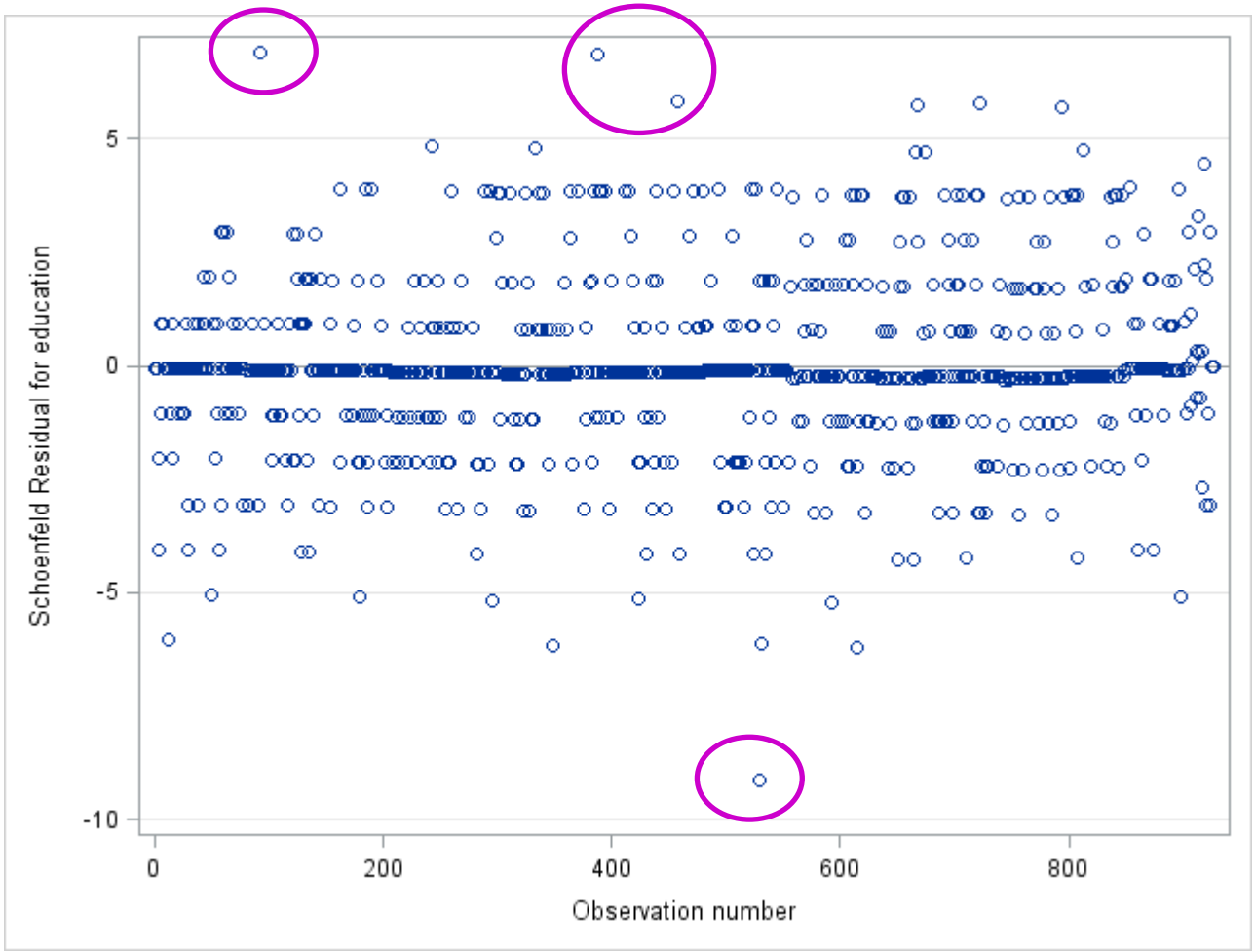
- **Cox-Snell residuals**, used to check the overall fit of a Cox model
- **Martingale residuals**, used to determine the functional form of a covariate to be included in a Cox model
- **Score residuals**, used for graphical checks of the proportional hazards assumption
- **Deviance residuals**, used to examine model accuracy for each individual
- **Schoenfeld residuals**, used to determine the influence of individual observations

Using Schoenfeld residuals to determine the influence of individual observations

Procedure:

1) Fit a Cox model including all covariates, and calculate the Schoenfeld residuals.
2) Plot the residuals vs. the individual number, or vs. the covariate Z_{jk} for each covariate.
3) This shows the influence of the jth observation for the kth covariate.

Example: Duration of breastfeeding



Observations with large influence on the model estimates should be examined further

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Program L7

- **Parametric regression for survival data**
  - Weibull distribution
  - Log logistic distribution
  - Diagnostic methods for parametric models

- **Declaration on professional ethics**

Recap (L1 p. 49): Modelling of covariate effects on survival

Two popular approaches:

1) Accelerated failure-time model, analogous to the classical linear regression approach.
2) Modelling the conditional hazard rate as a function of the covariates.

Recap (L1 p. 50): Accelerated failure-time model

$Y = \ln(X)$ is modelled.

A linear model is assumed for Y:

$$Y = \mu + \boldsymbol{\gamma}^{t}\mathbf{Z} + \sigma W$$

$\boldsymbol{\gamma}^{t} = (\gamma_1, \ldots, \gamma_p)$ = vector of regression coefficients

W = error distribution.

The regression coefficients are estimated using maximum likelihood methods.

Recap (L1 p. 51): Accelerated failure-time model

$$X = e^{Y}$$

The effect of the explanatory variables in the original time scale is accelerated (or degraded) by a constant factor, thereof the name of the model (further explanation in the book).

The use of the accelerated failure-time model is restricted by the error distributions one can assume.

Accelerated failure time model

$$S(x \mid \mathbf{Z}) = \Pr(X > x \mid \mathbf{Z}) = \Pr(Y > \ln x \mid \mathbf{Z})$$

$$= \Pr(\mu + \boldsymbol{\gamma}'\mathbf{Z} + \sigma W > \ln x \mid \mathbf{Z})$$

$$= \Pr(\mu + \sigma W > \ln x - \boldsymbol{\gamma}'\mathbf{Z} \mid \mathbf{Z})$$

$$= \Pr(\exp\{\mu + \sigma W\} > x \exp\{-\boldsymbol{\gamma}'\mathbf{Z}\} \mid \mathbf{Z})$$

Accelerated failure time model

If Z=0: $Y = \mu + \boldsymbol{\gamma}'\mathbf{Z} + \sigma W = \mu + \sigma W$

$$X = \exp\{Y\} = \exp\{\mu + \sigma W\}$$

$$S_0(x) = \Pr(\exp\{\mu + \sigma W\} > x)$$

$$S(x \mid \mathbf{Z}) = \Pr(\exp\{\mu + \sigma W\} > x \exp\{-\boldsymbol{\gamma}'\mathbf{Z}\} \mid \mathbf{Z})$$

$$= S_0(x \exp\{-\boldsymbol{\gamma}'\mathbf{Z}\})$$

$$= S_0(x \exp\{\boldsymbol{\theta}'\mathbf{Z}\}) \qquad (\boldsymbol{\theta} = -\boldsymbol{\gamma})$$

Accelerated failure time model

The accelerated failure time model is defined by the relationship

$$S(x \mid \mathbf{Z}) = S_0\left(x \exp\{\boldsymbol{\theta}'\mathbf{Z}\}\right)$$

Acceleration factor (constant for an individual)

Acc. factor = 2 ➡ An individual is ageing (approaching the event) twice as fast as the baseline

Acc. factor > 1 ➡ Individual ageing faster than baseline

Acc. factor < 1 ➡ Individual ageing slower than baseline

Example: dog vs. human life times

Survival function for dogs vs. humans:

$$S(x \mid Z) = S_0\left(x \exp\{\theta Z\}\right) = S_0\left(x \cdot 7 \mid Z = 1\right)$$

1 = dog
0 = human

Survival for humans

Acceleration factor = 7 for dogs

Dogs are ageing 7 times as fast as humans

The survival at time x for humans is the same as at time $x \cdot 7$ for dogs.

Accelerated failure time model vs. Cox model

Accelerated failure time model:

$$h(x \mid \mathbf{Z}) = h_0\left(x \exp\{\boldsymbol{\theta}'\mathbf{Z}\}\right)\exp\{\boldsymbol{\theta}'\mathbf{Z}\}$$

$\exp\{\boldsymbol{\theta}'\mathbf{Z}\} = 2$: Individual ageing twice as fast as baseline

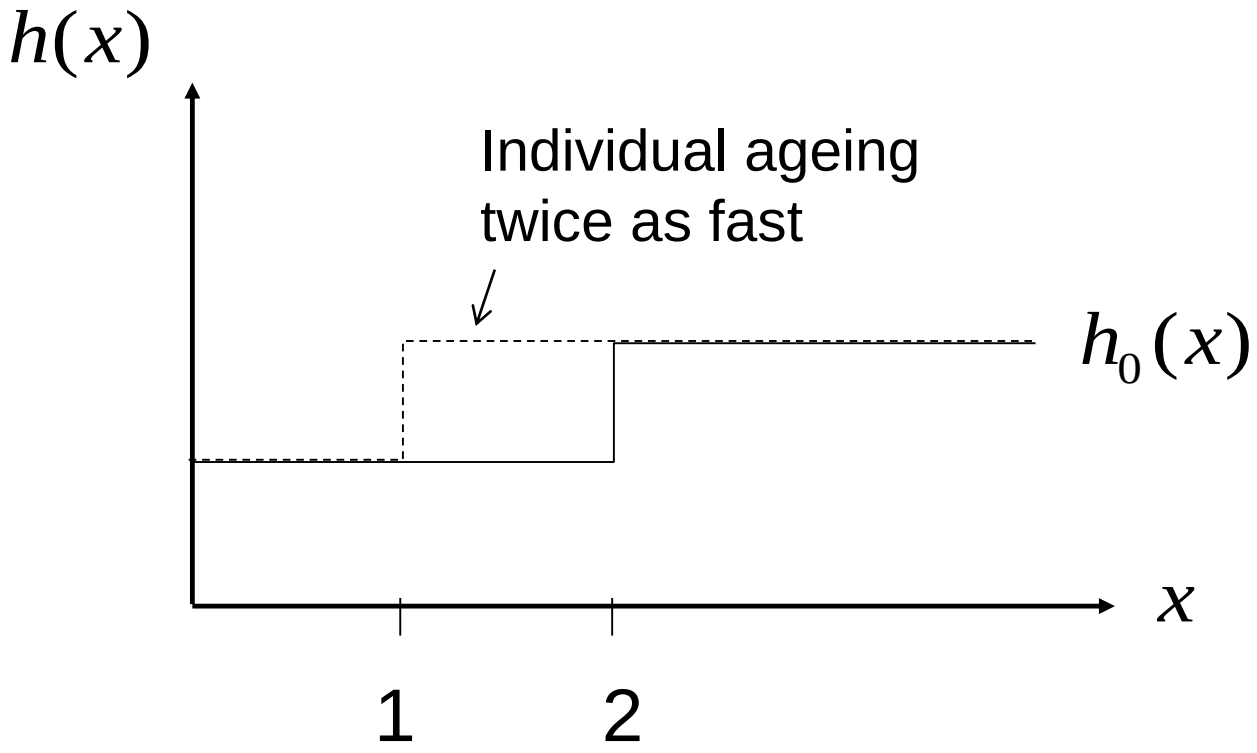
Cox model:

$$h(x \mid \mathbf{Z}) = h_0(x)\exp\{\boldsymbol{\beta}'\mathbf{Z}\}$$

$\exp\{\boldsymbol{\beta}'\mathbf{Z}\} = 2$: The hazard is twice as large

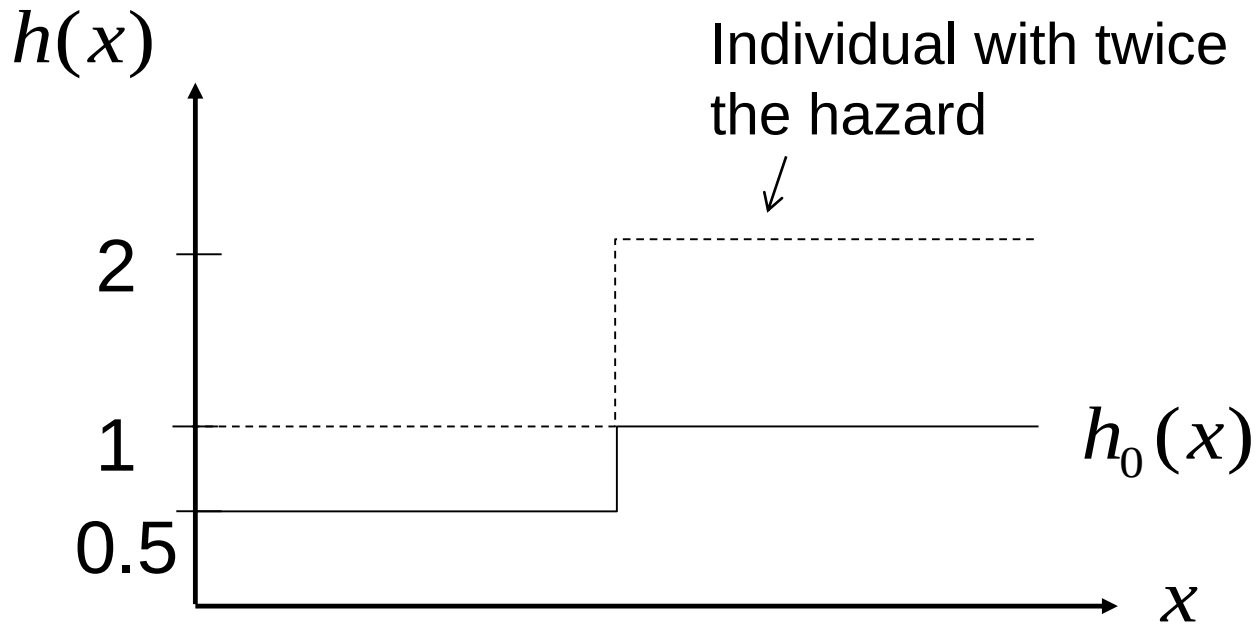
Accelerated failure time model vs. Cox model

Accelerated failure time model:



Accelerated failure time model vs. Cox model

Cox model:



Parametric regression models

Common choices for the error distribution:

- Standard ND which yields a log normal regression model
- Extreme value distribution which yields a Weibull regression model
- Logistic distribution which yields a log logistic regression model

Parametric regression models

When a parametric model fits the data well, it can give more precise estimates than the Cox model.

If the parametric model doesn't fit the data, the results can be misleading.

Important that the parametric model is chosen correctly!

Weibull distribution

$$S(x) = \mathrm{e}^{-\lambda x^{\alpha}}$$

$$S(x \mid \mathbf{Z}) = S_0\left(x \exp\{\boldsymbol{\theta}'\mathbf{Z}\}\right)$$

$$= \mathrm{e}^{-\lambda\left(x \exp\{\boldsymbol{\theta}'\mathbf{Z}\}\right)^{\alpha}}$$

$$= \mathrm{e}^{-\lambda x^{\alpha} \exp\{\alpha\, \boldsymbol{\theta}'\mathbf{Z}\}}$$

$$= \mathrm{e}^{-\lambda \exp\{\alpha\, \boldsymbol{\theta}'\mathbf{Z}\}\; x^{\alpha}}$$

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Weibull distribution

$$h(x) = -\frac{d \ln S(x)}{dx} = \frac{d}{dx} \lambda \exp\{\alpha \, \boldsymbol{\theta}' \mathbf{Z}\} \, x^{\alpha}$$

$$= \exp\{\alpha \, \boldsymbol{\theta}' \mathbf{Z}\} \underbrace{\lambda \, \alpha \, x^{\alpha - 1}}_{h_0(x)}$$

$$= h_0(x) \exp\{\boldsymbol{\beta}' \mathbf{Z}\}$$

Proportional hazards model!

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Weibull distribution

The Weibull distribution is the only parametric regression model with both a proportional hazards representation and an accelerated failure time representation.

Estimates of the regression coefficients are found numerically (maximum likelihood estimates), by using proc **lifereg**.

Weibull distribution

Accelerated failure time model:

$$Y = \mu + \gamma^{t}\mathbf{Z} + \sigma W$$

When W has the standard extreme value distribution this leads to a proportional hazards model for X with a Weibull baseline hazard.

$$h(x) = \lambda\, \alpha\, x^{\alpha-1} \exp\{\boldsymbol{\beta}'\mathbf{Z}\}$$

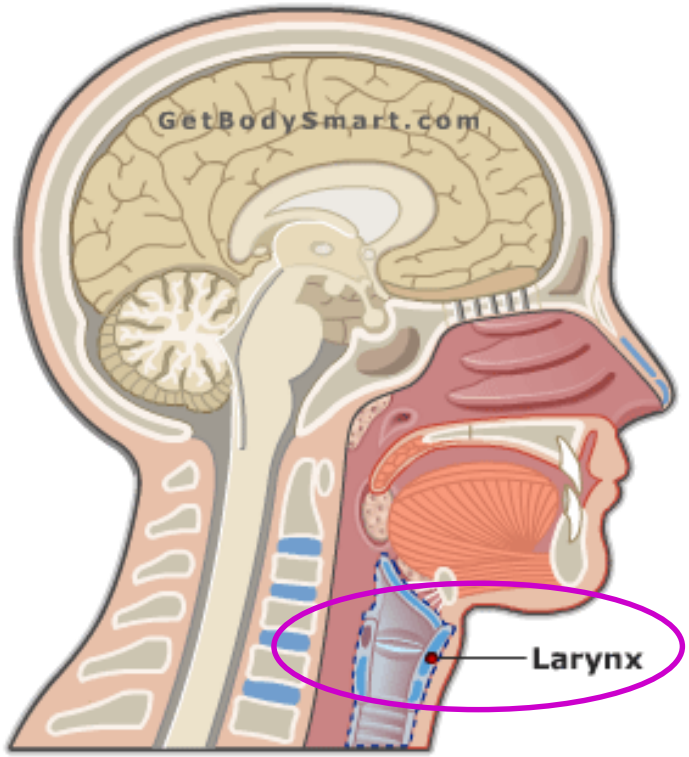
$\lambda = \exp(-\mu/\sigma)$

$\alpha = 1/\sigma$

$\beta_j = -\gamma_j/\sigma_j$

Example: larynx cancer

A study of 90 males with cancer of the larynx is described in section 1.8.



Example: larynx cancer

Variables:

X = survival time from first treatment (years)

death = death indicator

stage = Four stages of the disease, stage I – stage IV, ordered from least serious to most serious.

age = patient's age (years)

Example: larynx cancer

Accelerated failure time model:

$$Y = \mu + \gamma_1 \text{age} + \gamma_2 \text{stage2} + \gamma_3 \text{stage3} + \gamma_4 \text{stage4} + \sigma W$$

W~standard extreme value

Estimates of γ and β

proc **lifereg** gives estimates of **γ** (for $Y = \ln X$)

To get the estimates of **β**, in order to interpret the relationship between the covariates and X, we have to transform the model estimates and their standard errors (SAS code available at Studium).

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Example: larynx cancer

Estimates of γ (for Y = ln X)

Analysis of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept	1	3.5288	0.9041	1.7567	5.3008	15.23	<.0001
age	1	-0.0175	0.0128	-0.0425	0.0076	1.87	0.1717
stage2	1	-0.1477	0.4076	-0.9465	0.6511	0.13	0.7171
stage3	1	-0.5866	0.3199	-1.2136	0.0405	3.36	0.0668
stage4	1	-1.5441	0.3633	-2.2561	-0.8321	18.07	<.0001
Scale	1	0.8848	0.1084	0.6960	1.1250		
Weibull Shape	1	1.1301	0.1384	0.8889	1.4368		

Example: larynx cancer

Estimates of β (for X)

Transformed Parameter Estimate

	Parameter Estimate	Standard Error
Intercept	0.019	0.019
age	0.020	0.014
stage2	0.167	0.461
stage3	0.663	0.356
stage4	1.745	0.415
Scale	1.130	0.138

Can be interpreted as proportional hazards estimates

E.g. relative risk for Stage3, compared to Stage1

$$= e^{0.663} \quad = 1.94$$

Example: larynx cancer

Accelerated failure time model:

$$\exp\{\boldsymbol{\theta}'\mathbf{Z}\} = 1.94$$

Individuals in stage 3 age (approach death) 1.94 times faster than individuals in stage 1 (assuming a Weibull model)

Cox model:

$$\exp\{\boldsymbol{\beta}'\mathbf{Z}\} = 1.94$$

The hazard (risk of dying) is 1.94 times larger for individuals in stage 3 compared to stage 1

Medians with the accelerated failure time model

The median time to event with a covariate $\mathbf{Z}$ is the baseline median divided by the acceleration factor.

$$\text{Median} \mid \mathbf{Z} = \frac{\text{Baseline median} \mid \mathbf{Z} = 0}{\exp\{\boldsymbol{\theta}'\mathbf{Z}\}}$$

Where $\theta = -\gamma$ from the log model

In other words, the median time to event is $1/\exp\{\boldsymbol{\theta}'\mathbf{Z}\}$ times as large as the baseline median.

Example: larynx cancer

Median lifetime for Stage 3 compared to Stage 1 (baseline):

$1/\exp\{\boldsymbol{\theta}'\mathbf{Z}\}$

Where $\theta = -\gamma$ for the Stage 3 covariate can be found from the estimated model

Example: larynx cancer

Estimates of γ (for Y = ln X) $\theta = -\gamma$

Analysis of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept	1	3.5288	0.9041	1.7567	5.3008	15.23	<.0001
age	1	-0.0175					.717
stage2	1	-0.1477					0.7171
stage3	1	-0.5866					0.0668
stage4	1	-1.5441					<.0001
Scale	1	0.8848					
Weibull Shape	1	1.1301	0.1384	0.8889	1.4368		

$$\exp\{\theta\ \text{Stage3}\} = \exp(0.5866) = 1.8$$

Median lifetime for Stage3 is 0.56 times (1/1.8) the median lifetime for Stage1.

Log logistic distribution

$$S(x) = \frac{1}{1+\lambda x^{\alpha}}$$

$$S(x \mid \mathbf{Z}) = S_0\left(x \exp\{\boldsymbol{\theta}'\mathbf{Z}\}\right) = \frac{1}{1+\lambda\left(x \exp\{\boldsymbol{\theta}'\mathbf{Z}\}\right)^{\alpha}}$$

$$= \frac{1}{1+\lambda\, x^{\alpha} \exp\{\alpha\, \boldsymbol{\theta}'\mathbf{Z}\}}$$

$$= \frac{1}{1+\lambda\, x^{\alpha} \exp\{\boldsymbol{\beta}'\mathbf{Z}\}}$$

Log logistic distribution

The factor $e^{\boldsymbol{\beta}'\mathbf{Z}}$ for the log logistic model can be interpreted in terms of **odds ratios**.

The odds of survival beyond time t:

$$\frac{S(x\,|\,\mathbf{Z})}{1-S(x\,|\,\mathbf{Z})} = e^{-\boldsymbol{\beta}'\mathbf{Z}} \frac{S(x\,|\,\mathbf{Z}=0)}{1-S(x\,|\,\mathbf{Z}=0)}$$

The factor $e^{-\boldsymbol{\beta}'\mathbf{Z}}$ is thus an estimate of how much the baseline odds of survival changes for an individual with covariates $\mathbf{Z}$.

Log logistic distribution

The log logistic distribution is the only parametric regression model with both a proportional odds representation and an accelerated failure time representation.

Estimates of the regression coefficients γ are again found numerically (maximum likelihood estimates), by using proc **lifereg**.

Log logistic distribution

Accelerated failure time model:

$$Y = \mu + \gamma^{t}\mathbf{Z} + \sigma W$$

When W has the standard logistic distribution this leads to a log logistic proportional odds model for X.

$$S(x \mid \mathbf{Z}) = \frac{1}{1 + \lambda\, x^{\alpha} \exp\{\boldsymbol{\beta}'\mathbf{Z}\}}$$

$\lambda = \exp(-\mu/\sigma)$ $\quad \alpha = 1/\sigma$ $\quad \beta_j = -\gamma_j/\sigma_j$

Example: larynx cancer

Variables:

X = survival time from first treatment (years)

death = death indicator

stage = Four stages of the disease, stage I – stage IV, ordered from least serious to most serious.

age = patient's age (years)

Example: larynx cancer

Accelerated failure time model:

$$Y = \mu + \gamma_1 \text{age} + \gamma_2 \text{stage2} + \gamma_3 \text{stage3} + \gamma_4 \text{stage4} + \sigma W$$

W~standard logistic

proc **lifereg** again gives estimates of **γ** (for $Y = \ln X$).

Example: larynx cancer

Estimates of γ (for $Y = \ln X$)

Analysis of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept	1	3.1022	0.9527	1.2350	4.9694	10.60	0.0011
age	1	-0.0151	0.0138	-0.0421	0.0119	1.20	0.2734
stage2	1	-0.1257	0.4152	-0.9395	0.6881	0.09	0.7621
stage3	1	-0.8057	0.3539	-1.4993	-0.1122	5.18	0.0228
stage4	1	-1.7661	0.4257	-2.6004	-0.9319	17.22	<.0001
Scale	1	0.7152	0.0860	0.5651	0.9053		

$$\exp\{\theta\,\text{Stage3}\} = \exp(0.8057) = 2.2$$ (θ = - γ from the log model)

Median lifetime for Stage3 is 0.45 times (1/2.2) the median lifetime for Stage1 (assuming a log logistic model).

Example: larynx cancer

Estimates of β (for X)

Transformed Parameter Estimate

	Parameter Estimate	Standard Error
Intercept	0.013	0.018
age	0.021	0.019
stage2	0.176	0.581
stage3	1.127	0.498
stage4	2.469	0.632
Scale	1.398	0.168

Can be interpreted as odds ratio estimates

E.g. relative odds of Stage3, compared to Stage1

$$= e^{-1.127} = 0.32$$

Stage3 patients have an 0.32 times lesser **odds of survival** than Stage1 patients.

Example: larynx cancer

Accelerated failure time model:

$$\exp\{\boldsymbol{\theta}'\mathbf{Z}\} = \exp\{0.8057\} = 2.2$$

Individuals in stage 3 age (approach death) 2.2 times faster than individuals in stage 1 (assuming a log logistic model).

Proportional odds model:

$$\exp\{-\boldsymbol{\beta}'\mathbf{Z}\} = 0.32$$

The odds of surviving for individuals in stage 3 is 0.32 times the size of the odds for individuals in stage 1. (or: the odds of dying is 1 / 0.32=3.1 times as large)

Diagnostic methods for parametric models

Graphical checks of the appropriateness of the chosen parametric model can be used to find clearly inappropriate models (not to “prove” that a chosen model is correct).

If there are no covariates, the key tool is to find a function of the cumulative hazard rate which is linear in some function of time.

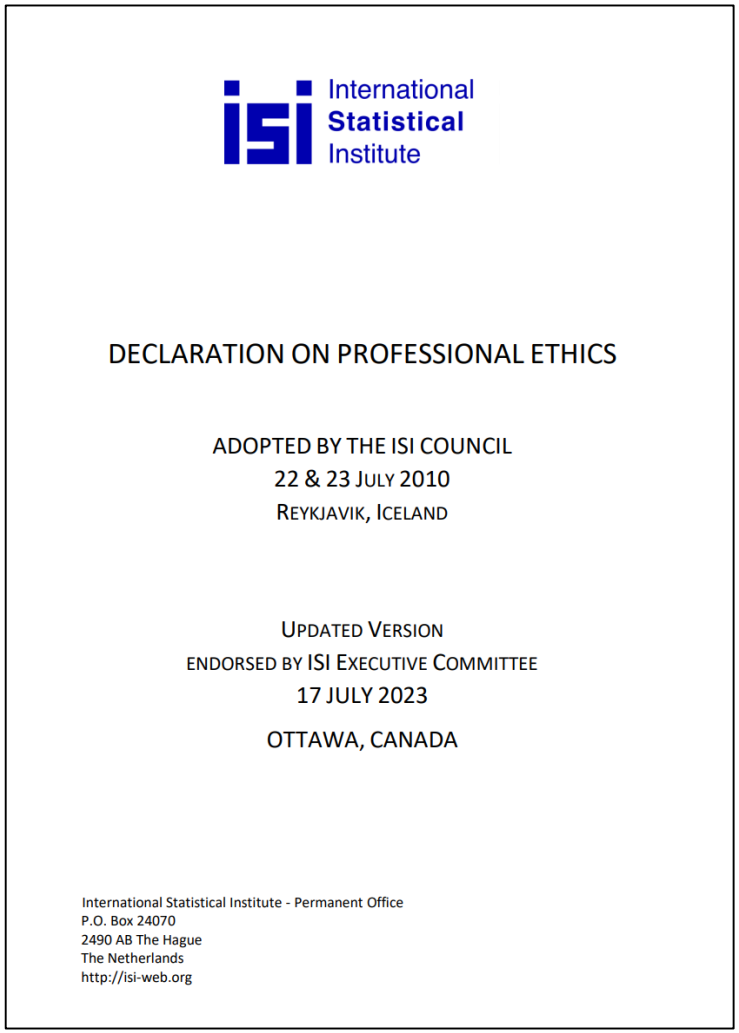
Different functions are used for different parametric models (described in section 12.5).

Diagnostic methods for parametric models

Analogs of the residual plots used for the Cox model can be used, with a redefinition of the residuals to incorporate the parametric form of the baseline hazard rates.

Different definitions are used for different parametric models.

Declaration on professional ethics (ISI)



International Statistical Institute

DECLARATION ON PROFESSIONAL ETHICS

ADOPTED BY THE ISI COUNCIL
22 & 23 JULY 2010
REYKJAVIK, ICELAND

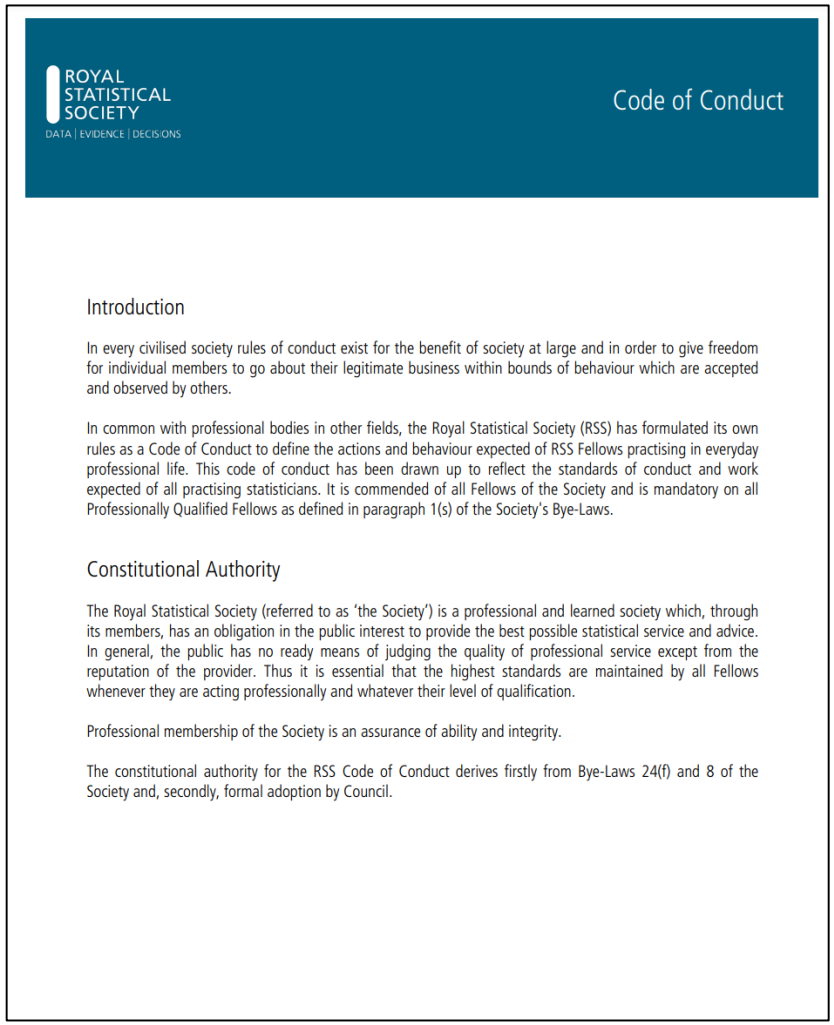
UPDATED VERSION
ENDORSED BY ISI EXECUTIVE COMMITTEE
17 JULY 2023
OTTAWA, CANADA

International Statistical Institute - Permanent Office
P.O. Box 24070
2490 AB The Hague
The Netherlands
http://isi-web.org

https://isi-web.org/sites/default/files/2024-01/isi-declaration-on-professional-ethics-English.pdf

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Code of Conduct (RSS)



ROYAL STATISTICAL SOCIETY
DATA | EVIDENCE | DECISIONS

Code of Conduct

Introduction

In every civilised society rules of conduct exist for the benefit of society at large and in order to give freedom for individual members to go about their legitimate business within bounds of behaviour which are accepted and observed by others.

In common with professional bodies in other fields, the Royal Statistical Society (RSS) has formulated its own rules as a Code of Conduct to define the actions and behaviour expected of RSS Fellows practising in everyday professional life. This code of conduct has been drawn up to reflect the standards of conduct and work expected of all practising statisticians. It is commended of all Fellows of the Society and is mandatory on all Professionally Qualified Fellows as defined in paragraph 1(s) of the Society's Bye-Laws.

Constitutional Authority

The Royal Statistical Society (referred to as 'the Society') is a professional and learned society which, through its members, has an obligation in the public interest to provide the best possible statistical service and advice. In general, the public has no ready means of judging the quality of professional service except from the reputation of the provider. Thus it is essential that the highest standards are maintained by all Fellows whenever they are acting professionally and whatever their level of qualification.

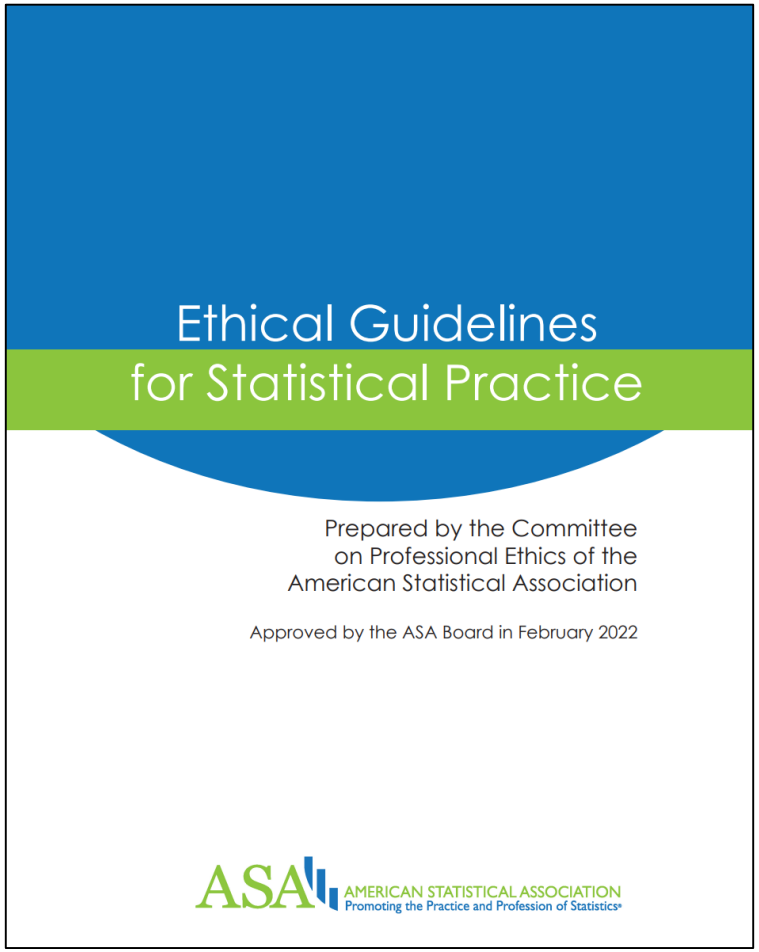
Professional membership of the Society is an assurance of ability and integrity.

The constitutional authority for the RSS Code of Conduct derives firstly from Bye-Laws 24(f) and 8 of the Society and, secondly, formal adoption by Council.

https://rss.org.uk/RSS/media/File-library/About/2019/RSS-Code-of-Conduct-2014.pdf

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Ethical Guidelines for Statistical Practice (ASA)



https://www.amstat.org/docs/default-source/amstat-documents/ethicalguidelines.pdf?Status=Master&sfvrsn=bdeeafdd_6/

Statistikfrämjandets etiska kod



https://statistikframjandet.se/wp-content/uploads/2023/06/Etisk-kod-Statistikframjandet-20230602.pdf

Declaration on professional ethics

What do you think is the most important part of the ethics document?

What would you do if you at a future job are asked to come up with a result that supports your boss' opinion (which may not be ethically correct).

How to use the ethics declaration in practice

The easy part is usually to agree with the declaration and work ethically.

The hard part is to stand up for the ethical way of working, without being influenced by pressure from e.g. politicians, funders, or job initiators.

“I’ll be happy to analyze this. During my work, I’ll follow the declaration on professional ethics for statisticians, which means that I might come up with results that support your thesis, but the analysis might also show results that don’t support your thesis.”